

Advanced Time Series Econometrics (ATSE)

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Preface

Foreword

These lecture notes are a follow-up to the ones for Time Series Analysis (TILM3541) concerning the course on Multivariate Time Series Analysis (TILM3586). The materials are largely based on the lecture materials for the course “Multivariate Time Series” (2011) by Professor Emeritus Pentti Saikkonen (University of Helsinki). In addition, the following sources have been utilized:

- Hamilton, J.D. (1994). Time Series Analysis. Princeton.
- Kilian, L. ja H. Lütkepohl (2017). Structural Vector Autoregressive Analysis. Cambridge University Press
- Lütkepohl, H. (2005). New Introduction to Multiple Time Series Analysis. Springer.
- Pena and Tsay (2021). Statistical Learning for Big Dependent Data. John Wiley & Sons Inc., Chapter 7.

I am grateful to MSc Samuel Rauhala and MSc Juho Pitkäranta for their assistance with this translation, as well as Professor Heikki Kauppi for his closely related translation. All remaining errors are the responsibility of the author.

At times the material references the prerequisite course [Time Series Econometrics](#).

Notation

On this course, we follow, for example, the following key notational selections:

- We use the same notation on random variables and their particular realized or observed values. One must understand the difference from the context.
- In the context of matrix algebra vectors are interpreted as column vectors. A vector $\mathbf{y}_t = (y_{1t}, \dots, y_{Kt})$ can be written in matrix notation as $\mathbf{y}_t = [y_{1t}, \dots, y_{Kt}]'$, where the symbol $'$ indicates a vector transpose (later also a matrix transpose).
- The above emphasize K -dimensional vector-valued time series and their values at time t . We have then observations of these vectors at time points $t = 1, \dots, T$.

- In the case of univariate time series (that is, $K = 1$), we can denote the whole time series data at times $t = 1, \dots, T$ as $(y_1, \dots, y_T)$. Furthermore, in the multiple time series case, we then have $(\mathbf{y}_1, \dots, \mathbf{y}_T)$ where $\mathbf{y}_t$ is defined as above.

1 Introduction

1.1 Background

In this course, we will consider statistical models for time series data. We concentrate, for example, the following questions:

- How such models can be applied to describe and forecast time series data?
- How are the model parameters estimated?
- How are hypotheses tested?

We start with a review of univariate time series models and methods. **The main objective is, however, in multiple time series case where multiple time series are analyzed simultaneously.** In addition to typical multivariate (multiple-equation) models, we also consider some essential tools for large (big) dependent data and introduce some ideas and machine learning-based techniques for time series.

Time series data are analyzed in various fields, including economics and finance, as well as social, natural, engineering, and medical sciences.

A (univariate) time series is a collection of data where observations correspond to consecutive time periods. A common notation for time series data is y_t , $t = 1, \dots, T$, where t denotes the time period and T is the total number of observations. The interval between observations y_{t-1} and y_t , or y_t and y_{t+1} , represents a unit of time, for example, an hour, a month, a quarter or one year. In many cases, it is natural to use more concrete labels for the time periods, such as $t = 2000, \dots, 2024$ for yearly data, or $t = 2000 : 1, 2000 : 2, \dots, 2024 : 12$ for monthly observations. Therefore, occasionally T may also refer to the final time label, such as 2024:12 (December 2024).

A vector-valued time series or **multiple time series** constitutes observations on a vector-valued variable in subsequent periods. We denote such a series by

$$\mathbf{y}_t = (y_{1t}, \dots, y_{Kt}), \quad t = 1, \dots, T,$$

where t indicates period or date, K is the number of variables and T is the number of observations.

- In this material, the bolded font in mathematical notations implies vectors and matrices (for the text it emphasizes important parts and concepts).
- For example, a classic three-variable (that is, $K = 3$) monetary policy model contains the (real) GDP (Gross Domestic Product) or unemployment rate, inflation and interest rate.
- If $K = 1$, then y_t is a **univariate** time series. Hereafter, when analysing the k th time series of the vector-valued $\mathbf{y}_t$, we may also denote it as y_{kt} .
- We assume throughout the course that the variables $y_{1t}, \dots, y_{Kt}$ are real-valued time series processes or can be at least treated like that. That is, observations are real numbers. In particular, this means that we do not allow clearly discrete such as binary-valued time series in this course. These discrete-valued cases are considered more detail in Nonlinear Time Series Analysis (University of Turku, TILM3589).

As described at the beginning in Notation, We use the same notation on random variables and their realized/observed values. One must understand the difference from the context. Moreover, **vectors are interpreted as column vectors**: A vector $\mathbf{y}_t = (y_{1t}, \dots, y_{Kt})$ can be written using matrix notation as

$$\mathbf{y}_t = [y_{1t} \cdots y_{Kt}]' = \begin{bmatrix} y_{1t} \\ \vdots \\ y_{Kt} \end{bmatrix}, \quad (K \times 1)$$

where the symbol $'$ indicates a vector transpose (later also matrix transpose).

Basics of modelling stationary univariate time series are briefly considered at first. In the univariate case y_t , the time series is scalar-valued, i.e. only one time series is analyzed at a time.

1.2 Empirical example

An example of a multivariate time series could be a collection of stocks from the Helsinki stock exchange in certain day or month in 2026. Another example and data that we will also later take a closer look on is a **bivariate model** with the **monthly change in the exchange rate of the euro and the USD** ($dexch_t$) between 1999:1–2009:12 as the other variable. The another variable is build upon the interest rates of the corresponding currencies (eurozone/US), which can be measured as **the difference in the average returns on the 10-year bonds in the eurozone and the US** ($rdif_t$).

Some economic motivation and remarks behind this application and data:

- According to Uncovered Interest Rate Parity (UIP), the expected future change in the exchange rate ($E_t(dexch_{t+1})$) should be driven by the interest rate differential. For empirical testing, assuming rational expectations, we typically replace the unobservable expectation with the realized future change ($dexch_{t+1}$) plus a random error term. However, in this course, we examine the dynamic interdependence between the current exchange rate change ($dexch_t$) and the interest differential ($rdif_t$) to analyze both predictive power and contemporaneous co-movement.
- While UIP provides a baseline theoretical condition for equilibrium, the relationship between exchange rates and long-term bond yields is theoretically multifaceted. The Sticky-Price Monetary Model suggests that interest rate shocks may cause immediate exchange rate overshooting (appreciation), generating short-run dynamics that contradict UIP. Furthermore, the Portfolio Balance Approach is particularly relevant for 10-year yields, as it accounts for risk premiums and imperfect asset substitutability. Finally, we must remember possible reverse causality driven by endogenous monetary policy, where central banks (e.g., the ECB or Fed) may adjust interest rates in response to significant exchange rate fluctuations to stabilize inflation
- Additionally, we utilize 10-year government bond yields rather than the theoretically standard short-term rates. While this implies a maturity mismatch regarding, e.g., strict UIP testing, long-term yields are used here to capture broader structural economic expectations (such as inflation and economic growth), providing a robust signal for demonstrating multivariate techniques.

The central point of interest here and for this course, as a general example of a multivariate time series, is the dependence between the examined series to each other (as motivated above), implying that their co-movements are not merely coincidental.

- As in the univariate case, stationarity and “trend-like” non-stationarity play a significant role in the analysis and techniques used for multivariate time series. In what follows, we will consider general topics relating to this in the case of multivariate time series. Later, we will consider the analysis of non-stationary time series.
- This bivariate example is selected partly because the two time series are visually clearly different, as we will see in Figure below and in the coming sections.

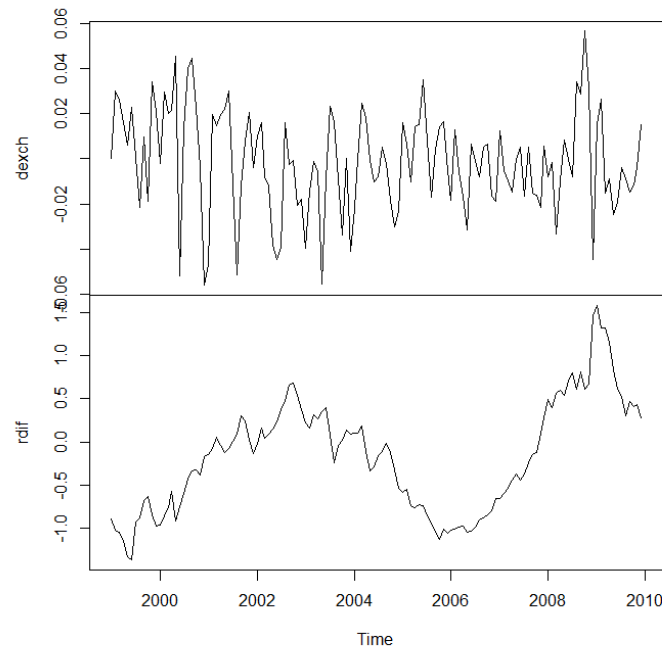


Figure: The change in the euro-USD exchange rate (EUR/USD) on the top and the difference of the 10-year bond yields in the eurozone and the US (bottom) between the time period (monthly data) 1999:1–2009:12.

1.3 Relation to macroeconometrics

The purpose of this course is to **introduce** some important statistical models and methods used in modern multiple (multivariate) time series analysis. The topics of this course are also closely related to the ones examined in a typical macroeconometrics (empirical macroeconomics) course. However, we introduce the methods **without} structural economic interpretations and identification assumptions**. These questions are left for the specific macroeconometrics courses such as **Macroeconometrics (TILM3609)**.

- The methods of this course are often used in the modern (applied) macroeconomic research, including, e.g., policy analysis and forecasting.
- The methods are also commonly used, e.g., in financial econometrics (empirical/applied finance).

Especially (applied) macroeconomic research can essentially be divided into two different areas:

- Long-run: Attempt to understand the determinants of long-run economic growth ($\Rightarrow$ increases in national income)
- Short-run: Understanding the causes and consequences of short-run fluctuations in economic activity ($\Rightarrow$ business cycles)

In this course, **we will focus primarily on Vector Autoregressive (VAR) models**, specifically examining statistical inference, forecasting techniques, and variable interdependence, as well as broader methodologies for handling large-scale time series

- Especially so called **structural VAR (SVAR) models** are considered more detail in the above mentioned **Macroeconometrics** course (not yet in this course).

Finally, even though the above emphasize (macro)economic and financial applications, multiple time series models (and methods in macroeconometrics) are also highly relevant in completely different areas of applications where (large) time series datasets are analyzed.

2 Some basic concepts for time series data

This section is a summarization of Sections 1–3 of Time Series Econometrics (TSE) material. Some additional details and illustrations can hence be obtained from those sections of the TSE material.

[Link to the TSE material](#)

2.1 General objective in time series modelling

The **aim** of time series analysis is often to build a **statistical model** that represents the observed time series and its fluctuation over time. This typically involves examining the dependence structure among consecutive observations.

Autocorrelation (serial correlation) is a key property of time series and must be accounted for. This contrasts with cross-sectional data introduced in basic statistics courses, where random sampling generally ensures mutually independent error terms.

When analyzing time series data, **a useful and natural first step is to plot the series and visually inspect its main characteristics** alongside any underlying theory. Important questions to consider include:

- Is there a **trend** (do values tend to increase or decrease over time)?
- Are there **seasonal effects** (regularly repeating patterns related to calendar time)?
- Are there major **outliers** (observations far away from other data points)?
- Is the **variability** (variance) relatively **constant over time**?
- Are there any **abrupt changes**, such as **breaks** or **regime switches**?

Overall, these visual findings suggest which **time series model** could be appropriate. Once the parameters are estimated and the model's compatibility with the data is assessed, the model serves for the three primary purposes:

- Providing a **summarized description** of the underlying data-generating process.
- **Forecasting** future values, potentially using information contained in other series (as later in this course).

- Exploring **temporal and contemporaneous dependencies** between time series (e.g., the relationship between inflation and stock market returns (examples in the TSE course) or the empirical illustrations above).

2.2 Some relevant transformations

As described above, in time series modelling a good first step is always to plot the time series under investigation. This provides an initial view of its main characteristics. If the observed series exhibits trends or seasonal patterns, appropriate transformations—such as **differencing** or **seasonal adjustment**—should be applied to mitigate them.

Common transformations in time series analysis include:

- **Lags:** The first lag of y_t is y_{t-1} , and the k th lag is y_{t-k} .
- **Differencing:** Perhaps the most vital transformation is the **first difference**, which represents the change between consecutive observations:

$$\Delta y_t = y_t - y_{t-1}.$$

- **Logarithms and log changes:** Differencing is often combined with natural logarithms to handle exponential growth or stabilizing variance. The **logarithmic change** approximates the relative change when changes are small (given $y_t > 0$):

$$\Delta \log(y_t) = \log(y_t) - \log(y_{t-1}) \approx \frac{y_t - y_{t-1}}{y_{t-1}}.$$

- **Financial returns:** A prominent application of differencing, particularly in finance, is in calculating returns of different assets. While the simple return is the percentage change in price P_t , the **continuously compounded return** (or log return) is exactly the first difference of the log prices $p_t = \log(P_t)$:

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right) = p_t - p_{t-1}.$$

Log returns possess advantageous properties, such as time-additivity, though dividend payments require specific adjustments in advanced financial applications.

- **Annualization:** Growth rates are often annualized for comparison. For quarterly data, the annualized rate is $400\Delta \log(y_t)$, and for monthly data, $1200\Delta \log(y_t)$.

Detrending and seasonal adjustment. Besides differencing, trends and seasonality can be removed by regressing the series against time or seasonal indicators. For example, a linear trend can be removed by estimating:

$$\hat{x}_t = y_t - \hat{\alpha} - \hat{\beta}t.$$

Seasonal adjustment can also be performed using seasonal dummy variables or trigonometric functions to isolate the non-seasonal component. Advanced detrending techniques, beyond these simple methods, are discussed in a separate and specialized econometrics literature.

2.3 Univariate stationary processes

The aim is to build a statistical model for the observed time series $y_1, \dots, y_T$.

- In this section, we assume that we have one, univariate, time series. Multiple time series case will be mainly considered in the next Sections and thereafter throughout the material.

We view the observed series as a **single realization** of a random vector $\mathbf{y} = (y_1, \dots, y_T)$. To build a model, we technically attempt to specify the T -dimensional joint probability distribution of this vector. Since observations in a time series are typically dependent (cf. conventional linear regression for cross-sectional data), specifying marginal distributions is insufficient; we must characterize the whole dependence structure.

A concrete and often used way to specify the joint distribution of $\mathbf{y}$ in time series analysis is to:

- specify a model equation that characterises the dependence structure of consecutive observations, and
- combine this with a “suitable” distributional assumption.

Before we proceed to such concrete model equations in univariate and multivariate time series cases, it is useful to first briefly study some basic concepts of probability theory that form the foundation of time series analysis.

Stochastic process

From a theoretical standpoint, it is convenient to extend the random vector to a **stochastic process** $\{y_t; t = 0, \pm 1, \pm 2, \dots\}$, which is a collection of random variables indexed by time.

- **Crucial distinction:** We effectively aim to learn the properties of the underlying process (the theoretical mechanism), but importantly **we only have access to one and partial realization of that process** (that is, the observed time series data at time points $t = 1, \dots, T$).
- To make statistical inference possible from a single realization, we must make restrictive assumptions about the process, most notably regarding its stability over time.

Weak stationarity

A fundamental stability assumption is **weak stationarity** (covariance stationarity). A process y_t is called weakly stationary if its first and second moments are time-invariant. Specifically:

1. **Constant mean:** $E(y_t) = \mu < \infty$ for all t .
2. **Constant autocovariances:** $\text{Cov}(y_t, y_{t+h}) = \gamma_h < \infty$ depends only on the lag h , not on the time index t .

To obtain autocovariances, recall the **covariance function** of the process y_t is in general

$$\gamma_{s,t} = \text{Cov}(y_s, y_t) = E[(y_s - \mu_s)(y_t - \mu_t)], \quad t = 0, \pm 1, \pm 2, \dots$$

In other words, a form of **time invariance** holds for weakly stationary processes: In addition to the constant expected value μ over time, and the covariance function $\gamma_{s,t}$ does not depend on the time indices s and t but only on their distance $t - s$:

$$\text{Cov}(y_t, y_{t+h}) = \gamma_{t,t+h} = \gamma_{0,h} \equiv \gamma_h < \infty \text{ for all } h, t = 0, \pm 1, \pm 2, \dots$$

Properties of the autocovariance function γ_h under weak stationarity:

- $\gamma_0 = \text{Var}(y_t) \geq 0$ (in principle only > 0).
- $|\gamma_h| \leq \gamma_0$.
- Symmetry: $\gamma_h = \gamma_{-h}$. That is, in this univariate (single time series process) case, we have (under weak stationary)

$$\text{Cov}(y_t, y_{t+h}) = \text{Cov}(y_{t-h}, y_t) = \text{Cov}(y_t, y_{t-h}),$$

which implies that $\gamma_h = \gamma_{-h}$. This expression shows where the term “**lag**” (“**lag length**”) h is coming from.

A useful **practical aspect** of weak stationarity is that the plausibility of this assumption can be investigated by looking at a graph of the observed time series: If the observations vary with constant variance around a fixed level, then weak stationarity appears a plausible assumption.

It is important to note that **weak stationarity does not imply the absence of random variability**. Even in a weakly stationary process, individual observations can vary substantially due to random variation (random shocks). What matters is that the statistical properties — the mean, variance, and autocovariance structure — remain constant over time.

2.4 Theoretical and sample autocorrelation

While autocovariance measures linear dependence, the **autocorrelation function (ACF)** is often more convenient as it is scale-independent. The theoretical autocorrelation at lag h is defined as:

$$\rho_h = \frac{\text{Cov}(y_t, y_{t+h})}{\sqrt{\text{Var}(y_t)\text{Var}(y_{t+h})}} = \frac{\gamma_h}{\gamma_0}.$$

The key properties include $\rho_0 = 1$, $|\rho_h| \leq 1$, and symmetry $\rho_h = \rho_{-h}$.

Typically, we also assume (if present and realistic) that autocorrelations decay as the distance between observations increases:

$$\gamma_h \rightarrow 0 \quad \text{as} \quad h \rightarrow \infty.$$

In this case, the random variables y_t and y_{t+h} become **nearly uncorrelated when the distance h is “large”** – that is, when y_t and y_{t+h} are “far” from each other in time. Therefore, in practice it suffices to estimate the autocorrelation function ρ_h when $h = 1, \dots, H$, and H is so large that $\rho_h \approx 0$ for $h > H$.

Sample estimators

Since we cannot observe the true means and autocovariances (for some concrete time series), we estimate them using the realized data.

- The **sample mean** $\bar{y} = \frac{1}{T} \sum_{t=1}^T y_t$ estimates μ .
- The **sample autocovariance** c_h estimates γ_h :

$$c_h = \frac{1}{T-h} \sum_{t=1}^{T-h} (y_t - \bar{y})(y_{t+h} - \bar{y}), \quad h = 1, \dots, H.$$

- The **sample autocorrelation coefficient** r_h estimates ρ_h :

$$r_h = \frac{c_h}{c_0}, \quad h = 1, \dots, H.$$

Visual inspection and confidence bands

When plotting the sample autocorrelation function (correlogram) of the (single) time series (see an example in this section), statistical software typically includes two horizontal lines representing **confidence bands**.

- If $\rho_h = 0$ for all $h > 0$ (i.e., no autocorrelation), then any individual estimator r_h has an approximately 95% probability of lying between the bands $(-1.96/\sqrt{T}, 1.96/\sqrt{T})$.
- If the individual sample autocorrelation coefficient r_h falls **outside** these bands, there is statistically significant autocorrelation at the 5% significance level at lag h .
- Significant spikes at specific lags (e.g., $h = 12, 24$ in monthly data) can suggest remaining seasonal patterns.

2.5 Univariate IID and white noise processes

To formalize econometric/statistical models for time series and, for example, derive relevant test statistics, we often need to define specific types of stationary processes.

The simplest building block in time series analysis/econometrics is the **(weak) white noise process**. A process u_t is called (weak) white noise, denoted $u_t \sim \text{wn}(\mu, \sigma^2)$, if:

- $E(u_t) = \mu$ (constant mean, usually 0 when u_t is used as an error term or “shock”),
- $\text{Var}(u_t) = \sigma^2$ (constant variance),
- $\text{Cov}(u_s, u_t) = 0$ for all $s \neq t$ (no autocorrelation).

Strict Stationarity

Weak stationarity only restricts the first two moments. A stronger condition, **strict stationarity**, requires that the **entire probability structure** of the process is time-invariant. Formally, the joint distribution of $(y_{t_1}, \dots, y_{t_m})$ must be identical to that of $(y_{t_1+h}, \dots, y_{t_m+h})$ for all time shifts h .

- Strict stationarity implies weak stationarity when assuming finite variance (and means and autocovariances).
- Crucially, strict stationarity is preserved under transformations (e.g., if x_t is strictly stationary, so is x_t^2).

- Stationarity in this material: Unless otherwise stated, the term “stationarity” refers to strict stationarity. In practice, however, visual inspection focuses on the observable characteristics of weak stationarity. We examine the plot to assess whether the mean, variance, and autocovariances appear stable, time invariant, over time, as these are the practical implications of a stationary process.

IID and NID processes

A specific, stronger form of white noise is the **Independent and Identically Distributed (IID)** process.

- If $u_t \sim \text{iid}(\mu, \sigma^2)$, the observations are statistically independent, not just uncorrelated. This implies strict stationarity.
- If we further assume that u_t follows a normal distribution, we denote this as **NID** (Normally and Independently Distributed):

$$u_t \sim \text{nid}(\mu, \sigma^2).$$

Below is a simulated realization of a $\text{nid}(0, 1)$ process. It exhibits purely random variation with no discernible patterns or trends. The length of the time series is 300 observations ($T = 300$).

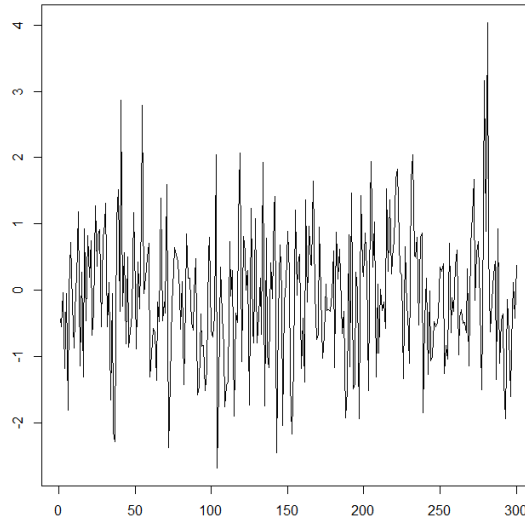


Figure: A simulated realization of $\text{nid}(0, 1)$ process ($T = 300$).

2.6 Testing autocorrelation

For the important special case of $y_t \sim \text{iid}(\mu, \sigma^2)$, it holds that

$$(r_1, \dots, r_H) \underset{as}{\sim} \mathbf{N}(\mathbf{0}, T^{-1} \mathbf{I}_H),$$

with a H -dimensional zero mean vector and $\mathbf{I}_H$ ($H \times H$) denotes an identity matrix. This result can be used to test whether it is realistic to consider an observed time series as an uncorrelated time series process. Under the hypothesis to be tested (uncorrelatedness), the estimators $r_1, \dots, r_H$ are approximately independent with distribution $\mathbf{N}(0, T^{-1})$. Based on this, we get

$$\mathbf{P}(|r_h| \geq 1.96/\sqrt{T}) \approx 0.05,$$

a result that can be used to evaluate the significance of individual sample autocorrelation coefficients.

To obtain a joint test for several autocorrelation coefficients, that is to test the null hypothesis of no autocorrelation $H_0 = \rho_1 = \dots = \rho_H = 0$, one can use the test statistic

$$Q = T \sum_{h=1}^H r_h^2 \underset{as}{\sim} \chi_H^2,$$

whose large values would lead to rejection. Note that the asymptotic χ_H^2 -distribution follows from the distribution result above for $(r_1, \dots, r_H)$ and the definition of the chi-squared distribution. In practice, an alternative and slightly different test statistic

$$Q_{LB} = T(T+2) \sum_{h=1}^H r_h^2 / (T-h) \underset{as}{\sim} \chi_H^2,$$

called the **Ljung-Box test statistic** is preferred because in small samples its distribution has been found to be closer to the χ_H^2 -distribution than that of the test statistic Q . It should be clear that both tests need H not to be too large compared to T to work well.

As an example, below we depict the sample autocorrelation function of the simulated and plotted `nid(0, 1)` series above. As expected, the sample autocorrelation coefficients are systematically inside the confidence intervals. Furthermore, when testing, say, the first 20 autocorrelation coefficients (that is selecting $H = 20$), the value of the Ljung-Box test statistic is 16.597 with the p-value of 0.6789. Therefore, as in line with the assumed `nid(0, 1)` process, we cannot reject the null hypothesis of no autocorrelation.

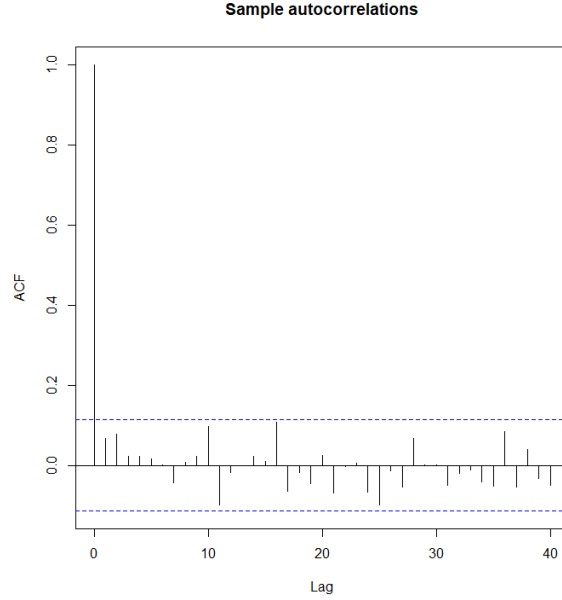


Figure: Sample autocorrelation function of the simulated realization of $\text{nid}(0, 1)$ process ($T = 300$). Notice that this function depicts also the uninteresting case $r_0 = 1$.

The autocorrelation function can be used to reveal linear dependences between the observations, but not nonlinear ones (with the exception of Gaussian processes). To investigate the presence of potential nonlinear dependence over time, one (somewhat limited) approach is to test for autocorrelation in the squared observations.

- Assuming $y_t \sim \text{iid}(\mu, \sigma^2)$ and $\mathbb{E}(y_t^4) < \infty$, what was said above about the sample autocorrelations also holds for sample autocorrelations computed from the squared observations y_t^2 . In particular, the asymptotic result(s) remain valid when the autocorrelations are computed from squared observations, and also the Ljung-Box test has its indicated asymptotic distribution. In this context, however, the test is usually called the **McLeod-Li test**.
- Investigating the autocorrelations between squared observations is of great interest, especially in the case of financial time series, which themselves often are (almost) uncorrelated.

3 Multivariate stationary processes

3.1 Multivariate stochastic process

As above for univariate processes, a multiple (vector-valued) time series can be interpreted as a realization of a corresponding **(multivariate) stochastic process**, that is a collection of random vectors

$$\mathbf{y}_t = (y_{1t}, \dots, y_{Kt}).$$

- At times this column vector is denoted (see Notation) as $\mathbf{y}_t = [y_{1t} \cdots y_{Kt}]'$.
- In practice, the dataset to be analyzed is then denoted as $\mathbf{y}_1, \dots, \mathbf{y}_T$.
- In what follows, we often denote a (multivariate) time series process by $\{\mathbf{y}_t\}$ or just $\mathbf{y}_t$. The notation $\{\mathbf{y}_t\}$ emphasizes that a stochastic process is generally an infinite-dimensional random variable. Throughout this material we consider real-valued discrete time series processes.

Extra: Some additional details on such stochastic process

For example, a real-valued discrete time process $\{\mathbf{y}_t\} = \{\dots, \mathbf{y}_{-1}, \mathbf{y}_0, \mathbf{y}_1, \dots\}$ takes values on the set $\mathbb{R}^\infty = \mathbb{R} \times \mathbb{R} \times \dots$. A continuous-time process in turn might take values on $C[0, 1]$, the set of continuous functions defined on the interval $[0, 1]$. In practice, an observed multiple time series is just one realization or value of such an infinite-dimensional process or usually a part of such value (like a sample $\mathbf{y}_1, \dots, \mathbf{y}_T$, $T < \infty$).

Given a stochastic process, an infinite-dimensional random variable, one may ask what is meant by its probability distribution, which is defined on a set like $\mathbb{R}^\infty$ or $C[0, 1]$.

The question about the distribution of a stochastic process is a topic of advanced probability theory and is not treated thoroughly here. It suffices to us to know that the theory says, to put it simple, that the distribution of an infinite-dimensional process can be defined on the basis of finite-dimensional distributions.

- It is obvious that the converse holds, i.e., all the finite-dimensional distributions follow uniquely from an infinite-dimensional distribution. Hence, we can characterize the distribution of the infinite-dimensional process by using finite-dimensional distributions.

- For example, a process $\{\mathbf{y}_t\}$ is normal when all of its *finite dimensional distributions* are normal, that is, $(\mathbf{y}_{t_1}, \dots, \mathbf{y}_{t_m})$ are multivariate normal for all $t_1, \dots, t_m$ and all $m \geq 1$ (here $m \in \mathbb{Z}$ while for a continuous-time process $t_1, \dots, t_m$ need not be integers).

The first and second moments of a stochastic process are defined (assuming they exist and are finite) as follows. The **mean (function) or expectation (function)** of $\mathbf{y}_t$ is

$$\boldsymbol{\mu}_t = \mathbb{E}(\mathbf{y}_t), \quad (K \times 1)$$

and the **covariance function** is

$$\boldsymbol{\Gamma}_{s,t} = \text{Cov}(\mathbf{y}_s, \mathbf{y}_t) = \mathbb{E}[(\mathbf{y}_s - \boldsymbol{\mu}_s)(\mathbf{y}_t - \boldsymbol{\mu}_t)'] \quad (K \times K)$$

By choosing $s = t$, we obtain the **covariance matrix** $\boldsymbol{\Sigma}_t = \boldsymbol{\Gamma}_{t,t}$, that is,

$$\boldsymbol{\Sigma}_t = \text{Cov}(\mathbf{y}_t) = \mathbb{E}[(\mathbf{y}_t - \boldsymbol{\mu}_t)(\mathbf{y}_t - \boldsymbol{\mu}_t)'] \quad (K \times K)$$

In practice, we often focus on analyzing these two moments (if $\mathbf{y}_t$ is normal, there is nothing else to analyze).

- In theoretical considerations we must nevertheless make assumptions on higher moments. Unless stated otherwise, the notations $\boldsymbol{\mu}_t = \mathbb{E}(\mathbf{y}_t)$ and $\boldsymbol{\Sigma}_t = \text{Cov}(\mathbf{y}_t)$ automatically assume that the underlying moments exist and are finite.

Example: Bivariate case ($K = 2$)

To illustrate these definitions concretely, consider a bivariate process composed of two time series (components), y_{1t} and y_{2t} . In this case, the random vector at time t is:

$$\mathbf{y}_t = \begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix}.$$

Mean vector. The mean vector $\boldsymbol{\mu}_t$ collects the expected values of the individual series:

$$\boldsymbol{\mu}_t = \begin{bmatrix} \mu_{1t} \\ \mu_{2t} \end{bmatrix} = \begin{bmatrix} \mathbb{E}(y_{1t}) \\ \mathbb{E}(y_{2t}) \end{bmatrix}.$$

Covariance matrix (at time t). The covariance matrix $\boldsymbol{\Sigma}_t$ captures the variance of each series and the contemporaneous covariance between them. It is obtained by the outer product

of the centered vector:

$$\begin{aligned}
\Sigma_t &= E \left[\begin{bmatrix} y_{1t} - \mu_{1t} \\ y_{2t} - \mu_{2t} \end{bmatrix} \begin{bmatrix} y_{1t} - \mu_{1t} & y_{2t} - \mu_{2t} \end{bmatrix} \right] \\
&= \begin{bmatrix} E[(y_{1t} - \mu_{1t})^2] & E[(y_{1t} - \mu_{1t})(y_{2t} - \mu_{2t})] \\ E[(y_{2t} - \mu_{2t})(y_{1t} - \mu_{1t})] & E[(y_{2t} - \mu_{2t})^2] \end{bmatrix} \\
&= \begin{bmatrix} \text{Var}(y_{1t}) & \text{Cov}(y_{1t}, y_{2t}) \\ \text{Cov}(y_{2t}, y_{1t}) & \text{Var}(y_{2t}) \end{bmatrix}
\end{aligned}$$

Here, the diagonal elements represent the variances of y_{1t} and y_{2t} , while the off-diagonal elements represent the covariance between the two series at time t . Note that the matrix is symmetric, so the off-diagonal elements $\text{Cov}(y_{1t}, y_{2t}) = \text{Cov}(y_{2t}, y_{1t})$.

3.2 Stationarity

In short and analogously to the univariate case, a stochastic process $\mathbf{y}_t$ ($K \times 1$) is (strictly) **stationary**, if the probability distribution of a stochastic process is time-invariant.

Extra: Strict stationarity in a multivariate context

A stochastic process $\mathbf{y}_t$ ($K \times 1$) is (strictly) **stationary**, if

$$P \{ \mathbf{y}_{t_1} \in \mathbf{A}_1, \dots, \mathbf{y}_{t_m} \in \mathbf{A}_m \} = P \{ \mathbf{y}_{t_1+h} \in \mathbf{A}_1, \dots, \mathbf{y}_{t_m+h} \in \mathbf{A}_m \}$$

for all $\mathbf{A}_1, \dots, \mathbf{A}_m$, $t_1, \dots, t_m$ and $m \geq 1$ (for a continuous time process only m must be integer). In words, the probability distribution of a stochastic process is time-invariant.

Under stationarity, $\mathbf{y}_t$ has the same distribution for all t and also $(\mathbf{y}_t, \mathbf{y}_{t+h})$ and $(\mathbf{y}_s, \mathbf{y}_{s+h})$ have the same distribution for all t and s and all (fixed) h . It is also convenient that transformations of strictly stationary time series remain strictly stationary.

- In particular, if $\mathbf{z}_t$ is strictly stationary (discrete time) process, then so is also $\mathbf{y}_t = \mathbf{g}(\mathbf{z}_{t+h}, \dots, \mathbf{z}_{t-l})$ ($h, l \geq 0$) provided $\mathbf{g}$ is well-behaving (say, continuous, or more generally, measurable) function. Here, either h or l or both can be replaced by ∞ . When h and l are finite, the result follows directly from the definition of strict stationarity.

A vector-valued process $\mathbf{y}_t$, ($K \times 1$) is **weakly stationary**, or **covariance stationary**, if its second (hence also first) moments exist, are finite and time invariant, that is

$$E(\mathbf{y}_t) = \boldsymbol{\mu} \quad (K \times 1) \quad \text{and} \quad \text{Cov}(\mathbf{y}_t, \mathbf{y}_{t+h}) = \boldsymbol{\Gamma}_h \quad \text{for all } t \text{ and } h,$$

where $\boldsymbol{\Gamma}_h$ are $(K \times K)$ matrices. As in the univariate case:

- Strict stationarity and the existence of finite first two moments imply weak stationarity. The converse holds under normality but not in general. Transformations of weakly stationary processes are not generally weakly stationary.
- In what follows in this course, “stationarity” refers to strict stationarity (although this terminology is not generally applied).

As we saw in the previous section, when $\mathbf{y}_t$ is univariate ($K = 1$), y_t , the covariance function is symmetric with respect to the lag length h (see below). This does not hold when $\mathbf{y}_t$ is vector-valued. Instead, it holds that

$$\begin{aligned}\mathbf{\Gamma}_{-h} &= \mathbf{E}[(\mathbf{y}_t - \boldsymbol{\mu})(\mathbf{y}_{t-h} - \boldsymbol{\mu})'] \\ &= (\mathbf{E}[(\mathbf{y}_{t-h} - \boldsymbol{\mu})(\mathbf{y}_t - \boldsymbol{\mu})'])' \\ &= \mathbf{\Gamma}'_h\end{aligned}$$

As in the univariate case, it suffices to know $\mathbf{\Gamma}_h$ for $h \geq 0$ (or $h \leq 0$).

The covariance function $\mathbf{\Gamma}_h$ contains the autocovariance and cross-covariance functions of the individual components of the process $\mathbf{y}_t$. As $\mathbf{y}_t = (y_{1t}, \dots, y_{Kt})$, then $\mathbf{\Gamma}_h = [\gamma_{ab,h}]$ ($a, b = 1, \dots, K$), and the autocovariance function of the process y_{at} (scalar-valued a th process) is

$$\gamma_{a,h} = \mathbf{E}\left((y_{at} - \mathbf{E}(y_{at}))(y_{a,t+h} - \mathbf{E}(y_{a,t+h}))\right).$$

Therefore, the **autocorrelation function** of the process y_{at} is defined as

$$\rho_{a,h} = \gamma_{a,h} / \gamma_{a,0}, \quad a = 1, \dots, K,$$

and the **cross-correlation function** of the processes y_{at} and y_{bt} is

$$\rho_{ab,h} = \gamma_{ab,h} / \sqrt{\gamma_{a,0}\gamma_{b,0}} \quad a, b = 1, \dots, K.$$

Here $\gamma_{a,h} = \gamma_{aa,h}$ and $\gamma_{ab,h}$ ($a \neq b$) is the **cross-autocovariance function**, respectively.

From a practical standpoint, and again analogously to the univariate case, a stationary time series can be assumed to satisfy the condition

$$\mathbf{\Gamma}_h \rightarrow \mathbf{0}, \text{ as } |h| \rightarrow \infty$$

where convergence can be interpreted to hold for each component. In this case, $\mathbf{y}_t$ and $\mathbf{y}_{t+h}$ are nearly uncorrelated when $|h|$ is large.

- Note that $\mathbf{\Gamma}_h \rightarrow \mathbf{0}$ does not follow from stationarity as such.
- If one wants to estimate the first- and second-order moments of $\mathbf{y}_t$, it suffices under $\mathbf{\Gamma}_h \rightarrow \mathbf{0}$ to estimate a finite number of parameters (that is, $\boldsymbol{\mu}$ and $\mathbf{\Gamma}_h$, $h = -H, \dots, H$, for some “large” H).

Example continues: Bivariate stationarity case ($K = 2$). To make these matrix definitions concrete, let us again consider the bivariate case, that is two time series y_{1t} and y_{2t} . For this vector process $\mathbf{y}_t = (y_{1t}, y_{2t})$ to be weakly stationary, two specific conditions must be “visible” in its moments:

- **Time invariance of the mean vector:** The expected value of each series must be constant over time, not depending on t :

$$\mathbb{E}(\mathbf{y}_t) = \boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \quad \text{for all } t.$$

Implication: Neither y_{1t} nor y_{2t} can have a trend. They must fluctuate around fixed levels μ_1 and μ_2 .

- **Time invariance of the covariance structure:** The covariance function $\boldsymbol{\Gamma}_h$ must depend only on the lag h , and not on the specific point in time t :

$$\boldsymbol{\Gamma}_h = \begin{bmatrix} \gamma_{11,h} & \gamma_{12,h} \\ \gamma_{21,h} & \gamma_{22,h} \end{bmatrix}.$$

This implies three time-invariant (stable) relationships simultaneously:

- Constant variance ($h = 0$): The volatility of y_{1t} ($\gamma_{11,0}$) and y_{2t} ($\gamma_{22,0}$) does not change over time.
- Stable cross-correlation ($h = 0$): The contemporaneous relationship between y_{1t} and y_{2t} (e.g., do they move together?) is constant.
- Stable lead-lag dynamics ($h \neq 0$): The way y_{1t} predicts $y_{2,t+h}$ (and vice versa) remains consistent throughout the sample. Interpretation the off-diagonal elements:
 - $\gamma_{12,h} = \text{Cov}(y_{1t}, y_{2,t+h})$: Measures how y_1 at time t covaries with y_2 at time $t + h$.
 - $\gamma_{21,h} = \text{Cov}(y_{2t}, y_{1,t+h})$: Measures how y_2 at time t covaries with y_1 at time $t + h$.

Important asymmetry: Unlike the autocovariance of a single series, cross-covariances are not generally symmetric in lag h . That is, usually $\gamma_{12,h} \neq \gamma_{21,h}$. For example, if y_1 is a “**leading indicator**” for y_2 , then $\gamma_{12,h}$ (current y_1 vs future y_2) might be large, while $\gamma_{21,h}$ (current y_2 vs future y_1) might be zero. However, stationarity ensures that $\boldsymbol{\Gamma}_{-h} = \boldsymbol{\Gamma}'_h$, meaning:

$$\boldsymbol{\Gamma}_{-h} = \begin{bmatrix} \gamma_{11,h} & \gamma_{21,h} \\ \gamma_{12,h} & \gamma_{22,h} \end{bmatrix}.$$

3.3 IID and white noise processes

A simple example of a stationary process is the sequence of independent and identically distributed vectors (stationarity can be deduced directly from the definition). If $\mathbf{y}_t$ ($K \times 1$) is such a sequence, $E(\mathbf{y}_t) = \boldsymbol{\mu}$ ($K \times 1$) and $\text{Cov}(\mathbf{y}_t) = \boldsymbol{\Sigma}$ ($K \times K$), and both are finite, we denote

$$\mathbf{y}_t \sim \text{iid}(\boldsymbol{\mu}, \boldsymbol{\Sigma}).$$

This process is called (strict) **white noise**. If $\mathbf{y}_t$ is also normally distributed, we denote

$$\mathbf{y}_t \sim \text{nid}(\boldsymbol{\mu}, \boldsymbol{\Sigma}).$$

If we replace independence by the weaker condition of zero correlation, that is $\text{Cov}(\mathbf{y}_t, \mathbf{y}_s) = \mathbf{0}$, $t \neq s$, then $\mathbf{y}_t$ is weakly stationary and may be called (multivariate) **weak white noise**.

3.4 Sample mean and sample autocovariances

Consistency of sample mean. Let $\mathbf{y}_t = (y_{1t}, \dots, y_{Kt})$ be a stationary process with expectation $E(\mathbf{y}_t) = \boldsymbol{\mu}$ and covariance matrix $\text{Cov}(\mathbf{y}_t, \mathbf{y}_{t+h}) = \boldsymbol{\Gamma}_h = [\gamma_{ij,h}]$. Assume the following condition (compare with $\boldsymbol{\Gamma}_h \rightarrow \mathbf{0}$)

$$\sum_{h=-\infty}^{\infty} \|\boldsymbol{\Gamma}_h\| < \infty.$$

Denote $\bar{\mathbf{y}} = (\bar{y}_1, \dots, \bar{y}_K)$ and $\boldsymbol{\mu} = (\mu_1, \dots, \mu_K)$. Consider the sample average $\bar{\mathbf{y}} = T^{-1} \sum_{t=1}^T \mathbf{y}_t$ as an estimator for $\boldsymbol{\mu}$. The estimator is

- unbiased

$$E(\bar{\mathbf{y}}) = T^{-1} \sum_{t=1}^T E(\mathbf{y}_t) = \boldsymbol{\mu}.$$

- consistent estimator for $\boldsymbol{\mu}$ and asymptotically normally distributed. See proof for consistency below

Moreover, the sample covariance function can be obtained as

$$\mathbf{C}_h = [c_{ij,h}] = (T - |h|)^{-1} \sum_{t=1}^{T-|h|} (\mathbf{y}_t - \bar{\mathbf{y}})(\mathbf{y}_{t+h} - \bar{\mathbf{y}})', \quad h = 0, 1, \dots, T-1,$$

where $\mathbf{C}_h = \mathbf{C}'_{-h}$, $h = -T+1, \dots, -1$. It also turns out that this estimator is consistent estimator of $\boldsymbol{\Gamma}_h$.

Extra: Proofs (consistency)

We have (for the time series i)

$$\begin{aligned}
\text{Var}(\bar{y}_i) &= \mathbb{E} \left[\left(T^{-1} \sum_{t=1}^T (y_{it} - \mu_i) \right)^2 \right] \\
&= T^{-2} \sum_{t=1}^T \sum_{s=1}^T \mathbb{E} [(y_{it} - \mu_i)(y_{is} - \mu_i)] \\
&= T^{-1} \sum_{t-s=T}^T (T - |t-s|) \gamma_{i,t-s} \\
&= T^{-1} \sum_{h=-T}^T \left(1 - \frac{|h|}{T} \right) \gamma_{i,h}.
\end{aligned}$$

Here, the third equation can be verified by noting that the previous double sum is the sum of the elements of the matrix $[\gamma_{i,t-s}]_{t,s=1,\dots,T}$. The triangular inequality yields the result

$$\text{Var}(\bar{y}_i) = \mathbb{E} [(\bar{y}_i - \mu_i)^2] \leq T^{-1} \sum_{h=-T}^T \left(1 - \frac{|h|}{T} \right) |\gamma_{i,h}| \rightarrow 0, \text{ as } T \rightarrow \infty.$$

Combining this with the above results, we can conclude (using the Markov inequality) that the sample mean $\bar{y}_i$ is a consistent estimator of the mean μ_i and further that the same holds for $\bar{y}$. That is, we have the stochastic convergence

$$\bar{\mathbf{y}} \xrightarrow{p} \boldsymbol{\mu}.$$

Consistency of Sample Covariance. Consider the sample covariance

$$\mathbf{C}_h = [c_{ij,h}] = (T-h)^{-1} \sum_{t=1}^{T-h} (\mathbf{y}_t - \bar{\mathbf{y}})(\mathbf{y}_{t+h} - \bar{\mathbf{y}})', \quad h = 0, 1, \dots, T-1,$$

$\mathbf{C}_h = \mathbf{C}'_{-h}$, $h = -T+1, \dots, -1$. Fix the indices i, j and $h \geq 0$ and define the variables $x_t = y_{it}y_{j,t+h}$, which is the central term of the estimator $c_{ij,h}$. Assume $E(y_{it}^4) < \infty$, which implies that x_t is stationary and has a finite second-order moment, and using arguments from the previous section, we have the result

$$\bar{x} = (T-h)^{-1} \sum_{t=1}^{T-h} y_{it}y_{j,t+h} \xrightarrow{p} \mathbb{E}(y_{it}y_{j,t+h}),$$

where by the definition of covariance

$$\mathbb{E}(y_{it}y_{j,t+h}) = \gamma_{ij,h} + \mathbb{E}(y_{it})\mathbb{E}(y_{j,t+h}) = \gamma_{ij,h} + \mu_i\mu_j$$

If the above assumptions hold for all i, j and $h \geq 0$, we can conclude from the consistency of $\bar{x}$ and the result that $c_{ij,h} \xrightarrow{p} \gamma_{ij,h}$, and

$$\mathbf{C}_h \xrightarrow{p} \mathbf{\Gamma}_h$$

As seen, the sample covariance function is consistent with reasonable assumptions. (Note however, that above h was fixed.) The necessary assumptions are easy to check from the special case $\mathbf{y}_t \sim \text{iid}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and then further with the case of a VMA(q) process ($q < \infty$). With common VARMA(p, q) processes the consistency result can be justified by using approximation VARMA(p, q) $\approx$ VMA(l) with l “large enough”. The details are, however, a bit technical and will be ignored.

In practice and as in the univariate case, covariances are replaced by correlations, that is, we typically consider the sample quantities

$$r_{ab,h} = \frac{c_{ab,h}}{\sqrt{c_{a,0}c_{b,0}}}, \quad c_{a,0} = c_{aa,0}.$$

It can be proved that when $\mathbf{y}_t \sim \text{iid}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then we have the asymptotic result

$$r_{ab,h} \stackrel{as}{\sim} \mathbf{N}(0, 1/T), \quad k \neq 0$$

In addition, if $R_h = [r_{ab,h}]$, then $\mathbf{R}_h$ and $\mathbf{R}_l$ are asymptotically independent, when $h \neq l$.

Example. Below we can see the autocorrelation and cross-correlation functions from the exchange rate-interest rate data as depicted in Section 1. The critical boundaries $\pm 1.96/\sqrt{T}$ inside which single correlations should stay with 95% likelihood if this was a realization of a iid process are displayed as well. Estimated autocorrelations confirm the clear autocorrelation we could see already in time series graphs. Estimated cross-correlations, however, seem to be relatively small. Notice, however, that due to the autocorrelation exhibited by the component series, the critical boundaries cannot be applied to the cross-correlation estimates in the same way as above.

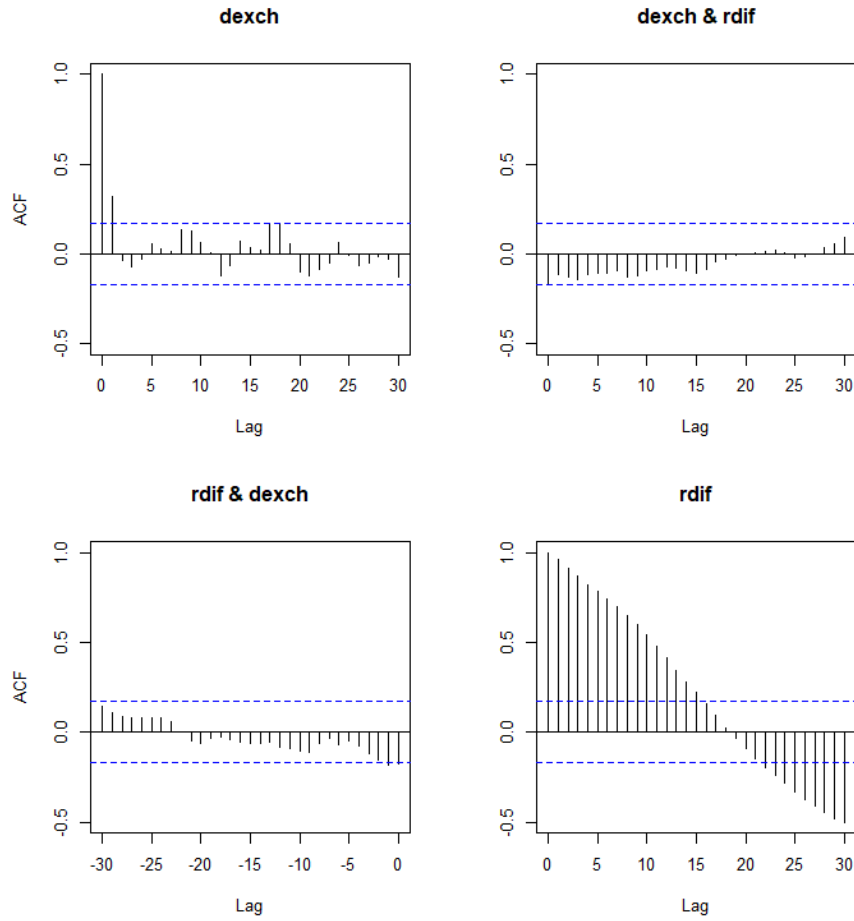


Figure: The estimated auto- and cross-correlation functions of the monthly change in the exchange rate between the euro and the USD and the difference between the returns of the average 10-year bond yield in the US and the eurozone (1999:1–2009:12, $T = 132$). The figures are based on the acf-function in R.

4 Univariate time series processes and models

In this chapter, we briefly introduce and review univariate time series models. We will concentrate on the autoregressive (AR) models, but more general ARMA models will be also briefly discussed.

- Notice that ‘process’ and ‘model’ are often used interchangeably. The term ‘**process**’ emphasizes that a time series is a realization of the underlying (stochastic) process, whereas ‘**model**’ emphasizes the specific statistical specification selected for the observed data.

A more extensive description of the univariate models is provided in the Time Series Econometrics:

[Link to the TSE material](#)

4.1 Autoregressive process

At a general level, the structure of a (univariate) time series y_t can be thought of as consisting of two (additive) parts: a systematic part (‘predictable’ or ‘structural’ part or ‘skeleton’) and an unpredictable part.

An autoregressive process (model) of order p , $\text{AR}(p)$ process, is defined by the equation

$$y_t = \nu + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + u_t, \quad u_t \sim \text{iid}(0, \sigma^2).$$

That is, the observation y_t is explained by its own lags $y_{t-1}, \dots, y_{t-p}$ and the constant term ν (cf. a linear regression model). Additionally, u_t is a zero mean error term. In the following, this assumption is strengthened (unless otherwise stated) so that $u_t \sim \text{iid}(0, \sigma^2)$ and occasionally $u_t \sim \text{nid}(0, \sigma^2)$, that is assuming normality of the error term.

- In this material, the process/model is written with the constant term ν . In an alternative notation, the process y_t will be **demeaned** around the mean, that is $y_t - \mu$, where $E(y_t) = \mu$, indicating a demeaned (centered) process without the constant term:

$$(y_t - \mu) = \phi_1 (y_{t-1} - \mu) + \cdots + \phi_p (y_{t-p} - \mu) + u_t, \quad u_t \sim \text{iid}(0, \sigma^2).$$

Stationarity. For the process defined in this way to be stationary, or even well-defined for all $t = 0, \pm 1, \dots$, conditions must be imposed on the coefficients $\phi_1, \dots, \phi_p$. In the case of $p = 1$, that is when considering an **AR(1) process**, we specify $|\phi_1| < 1$. Then

$$\begin{aligned} y_t &= \nu + \phi_1 y_{t-1} + u_t \\ &= \nu + \phi_1 \nu + u_t + \phi_1 u_{t-1} + \phi_1^2 y_{t-2} \\ &\vdots \\ &= \nu \sum_{j=0}^k \phi_1^j + \sum_{j=0}^k \phi_1^j u_{t-j} + \phi_1^{k+1} y_{t-k-1}, \end{aligned}$$

which leads to the representation (given $|\phi_1| < 1$)

$$y_t = \nu \sum_{j=0}^{\infty} \phi_1^j + \sum_{j=0}^{\infty} \phi_1^j u_{t-j}.$$

Since u_t is strictly (strongly) stationary, the equation above shows that the same applies to the process y_t under the condition $|\phi_1| < 1$. Weak stationarity also holds. To see this, let us compute the expectation

$$\begin{aligned} \mathbb{E}(y_t) &\equiv \mu \\ &= \mathbb{E} \left(\nu \sum_{j=0}^{\infty} \phi_1^j + \sum_{j=0}^{\infty} \phi_1^j u_{t-j} \right) \\ &= \nu \sum_{j=0}^{\infty} \phi_1^j + \sum_{j=0}^{\infty} \phi_1^j \mathbb{E}(u_{t-j}) \\ &= \nu \sum_{j=0}^{\infty} \phi_1^j \\ &= \nu / (1 - \phi_1) < \infty, \end{aligned}$$

and variance

$$\text{Var}(y_t) \stackrel{!}{=} \sum_{j=0}^{\infty} \text{Var}(\phi_1^j u_{t-j}) = \sum_{j=0}^{\infty} \phi_1^{2j} \text{Var}(u_{t-j}) = \frac{\sigma^2}{1 - \phi_1^2} < \infty.$$

We can also notice that $u_t \perp\!\!\!\perp y_{t-h}$, $h > 0$, from which it follows $\mathbb{E}(u_t(y_{t-h} - \mu)) = 0$, and

$$\mathbb{E}((y_t - \mu)(y_{t-h} - \mu)) = \phi_1 \mathbb{E}((y_{t-1} - \mu)(y_{t-h} - \mu)),$$

and therefore autocorrelation coefficients (see the definition in Section 2) in this special case of $p = 1$ follow a recursion

$$\rho_h = \phi_1 \rho_{h-1}, \quad h = 1, 2, \dots$$

Since $\rho_0 = 1$, the solution to the AR(1) process is

$$\rho_h = \phi_1^h, \quad h = 0, 1, 2, \dots,$$

from which it is seen that $\rho_h \rightarrow 0$ geometrically as $h \rightarrow \infty$. The fact that mean and autocovariances (autocorrelations) are not dependent on time confirms weak stationarity of an AR(1) process.

Stationarity of the AR(p) process. Now, let us return to the general AR(p) case and introduce the **lag (backshift) operator** B , which has the properties

$$B^1 x_t = x_{t-1}, \quad B^k x_t = x_{t-k} \quad \text{and} \quad B^0 = 1,$$

where, in general, x_t can be scalar or vector-valued process. Then the AR(p) process can be written as

$$\phi(B)y_t = \nu + u_t,$$

where $\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p$.

A sufficient condition for the stationarity of an AR(p) process. A sufficient condition for the stationarity of the AR(p) process is that the roots of the polynomial (characteristic equation) $\phi(z)$ ($z \in \mathbb{C}$) lie outside the unit circle/disk in the complex plane, or equivalently

$$\phi(z) \neq 0 \text{ for } |z| \leq 1.$$

- As an example, in the stationary AR(1) case ($|\phi_1| < 1$), the characteristic equation yields $\phi(z) = 0 \Leftrightarrow 1 - \phi_1 z = 0$ when $|z| > 1$. This shows why in the case $p = 1$, the condition $|\phi_1| < 1$ was found sufficient to ensure the existence of a (causal) stationary MA(∞)-representation.
- Notice that the resulting roots can also be complex numbers. Related to the above stationarity condition, the absolute value or norm of a complex number $z = x + iy$ ($i = \sqrt{-1}$) is defined as $|z| = \sqrt{x^2 + y^2}$.
- This stationarity condition is not necessary. However, in what follows, we will focus on causal processes only, and in this case this stationarity condition is also a necessary condition.

Extra: More details on complex roots and the stationarity condition

Still more details about the possibility of getting complex roots. The absolute value or norm of a complex number $z = x + iy$ ($i = \sqrt{-1}$) can be identified with the norm of the vector (x, y) . The latter form of the stationarity condition can be further expressed as $\phi(z) = 0 \Rightarrow |z| > 1$. Because the coefficients of the polynomial $\phi(z)$ are real numbers, the potential complex roots

always appear as conjugate roots, that is, if $\zeta = x + iy$ is a root, then also $\tilde{\zeta} = x - iy$ is a root.

Furthermore, one way to illustrate this stationarity condition makes use of the fundamental theorem of algebra. As a consequence of this result, the polynomial $\phi(z)$ can be written as (assuming that $\phi_p \neq 0$)

$$\phi(z) = (1 - \zeta_1^{-1}z) \cdots (1 - \zeta_p^{-1}z),$$

where for the roots ζ_i it therefore holds that $\phi(\zeta_i) = 0$ and $|\zeta_i| > 1$ ($i = 1, \dots, p$). Therefore, an $\text{AR}(p)$ process can be expressed as $(1 - \zeta_1^{-1}B) \cdots (1 - \zeta_p^{-1}B) y_t = u_t$. If all the roots of $\phi(z)$ are real, this equation can be divided with the polynomials $(1 - \zeta_i^{-1}B)$, $i = 1, \dots, p$, one at a time, and in this way it can be seen (similarly as in the case $p = 1$) that the resulting expression is a well-defined linear process. This procedure can also be generalized to the case of complex roots.

Based on the above-mentioned, a (causal) stationary $\text{AR}(p)$ process have a (causal) **MA(∞) representation**

$$y_t = \phi(B)^{-1} \nu + \phi(B)^{-1} u_t = \phi(1)^{-1} \nu + \psi(B) u_t = \mu + \sum_{j=0}^{\infty} \psi_j u_{t-j},$$

where $\psi(B) = \sum_{j=0}^{\infty} \psi_j B^j = \phi(B)^{-1}$ and $\mu = \phi(1)^{-1} \nu$. The coefficients ψ_j can be solved as a function of the parameters $\phi_1, \dots, \phi_p$ from equation

$$(1 - \phi_1 B - \cdots - \phi_p B^p) (\psi_0 + \psi_1 B + \psi_2 B^2 + \cdots) = 1$$

by interpreting the right hand side as a power series in B , and setting the coefficients of B^j equal to each other on both sides of the equation.

- Recall that two polynomials are the same if their coefficients are the same.
- For instance, the coefficient of B^0 is 1, so that $\psi_0 = 1$. Next, the coefficient of B^1 is zero, so that $\psi_1 - \phi_1 \psi_0 = 0$ from which $\psi_1 = \phi_1$ follows. The general solution is left as an exercise.

Autocorrelation function. The autocorrelation function of an $\text{AR}(p)$ process could be derived by making use of its $\text{MA}(\infty)$ representation. Instead, we present an often used and rather practical alternative approach based on the model equation.

Multiplying both sides of the $\text{AR}(p)$ process with $y_{t-h} - \mu$ ($h \geq 0$) and taking expectations, we obtain

$$\mathbb{E}((y_t - \mu)(y_{t-h} - \mu)) = \phi_1 \mathbb{E}((y_{t-1} - \mu)(y_{t-h} - \mu)) + \cdots + \phi_p \mathbb{E}((y_{t-p} - \mu)(y_{t-h} - \mu)) + \mathbb{E}(u_t(y_{t-h} - \mu)).$$

Because y_{t-h} is a linear function of the innovation terms $u_{t-h}, u_{t-h-1}, \dots$, the variables y_{t-h} and u_t are independent when $h > 0$. Therefore, we get

- $E(u_t(y_{t-h} - \mu)) = 0$ ($h > 0$)
- For $h = 0$, it can be seen that $E(u_t(y_t - \mu)) = E(u_t^2) = \sigma^2$.

As $\gamma_h = \gamma_{-h}$, we get

$$\gamma_h = \begin{cases} \phi_1 \gamma_1 + \dots + \phi_p \gamma_p + \sigma^2, & h = 0, \\ \phi_1 \gamma_{h-1} + \dots + \phi_p \gamma_{h-p}, & h > 0. \end{cases}$$

Dividing this (in the case $h > 0$) with the variance γ_0 leads to

$$\rho_h = \phi_1 \rho_{h-1} + \dots + \phi_p \rho_{h-p}, \quad h > 0,$$

so that the autocorrelation function of an $AR(p)$ process satisfies a difference equation similar to the one the $AR(p)$ process itself satisfies.

- When the roots of $\phi(z)$ lie outside the unit disk on the complex plane, the solution ρ_h to this difference equation decays exponentially to zero as the lag length h increases.
- The solution ϕ_1^h for the $AR(1)$ is an example of this (this solution can also be easily obtained by solving the difference equation in the case $p = 1$ using the initial value $\rho_0 = 1$).

Partial autocorrelation function. We next define the partial autocorrelation function, which is, among other things, a useful tool for model selection in univariate time series analysis. In general, α_h equals the conventional **partial correlation coefficient** that measures the correlation between the random variables y_t and y_{t-h} when the linear effect of the random variables $y_{t-1}, \dots, y_{t-h+1}$ has been first eliminated.

- Therefore, a particular consequence is that $|\alpha_h| \leq 1$.

In the case of an $AR(p)$ process, the partial autocorrelation function α_h has a special useful feature. For $m > p$, an $AR(p)$ process can be interpreted as an $AR(m)$ process with $\phi_{p+1} = \dots = \phi_m = 0$, which makes it clear that the partial autocorrelation function of an $AR(p)$ process satisfies (also $\alpha_0 = 1$):

$$y_t \sim AR(p) \Rightarrow \alpha_p = \phi_p \quad \text{and} \quad \alpha_h = 0 \text{ for } h > p.$$

In other words, the partial autocorrelation function of an $AR(p)$ process drops to zero after lag p (assuming $\phi_p \neq 0$).

- The (sample) partial autocorrelation function can generally be based on so called Yule-Walker equations.
- The sample counterpart of the (population) partial autocorrelation function, **sample partial autocorrelation function** (cf. the sample autocorrelation function defined in Section 2) is obtained by using sample autocovariances c_h .

An important detail is that if an observed time series has been generated by an $\text{AR}(p)$ process, its **estimated autocorrelation function** should decay to zero as the lag length increases **with no apparent breaks**, whereas **the estimated partial autocorrelation function** should have a **visible break after lag p** . These features can be made use of when judging whether an $\text{AR}(p)$ process would be a good choice for an observed time series.

To identify the location of the break point in the estimated partial autocorrelation function, one can make use of the following result. In the case of an $\text{AR}(p)$ process, the estimators $\hat{\alpha}_h$, $h > p$, are approximately independent with $\text{N}(0, T^{-1})$ -distribution. Therefore,

$$P(|\hat{\alpha}_h| \geq 1.96/\sqrt{T}) \approx 0.05$$

when $h > p$. On the other hand, because the estimators $c_1, \dots, c_p$ are consistent, the estimator $\hat{\alpha}_p$ converges in probability to the value of the theoretical partial autocorrelation coefficient α_p which for an $\text{AR}(p)$ process is nonzero. Therefore, for $h = p$, $P(|\hat{\alpha}_h| \geq 1.96/\sqrt{T})$ approaches one as the sample size increases.

These remarks justify depicting the sample partial autocorrelation coefficients similar to that of the usual sample autocorrelation coefficients, and adding to it horizontal critical/confidence bands to make its interpretation easier.

4.2 ARMA processes

Already above we have seen that when the $\text{AR}(p)$ process is stationary, it also has the following **univariate (causal) linear process**

$$y_t = \mu + \sum_{j=0}^{\infty} \psi_j u_{t-j}, \quad u_t \sim \text{iid}(0, \sigma^2),$$

where $\psi_0 = 1$ and coefficients ψ_j are functions of the AR parameters, and

$$\sum_{j=0}^{\infty} \psi_j^2 < \infty.$$

As the process $\{u_t\}$ is (strictly) stationary, so is the process $\{y_t\}$.

- A linear process can be defined more generally by allowing $j < 0$. This type of linear process is called non-causal. Non-causal models won't be considered in this course.
- **As two special cases** (cf. and in addition to the $\text{AR}(p)$ process) of the linear process, we consider **MA(q)** and **ARMA(p, q) processes**.

MA(q) process. The MA(q) process is defined as

$$y_t = \mu + u_t + \theta_1 u_{t-1} + \cdots + \theta_q u_{t-q}, \quad u_t \sim \text{iid}(0, \sigma^2),$$

or alternatively using the lag operator as

$$y_t = \mu + \theta(B) u_t, \quad u_t \sim \text{iid}(0, \sigma^2)$$

with the lag-polynomial $\theta(B) = 1 + \theta_1 B + \cdots + \theta_q B^q$. Therefore, the current value of the process is assumed to depend linearly on the present and q last unobservable random shocks (or error terms or innovations). Because an MA(q) process is a special case of the (causal) linear process, it is always (both weakly and strictly) stationary. Moreover, we get

- $E(y_t) = \mu$,
- $\text{Var}(y_t) = \gamma_0 = (1 + \theta_1^2 + \cdots + \theta_q^2) \sigma^2$.

Autocorrelation function. By making use of the general results for linear process, the MA(q) process has the autocovariance function given by

$$\gamma_h = \begin{cases} \sigma^2 \sum_{j=0}^{q-h} \theta_j \theta_{j+h}, & \text{for } 0 \leq h \leq q, \\ 0, & \text{for } h > q, \end{cases}$$

where $\theta_0 = 1$. The autocorrelation function is then obtained via the above-mentioned and discussed formula $\rho_h = \gamma_h / \gamma_0$. Therefore, the autocorrelation function of an MA(q) process drops to zero after lag q (assuming that $\theta_q \neq 0$). If one observes a similar feature in a sample autocorrelation function, one can consider an MA(q) process as a good candidate to model the time series.

When considering the suitability of an MA(q) process, one can make use of the following result: In the case of an MA(q) process, the estimators r_h , $h > q$, are approximately normally distributed with mean zero and variance $(1 + 2\rho_1^2 + \cdots + 2\rho_q^2) / T$. Therefore, if one wants to test at the 5% significance level whether an individual autocorrelation coefficient is different from zero, the estimates r_h , $h > q$, should be compared with the critical bounds

$$\pm 1.96 \sqrt{\hat{w}_q / T}, \quad \hat{w}_q = (1 + 2r_1^2 + \cdots + 2r_q^2).$$

In this case, $P(|r_h| \geq 1.96 \sqrt{\hat{w}_q / T}) \approx 0.05$. Note also that the bounds above are wider than in the case $y_t \sim \text{iid}(0, \sigma^2)$ when $\hat{w}_q$ is replaced by 1.

Invertibility. It can be shown that for the (causal) AR(p) process, there exists a one to one correspondence between the parameters $\phi = (\phi_1, \dots, \phi_p)$ and σ^2 , and the autocovariance function.

- For an MA(q) process, a similar result does not hold. To see this, consider as an example the MA(1) process for which it holds that (assume that $\theta_1 \neq 0$)

$$\gamma_0 = \sigma^2 (1 + \theta_1^2) = \sigma^2 \theta_1^2 (1 + 1/\theta_1^2) \quad \text{and} \quad \gamma_1 = \sigma^2 \theta_1 = \theta_1^2 \sigma^2 (1/\theta_1).$$

Define $\theta_1^* = 1/\theta_1$ and $\sigma_*^2 = \theta_1^2 \sigma^2$, so that $u_t^* = \theta_1 u_t \sim \text{iid}(0, \sigma_*^2)$. Now, it can be seen that the two MA(1) processes $y_t = \mu + u_t + \theta_1 u_{t-1}$ and $y_t^* = \mu + u_t^* + \theta_1^* u_{t-1}^*$ have exactly the same autocovariance function, so that they cannot be distinguished from each other based on autocovariances. If in addition $u_t \sim \text{iid}(0, \sigma^2)$, the entire probability structures of these two processes are indistinguishable.

- A consequence is that estimating the parameters based on an observed time series, unless one “tells” the method which of the parameter combinations (θ_1, σ^2) and (θ_1^*, σ_*^2) (that fit the data equally well), should be chosen.

The typical solution to the above problem in the MA(1) process is to set the condition $|\theta_1| < 1$. Then, one can use the equation $u_t = y_t - \mu - \theta_1 u_{t-1}$ and repetitive substitutions to first obtain $u_t = y_t - \mu - \theta_1(y_{t-1} - \mu) + \theta_1^2 u_{t-2}$, and eventually

$$(y_t - \mu) = - \sum_{j=1}^k (-\theta_1)^j (y_{t-j} - \mu) - (-\theta_1)^{k+1} u_{t-k-1} + u_t.$$

The condition $|\theta_1| < 1$ ensures that the summation on the right hand side converges in mean square and therefore we obtain the representation

$$y_t = - \sum_{j=1}^{\infty} (-\theta_1)^j y_{t-j} + u_t.$$

That is, when $|\theta_1| < 1$ holds, an MA(1) process can be “inverted” to an AR(∞) process, which explains why in the case $|\theta_1| < 1$ an MA(1) process is called **invertible**.

Invertibility condition of an MA(q) process . An MA(q) process, including the above case $q = 1$, is invertible, if the roots of the polynomial $\theta(z)$ ($z \in \mathbb{C}$) lie outside the unit circle/disk on the complex plane or, equivalently, if

$$\theta(z) \neq 0 \quad \text{for} \quad |z| \leq 1.$$

When the invertibility condition holds, one can formally solve u_t from equation $(y_t - \mu) = \theta(B) u_t$ by dividing both sides by the polynomial $\theta(B)^{-1}$. The solution is

$$\pi(B) (y_t - \mu) = u_t \quad \text{or} \quad \sum_{j=0}^{\infty} \pi_j (y_{t-j} - \mu) = u_t,$$

where $\pi(B) = \sum_{j=0}^{\infty} \pi_j B^j = \theta(B)^{-1}$ and the coefficients π_j can be solved as a function of the parameters $\theta_1, \dots, \theta_q$ from the equation

$$(1 + \theta_1 B + \dots + \theta_q B^q)(\pi_0 + \pi_1 B + \pi_2 B^2 + \dots) = 1$$

by interpreting the right hand side as a power series in B , setting the coefficients of B^j equal on both sides of the equation (and $\pi_0 = 1$).

The general definition of a **partial autocorrelation function** (PACF) can also be applied in the case of an $MA(q)$ process, but the calculations involved would become rather complicated. However, because an $MA(q)$ process has (due to invertibility and assuming here $\theta_q \neq 0$) an $AR(\infty)$ representation (see above), and based on what was said above about the partial autocorrelation function of an $AR(p)$ process, it is intuitively clear that the **partial autocorrelation function of an $MA(q)$ process does not drop to zero at any point**, but rather smoothly decays towards zero.

ARMA(p, q) process can be characterized as a combination of $AR(p)$ and $MA(q)$ processes

$$y_t = \nu + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} + u_t + \theta_1 u_{t-1} + \dots + \theta_q u_{t-q},$$

where $u_t \sim \text{iid}(0, \sigma^2)$. Unlike the $AR(p)$ process, the error term is now (generally) not an independent random shock, but instead an autocorrelated $MA(q)$ process.

Using the lag operator and the lag-polynomials $\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p$ and $\theta(B) = 1 + \theta_1 B + \dots + \theta_q B^q$, the $ARMA(p, q)$ process can be expressed as

$$\phi(B) y_t = \nu + \theta(B) u_t.$$

- If $\theta(B) = 1$ (that is, $q = 0$), one obtains the $AR(p)$ process as a special case.
- If $\phi(B) = 1$ (that is, $p = 0$), one obtains the $MA(q)$ process.

Stationarity and invertibility. Based on what was said for linear process, it is clear than an $ARMA(p, q)$ process has a well-defined (causal) $MA(\infty)$ representation if the polynomial $\phi(z)$ satisfies the sufficient stationarity condition of an $AR(p)$ process (apply what was said in the case of an $AR(p)$ process, but now with $MA(q)$ error term).

- A sufficient condition for the stationarity of an $ARMA(p, q)$ process is that the roots of the polynomial $\phi(z)$ ($z \in \mathbb{C}$) lie outside the unit circle/disk on the complex plane, or equivalently that

$$\phi(z) \neq 0 \text{ for } |z| \leq 1.$$

Based on the above discussion, the **invertibility condition** of an $\text{MA}(q)$ process will also suffice to ensure the invertibility of an $\text{ARMA}(p,q)$ process

$$\theta(z) \neq 0 \text{ for } |z| \leq 1.$$

Furthermore, as can be deduced from what was discussed above, when the stationarity condition holds, an $\text{ARMA}(p,q)$ process has an $\text{MA}(\infty)$ representation (cf. the $\text{AR}(p)$ case)

$$(y_t - \mu) = \frac{\theta(B)}{\phi(B)} u_t = \psi(B) u_t,$$

where $\psi(B) = \sum_{j=0}^{\infty} \psi_j B^j = \theta(B) \phi(B)^{-1}$ and the coefficients ψ_j can be solved as a function of the parameters $\phi_1, \dots, \phi_p$ and $\theta_1, \dots, \theta_q$ by equating the coefficients of B^j on both sides of the equation

$$(1 - \phi_1 B - \dots - \phi_p B^p) (\psi_0 + \psi_1 B + \psi_2 B^2 + \dots) = 1 + \theta_1 B + \dots + \theta_q B^q.$$

- From this equation, it can be solved that $\psi_0 = 1$, $\theta_1 = \psi_1 - \psi_0 \phi_1$, and in general,

$$\psi_j = \sum_{i=1}^p \phi_i \psi_{j-i} + \theta_j, \quad j = 0, 1, 2, \dots,$$

where $\theta_0 = 1$, $\theta_j = 0$, $j > q$, and $\psi_j = 0$, $j < 0$.

On the other hand, when the invertibility condition holds, an $\text{ARMA}(p,q)$ process also has an $\text{AR}(\infty)$ representation

$$\frac{\phi(B)}{\theta(B)} (y_t - \mu) = \pi(B) (y_t - \mu) = u_t,$$

where $\pi(B) = \sum_{j=0}^{\infty} \pi_j B^j = \phi(B) \theta(B)^{-1}$. The coefficients π_j can be solved similarly as the ψ_j 's above (just exchange the roles of the polynomials $\phi(B)$ and $\theta(B)$). The end result can be expressed as

$$\pi_j = - \sum_{i=1}^p \theta_i \pi_{j-i} - \phi_j, \quad j = 0, 1, 2, \dots,$$

where $\phi_0 = -1$, $\phi_j = 0$, $j > p$, and $\pi_j = 0$, $j < 0$. In particular, it holds that $\pi_0 = 1$ and, as in the case of the coefficients ψ_j , $\pi_j \rightarrow 0$ at an exponential rate as $j \rightarrow \infty$.

The general discussion concerning invertibility in the previous subsection in connection with the $\text{MA}(1)$ process also holds here and generalizes to the case of an $\text{ARMA}(p,q)$ process.

Identification (or uniqueness) condition of an $\text{ARMA}(p,q)$ process. Earlier in connection to MA processes, the invertibility condition was motivated by the fact that it guarantees the existence of a one-to-one correspondence between the autocovariance function of a process and its model parameters. For general $\text{ARMA}(p,q)$ processes, invertibility alone is not sufficient to guarantee the existence of such a correspondence, which is also required for maximum likelihood estimation of the parameters.

- Therefore, the polynomials $\phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p$ and $\theta(z) = 1 + \theta_1 z + \dots + \theta_q z^q$ of a stationary and invertible ARMA(p, q) process are assumed to have **no common roots** and, moreover, $\phi_p \neq 0$ or $\theta_q \neq 0$.
- Quite often, the requirement that $\phi_p \neq 0$ or $\theta_q \neq 0$ is not explicitly mentioned but is understood to be contained in the condition ruling out common roots. Unless otherwise mentioned, in what follows, we assume that this identification condition holds.
- As an example of non-identifiable process, consider an ARMA(1,1) process, with the linear representation

$$(y_t - \mu) = \frac{1 + \theta_1 B}{1 - \phi_1 B} u_t,$$

in the special case $\phi_1 = -\theta_1$, it is clear that the polynomials on the right hand side can be cancelled out, resulting in the equation $y_t - \mu = u_t$ so that the process is not a “real” ARMA(1,1) process, but instead just white noise.

The **autocorrelation function of an ARMA(p, q) process** can be derived in a manner similar to the AR(p) case. In what follows, we only present the general principle of the solution.

Multiplying both sides of the (demeaned) ARMA(p, q) process with y_{t-h} ($h \geq 0$) and taking expectations yields

$$\begin{aligned} \mathbb{E}\left((y_t - \mu)(y_{t-h} - \mu)\right) &= \phi_1 \mathbb{E}\left((y_{t-1} - \mu)(y_{t-h} - \mu)\right) + \dots + \phi_p \mathbb{E}\left((y_{t-p} - \mu)(y_{t-h} - \mu)\right) + \mathbb{E}\left(u_t(y_{t-h} - \mu)\right) \\ &\quad + \theta_1 \mathbb{E}\left(u_{t-1}(y_{t-h} - \mu)\right) + \dots + \theta_q \mathbb{E}\left(u_{t-q}(y_{t-h} - \mu)\right). \end{aligned}$$

Making use of the MA(∞) representation and the definition of autocovariance this further leads to

$$\gamma_h = \begin{cases} \phi_1 \gamma_{h-1} + \dots + \phi_p \gamma_{h-p} + \sigma^2 \sum_{j=0}^{\infty} \theta_{h+j} \psi_j, & 0 \leq h < \max\{p, q+1\}, \\ \phi_1 \gamma_{h-1} + \dots + \phi_p \gamma_{h-p}, & h \geq \max\{p, q+1\}, \end{cases}$$

where $\theta_0 = 1$ and $\theta_j = 0$ for $j \notin \{0, \dots, q\}$. Moreover, the coefficients ψ_j can be expressed as function of the parameters $\phi_1, \dots, \phi_p$ and $\theta_1, \dots, \theta_q$.

- It is rather clear that as the lag length h increases, the autocorrelation function of an ARMA(p, q) process decays exponentially to zero, potentially following the shape of a dampening sine wave (in the case $p = 1$ this is rather easy to verify).
- Because an ARMA(p, q) process has an AR(∞) representation when the invertibility condition holds, one can use the general definition of the partial autocorrelation function to conclude that the general shape of the partial autocorrelation function of an ARMA(p, q) process is comparable to that of the autocorrelation function.

To conclude, importantly, neither the autocorrelation nor the partial autocorrelation function of an ARMA(p, q) process **ever drop to zero** but instead both steadily decay towards zero.

4.3 Parameter estimation of AR and ARMA models

Consider the AR(p) model and normally distributed errors where $u_t \sim \text{nid}(0, \sigma^2)$ and the parameters $\phi_1, \dots, \phi_p$ satisfy the stationarity condition.

Denote

$$\underline{\mathbf{Y}}_t = (y_{-p+1}, \dots, y_0, y_1, \dots, y_t),$$

then the density function of the random vector $\underline{\mathbf{Y}}_t$ determines the likelihood function of the model parameters. By omitting the arguments of the density function, we can write using the definition of the conditional density function

$$\begin{aligned} f_{\underline{\mathbf{Y}}_T} &= f_{y_T | \underline{\mathbf{Y}}_{T-1}} \cdot f_{\underline{\mathbf{Y}}_{T-1}} \\ &= f_{y_T | \underline{\mathbf{Y}}_{T-1}} \cdot f_{y_{T-1} | \underline{\mathbf{Y}}_{T-2}} \cdot f_{\underline{\mathbf{Y}}_{T-2}} \\ &\vdots \\ &= \prod_{t=1}^T f_{y_t | \underline{\mathbf{Y}}_{t-1}} \cdot f_{y_0}. \end{aligned}$$

We can denote

$$\mathbf{x}_t = [1 \quad y_{t-1} \quad \dots \quad y_{t-p}]' \quad \text{and} \quad \boldsymbol{\beta} = [\nu \quad \phi_1 \quad \dots \quad \phi_p]',$$

where $\mathbf{x}_t$ is now the vector of predictive/explanatory variables, which includes the lags of y_t in the AR(p) case. Since $\mathbf{x}_t \perp\!\!\!\perp u_t$, it follows from the assumptions made that

$$y_t | \underline{\mathbf{Y}}_{t-1} \sim \text{N}(\mathbf{x}_t' \boldsymbol{\beta}, \sigma^2).$$

Thus,

$$f_{y_t | \underline{\mathbf{Y}}_{t-1}}(y_t | \mathbf{x}_t) = (2\pi)^{-1/2} (\sigma^2)^{-1/2} \exp \left(-\frac{1}{2\sigma^2} (y_t - \mathbf{x}_t' \boldsymbol{\beta})^2 \right).$$

Assuming that the initial values $y_{-p+1}, \dots, y_0$ are available, the expressions for the factors of the density function $f_{\underline{\mathbf{Y}}_T}$ can be obtained from this, except for the factor f_{y_0} .

- In the exact maximum likelihood estimation method, the exact expression for this factor is also used. Since the corresponding possibility is rather complex, we consider the conditional maximum likelihood estimation conditional on the impact of the initial values f_{y_0} .

The conditional log-likelihood function is then obtained as

$$l(\boldsymbol{\beta}, \sigma^2) = -\frac{T}{2} \log(2\pi) - \frac{T}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=1}^T (y_t - \mathbf{x}_t' \boldsymbol{\beta})^2.$$

Since conditioning only loses the information of the first p observations, it has no effect in large samples (asymptotically). Conditional maximum likelihood estimation thus clearly reduces to least squares (LS) estimation, yielding the maximum likelihood estimators as

$$\hat{\boldsymbol{\beta}} = \left(\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right)^{-1} \sum_{t=1}^T \mathbf{x}_t y_t$$

and

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T (y_t - \mathbf{x}_t' \hat{\boldsymbol{\beta}})^2.$$

Since

$$\frac{\partial l(\boldsymbol{\beta}, \sigma^2)}{\partial \boldsymbol{\beta}} = \frac{1}{\sigma^2} \sum_{t=1}^T (y_t - \mathbf{x}_t' \boldsymbol{\beta}) \mathbf{x}_t,$$

it holds that

$$\mathbf{E}(\partial^2 l(\boldsymbol{\beta}, \sigma^2) / \partial \boldsymbol{\beta} \partial \sigma^2) = \mathbf{0}$$

and

$$-\mathbf{E}(\partial^2 l(\boldsymbol{\beta}, \sigma^2) / \partial \boldsymbol{\beta} \partial \boldsymbol{\beta}') = \sigma^{-2} \sum_{t=1}^T \mathbf{E}(\mathbf{x}_t \mathbf{x}_t') = T \sigma^{-2} \mathbf{E}(\mathbf{x}_t \mathbf{x}_t')$$

(stationarity!). Referring to the general maximum likelihood estimation theory, the asymptotic results are thus obtained as

$$\hat{\boldsymbol{\beta}}_{\text{as.}} \sim \mathbf{N}\left(\boldsymbol{\beta}, T^{-1} \sigma^2 (\mathbf{E}(\mathbf{x}_t \mathbf{x}_t'))^{-1}\right)$$

and

$$\hat{\boldsymbol{\beta}}_{\text{as.}} \perp\!\!\!\perp \hat{\sigma}^2.$$

Hypotheses concerning the parameters $\phi_1, \dots, \phi_p$ can be tested using conventional likelihood theory tests.

- In particular, the hypothesis $\phi_p = 0$ can be tested by comparing the estimate $\hat{\phi}_p$ to its (approximate) standard error, which is obtained by calculating the square root of the p th (i.e., in this case, the last) diagonal element of the matrix $\hat{\sigma}^2 \left(\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right)^{-1}$.
- If $|\hat{\phi}_p|$ is greater than $1.96 \cdot \hat{\phi}_p$'s standard error, the hypothesis is rejected at approximately the 5% significance level.

The aforementioned estimate $\hat{\phi}_p$ can be interpreted as the estimate of the PACF of the process y_t at the lag p .

- For an $AR(p)$ process, the PACF has a cutoff at lag p , which can be used to determine the order of the $AR(p)$ model or to examine whether the AR model is appropriate.

4.4 Model selection of the $ARMA(p,q)$ model

Sample autocorrelations and partial autocorrelations. After visualizing the series, a logical next step is to investigate the autocorrelation structure using the estimated (sample) autocorrelation and partial autocorrelation functions. In many cases, these functions provide clues regarding the suitable lag lengths p and q for an $ARMA(p,q)$ model.

In particular, **given what we have learned** about the (theoretical) properties of AR, MA and ARMA processes, **one should look for potential breaks (“sudden drops to zero”) in the sample autocorrelation and sample partial autocorrelation functions.**

- A break (cutoff) in the sample autocorrelation function, but not in the sample partial autocorrelation function, suggests that an MA model may be a suitable candidate model.
- In the AR case, the properties are reversed. That is, we observe a break in the sample partial autocorrelation function but not in the sample autocorrelation function.
- If there are no clear breaks in either function (i.e., both tail off gradually), an ARMA model appears to be a likely candidate.

This is the so-called **Box-Jenkins model selection procedure**. Although this procedure often does not point to a single unique candidate model, it helps rule out certain alternatives and serves as a valuable tool when used in conjunction with the model selection criteria introduced below.

Sequential tests. One way to choose the $AR(p)$ or $ARMA(p,q)$ model is based on sequential tests. In what follows, we will consider an $AR(p)$ model for simplicity, but the following procedure can be generalized into ARMA models too.

- Start by choosing a relatively large model (that is, a long lag length p^*).
- Estimate an $AR(p^*)$ model

$$y_t = \nu + \phi_1 y_{t-1} + \cdots + \phi_{p^*-1} y_{t-p^*+1} + \phi_{p^*} y_{t-p^*} + u_t.$$

- Test (evaluate) constraint $H_0 : \phi_{p^*} = 0$
- If the null hypothesis cannot be rejected, estimate $AR(p^*-1)$ model, and test for $H_0 : \phi_{p^*-1} = 0$

- This sequential testing procedure is carried out until we can reject the null hypothesis (get the first rejection).

The sequential testing is often used quite mechanically. Notice the obvious multiple testing problem and its impact on p-values, but this is often ignored in this context on t -test statistics, the Wald test or using information criteria.

Information criteria. In modern practice, model selection criteria are widely used. The most common of these take the form

$$C(m) = -\frac{2}{T}l(\widehat{\beta}(m), \widehat{\sigma}^2(m)) + \frac{(m+1)g(T)}{T},$$

where $\widehat{\beta}(m)$ and $\widehat{\sigma}^2(m)$ are the estimators (estimates) of the parameters β and σ^2 based on the $\text{AR}(m)$ model (including constant term), and $g(T)$ is a so-called penalty function that penalizes the use of “unnecessary” parameters. “Good” models have a small value of $C(m)$. Note that these criteria can sometimes be represented as

$$C(m) - \log(2\pi) - 1 = \log(\widehat{\sigma}^2(m)) + \frac{(m+1)g(T)}{T}.$$

Three commonly used penalty functions are (i.e., **Akaike (AIC)**, **Schwarz (BIC)**, also Bayesian Information Criterion), and Hannan-Quinn (HQ) information criteria)

$$\begin{aligned} g(T) &= 2 \text{ (AIC) ,} \\ g(T) &= \log(T) \text{ (BIC), and} \\ g(T) &= 2 \log \log(T) \text{ (HQ).} \end{aligned}$$

Occasionally, the expression for the information criteria is multiplied by a factor, such as T . This does not change the relative ranking of the models, which is the only factor of interest for model selection.

Residuals and diagnostic checking. In addition to the above, we can examine the estimated autocorrelation function (ACF) of the residuals $\widehat{u}_t$ (defined as $\widehat{u}_t = y_t - \mathbf{x}_t' \widehat{\beta}$ in the context and above notation of AR models). Ideally, no significantly large values should appear. Note that while the standard error is often roughly approximated as $1/\sqrt{T}$, this is not strictly correct for residuals.

Additionally, the Ljung-Box test statistic can be used to test for remaining residual autocorrelation, provided that the lag length H is sufficiently large. In this context, the appropriate asymptotic distribution is χ^2_{H-p-1} , where $p+1$ represents the number of estimated parameters in the $\text{AR}(p)$ model.

- Note that for $\text{ARMA}(p, q)$ models, the degrees of freedom must be adjusted accordingly to account for all estimated parameters (that is, $p + q + 1$).

Example (continued). Let us continue with the interest rate differential and exchange rate example. In this part, we specify univariate AR and ARMA models for each time series separately.

Starting with the (differenced) exchange rate series, we observe that it is clearly less persistent than the interest rate differential. For instance, the reported Ljung-Box test p-values for various lag lengths H are typically around the 0.05 level (5%), but not lower. The sample autocorrelation function was reported previously in Section 3; below, we depict the sample partial autocorrelation function:

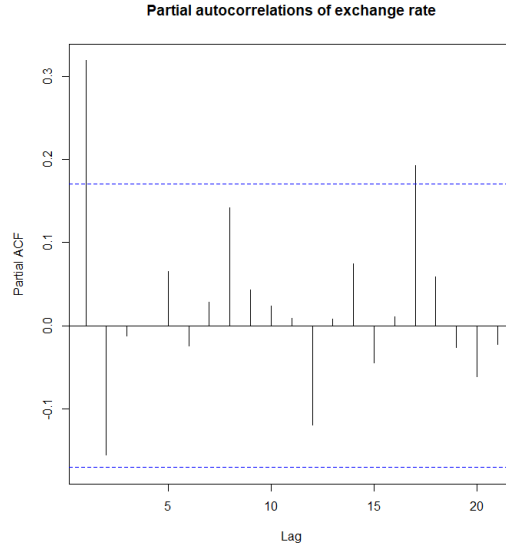


Figure: Sample partial autocorrelation function for $dexh_t$.

This, considered alongside the sample autocorrelation function, suggests that an $\text{AR}(1)$ or $\text{MA}(1)$ model might be appropriate. Specifically, the cutoff point in both the sample ACF and PACF appears to occur after the first lag. If we interpret the break as occurring in the PACF but not the ACF, we would select an $\text{AR}(1)$ model. Conversely, if we interpret the cutoff as being distinct only in the sample ACF, this implies an $\text{MA}(1)$ process.

- Information criteria: When restricting the analysis to AR models, the AIC suggests an $\text{AR}(2)$ specification, whereas the BIC suggests an $\text{AR}(1)$. However, when considering

ARMA models (i.e., allowing for an MA component), both information criteria favor an MA(1) model.

For the interest rate differential, as previously noted, the autocorrelation is substantially stronger; the p-values of the Ljung-Box test are 0.000 across various lag lengths. A comparison of the estimated partial autocorrelation function with the sample ACF strongly suggests that an AR(1)—or perhaps an AR(2)—model is adequate.

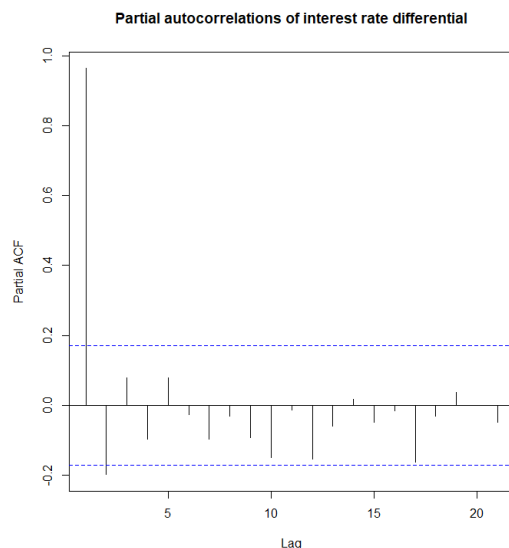


Figure: Sample partial autocorrelation function for $rdif_t$.

Automatic information criteria (for $rdif_t$) suggest taking the first difference of the time series—that is, moving to an ARIMA($p, 1, q$) model, as described in more detail in the following section. While the relatively strong persistence implies that this might be reasonable, given the relatively short time series, we can still consider an AR(2) model for the series without differencing.

When restricting the analysis to autoregressive models—specifically an AR(1) for $dexch_t$ and an AR(2) for $rdif_t$ —we obtain the following estimation results:

dexch		Estimate	Std. Error	t value	Pr(> t)
constant		-0.0009	0.002	-0.48	0.633
ar1		0.318	0.082	3.86	0.000

Ljung-Box tests:

Lag	Q.statistic	p.value
8	6.134	0.408
12	9.622	0.474
16	12.110	0.597
20	19.527	0.360

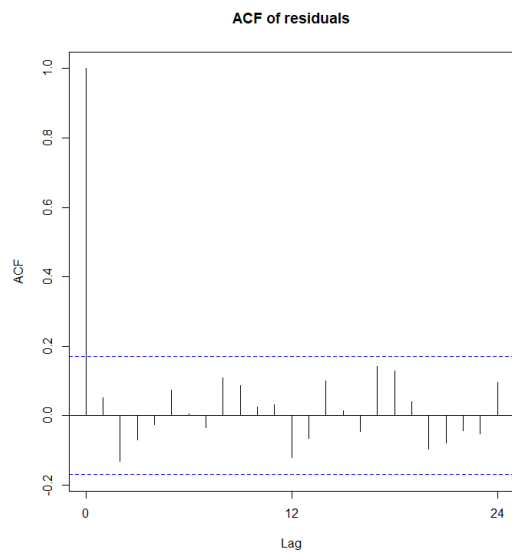
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rdif

	Estimate	Std. Error	t value	Pr(> t)
constant	-0.007	0.012	-0.643	0.520
ar1	1.151	0.085	13.482	0.000
ar2	-0.187	0.086	-2.175	0.030

Ljung-Box tests:

	Lag	Q.statistic	p.value
1	8	10.450	0.063
2	12	16.617	0.055
3	16	17.036	0.198
4	20	18.371	0.366



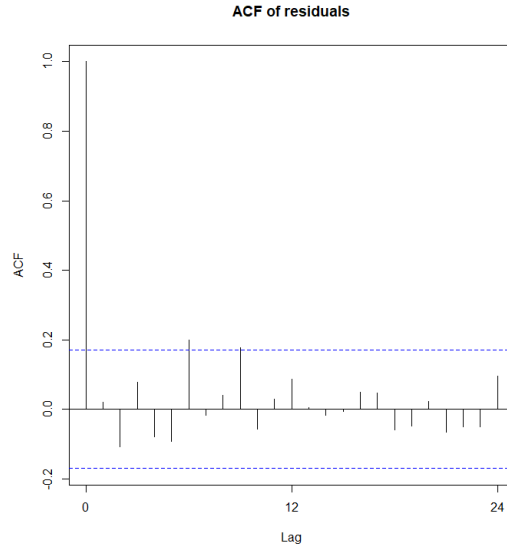


Figure: Sample autocorrelations for residuals of the estimated AR(1) and AR(2) models for $dexch_t$ and $rdif_t$, respectively.

In both cases—and particularly for $dexch$ —the selected models appear adequate, as evidenced by the absence of remaining residual autocorrelation. Furthermore, the AR coefficients are statistically significant at the 5% level.

- The question of whether the interest rate differential series should be first differenced is discussed further in the next section

Note that we do not consider possible residual autocorrelation in the squared residuals. If such dependency is present, this may indicate nonlinear dependencies such as conditional heteroskedasticity. The TSE material contains details on ARCH and GARCH models (combined with an AR model for the conditional mean)

[Link to the TSE material](#)

4.5 Forecasting with AR and ARMA models

Let us consider the forecasting formulas for the AR(p) model. First, we recall the general properties of the conditional expectation (see also the Appendix A). In the following, we aim

to predict the value of the random variable Y using the value $X = x$ obtained by the random variable, or random vector, X :

- **CEV1:** $E(aY_1 + bY_2 | X) = aE(Y_1 | X) + bE(Y_2 | X)$, when a and b are constants.
- **CEV2:** $E(Y | X) = E(Y)$, when Y and X are independent random variables.
- **CEV3:** $E(Y) = E[E(Y | X)]$ (so-called law of iterated expectations)
- **CEV4:** $E[h(X)Y | X] = h(X)E(Y | X)$ for any function h (assuming the expected value of $h(X)Y$ exists and is finite).

When applied to time series processes, in terms of the mean squared forecast error, the best (and otherwise natural) forecast for the observation y_{t+1} at time t is the conditional expectation

$$\begin{aligned} E_t(y_{t+1}) &= E(y_{t+1} | y_{t-j}, j \geq 0) \\ &= E(y_{t+1} | \mathcal{F}_t) \\ &\stackrel{\text{notation}}{=} y_t(1), \end{aligned}$$

where $\mathcal{F}_t = \{y_t, \dots, y_{t-p+1}\}$ is the **information set** used for forecasting at time t (including, among other things, the lags of y_t).

For the AR(p) model, it follows that

$$\begin{aligned} y_t(1) &= \nu + \phi_1 E_t(y_t) + \dots + \phi_p E_t(y_{t-p+1}) + E_t(u_{t+1}) \\ &= \nu + \phi_1 y_t + \dots + \phi_p y_{t-p+1}, \end{aligned}$$

where **based on the CEV2** (and, e.g., the MA(∞) presentation of AR(MA) model), we get $E_t(u_{t+1}) = E(u_{t+1}) = 0$. - This generalizes also to the case of $E_t(u_{t+h}) = E(u_{t+h}) = 0$.

For more general, when the forecast horizon $h > 1$, we get

$$E_t(y_{t+h}) = \nu + \phi_1 E_t(y_{t+h-1}) + \dots + \phi_p E_t(y_{t+h-p}),$$

that is

$$y_t(h) = \nu + \phi_1 y_t(h-1) + \dots + \phi_p y_t(h-p),$$

where $y_t(h-j) = E_t(y_{t+h-j})$ and $y_t(h-j) = y_{t+h-j}$, when $h \leq j$. That is, which values of y_t are already included in the information set and which ones need to be predicted as a part of this iterative multiperiod forecasting.

From the MA(∞) representation of the model (process), it can be seen that

$$y_t(h) = \mu + \sum_{j=1}^{\infty} \psi_j u_{t+h-j} \quad (\psi_0 = 1)$$

and when assuming $u_t \sim \text{nid}(0, \sigma^2)$, we get the result

$$y_{t+h} - y_t(h) \sim \text{N} \left(0, \sigma^2 \sum_{j=0}^{h-1} \psi_j^2 \right).$$

This result can be used to construct confidence intervals. For example, a 95% confidence interval is obtained as $y_t(h) \pm 1.96 \left(\sigma^2 \sum_{j=0}^{h-1} \psi_j^2 \right)^{1/2}$.

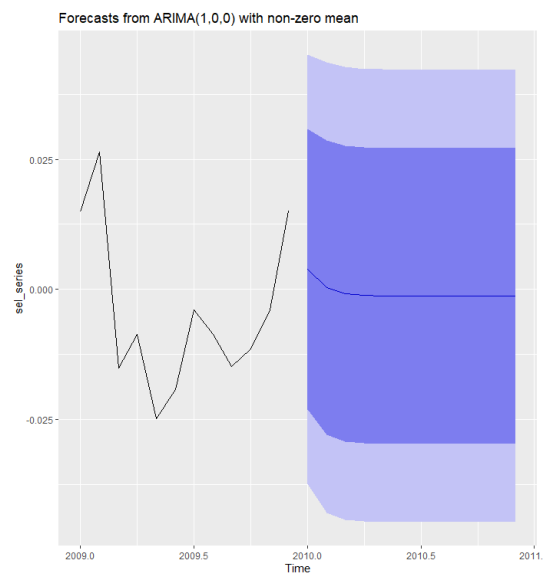
- In practice, unknown parameters are, of course, replaced by estimates, so the above applies only approximately.
- Note also that in the confidence interval, h was assumed to be fixed. When examining multiple such confidence intervals simultaneously, one encounters the usual problems of combined confidence intervals.

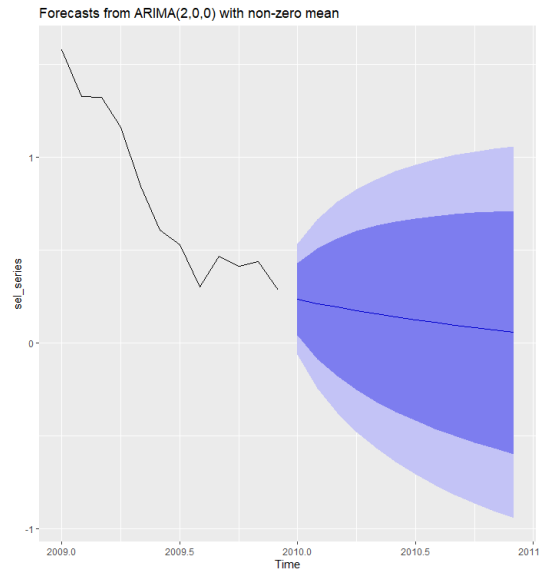
Forecasts can naturally be computed also for the ARMA models in analogous way. However, the fact that the past error terms (i.e. the MA part) are not observable somewhat complicates the forecast construction.

Example (continued). As illustrations, below we report (point) forecasts 12-months-ahead ($h = 12$) from the end of the *dexch* and *rdif* time series (Dec 2009). Furthermore, we depict the constructed point forecasts with 80% and 95% confidence intervals:

dexch					
Month	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan 2010	0.0039	-0.0230	0.0309	-0.0373	0.0451
Feb 2010	0.0004	-0.0279	0.0286	-0.0429	0.0436
Mar 2010	-0.0008	-0.0291	0.0276	-0.0442	0.0427
Apr 2010	-0.0011	-0.0295	0.0273	-0.0446	0.0423
May 2010	-0.0012	-0.0296	0.0272	-0.0447	0.0422
Jun 2010	-0.0013	-0.0297	0.0272	-0.0447	0.0422
Jul 2010	-0.0013	-0.0297	0.0271	-0.0447	0.0422
Aug 2010	-0.0013	-0.0297	0.0271	-0.0447	0.0422
Sep 2010	-0.0013	-0.0297	0.0271	-0.0447	0.0422
Oct 2010	-0.0013	-0.0297	0.0271	-0.0447	0.0422
Nov 2010	-0.0013	-0.0297	0.0271	-0.0447	0.0422
Dec 2010	-0.0013	-0.0297	0.0271	-0.0447	0.0422
=====					
rdif					
Month	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan 2010	0.2363	0.0414	0.4311	-0.0617	0.5342

Feb 2010	0.2116	-0.0855	0.5087	-0.2428	0.6660
Mar 2010	0.1919	-0.1788	0.5626	-0.3750	0.7588
Apr 2010	0.1739	-0.2537	0.6015	-0.4801	0.8278
May 2010	0.1568	-0.3169	0.6305	-0.5676	0.8812
Jun 2010	0.1405	-0.3716	0.6527	-0.6427	0.9238
Jul 2010	0.1250	-0.4199	0.6698	-0.7083	0.9582
Aug 2010	0.1101	-0.4629	0.6832	-0.7663	0.9865
Sep 2010	0.0960	-0.5016	0.6936	-0.8180	1.0099
Oct 2010	0.0824	-0.5367	0.7016	-0.8645	1.0293
Nov 2010	0.0695	-0.5687	0.7077	-0.9065	1.0455
Dec 2010	0.0572	-0.5979	0.7122	-0.9446	1.0589





4.6 Out-of-sample forecasting

In various fields, particularly in economics and finance, it is crucial to assess the reliability and accuracy of forecasts generated by estimated time series models. A popular approach for this evaluation is (pseudo) out-of-sample forecasting aiming to mimic a real-time forecasting scenario as closely as possible, using only information that would have been available at the time the forecast was made.

The fundamental idea is to evaluate forecast performance by systematically reserving a portion of the data— withheld from the model during estimation—to serve as a “future” data against which forecasts are compared. This method is at times termed “pseudo” out-of-sample forecasting because, while we simulate real-time forecasting, we typically possess full knowledge of the entire dataset beforehand. In “true” real-time forecasting, future observations are, by definition, unknown.

See details in the TSE course material:

[Link to the TSE material](#)

The principles of out-of-sample forecasting are valid, not just for univariate models, but also for multivariate/multiple-equation models such as VAR models to be considered in the coming sections.

5 Univariate nonstationary processes

5.1 Integrated process and random walk

In many applications, time series often exhibit trends, implying that their properties cannot be modeled by stationary processes. (Note that the term “trend” is used quite loosely here.) However, quite often the differences of such time series $y_t - y_{t-1} \equiv \Delta y_t$ (that is, $\Delta = 1 - B$) look stationary and hence could be modeled as stationary processes.

- If the differenced series is trending, then one may consider modeling the change of the difference $\Delta(\Delta y_t) = y_t - 2y_{t-1} + y_{t+2} \equiv \Delta^2 y_t$.
- One might apply the same idea to justify using higher order differences, but this is very rare in practice.

Integrated process. In line with the aforementioned, a stochastic (possibly even vector-valued) process $\{y_t, t = 1, 2, \dots\}$ is said to be integrated of order d or $I(d)$ process, if the process

$$\{\Delta^d y_t, t = 1, 2, \dots\}$$

is (asymptotically) stationary, but the process $\{\Delta^{d-1} y_t, t = 1, 2, \dots\}$ is not (asymptotically) stationary. If y_t is an $I(d)$ process, we write $y_t \sim I(d)$ ($d \geq 1$).

- In what follows, we focus on the case $d = 1$ (and naturally $d = 0$) in this course.
- At times the above definition is formulated as $\{\Delta^d(y_t - E(y_t)), t = 1, 2, \dots\}$ if emphasizing demeaning.

Random walk. The simplest example of an $I(1)$ process is the random walk, which is defined by

$$\Delta y_t = \nu + u_t, \quad t = 1, 2, \dots$$

where $u_t \sim \text{iid}(0, \omega^2)$. By sequential substitution, we obtain

$$y_t = y_0 + \nu t + \sum_{j=1}^t u_j, \quad t = 1, 2, \dots$$

Assume the initial value y_0 is fixed. Then

$$\mathbb{E}(y_t) = y_0 + \nu t$$

and

$$\text{Var}(y_t) = \text{Var}\left(\sum_{j=1}^t u_j\right) = \omega^2 t,$$

and further

$$\text{Corr}(y_t, y_{t-h}) = \frac{1}{\sqrt{1 + h/t}}, \quad h \geq 0.$$

We observe that the random walk is nonstationary and this holds independent of the assumption on the initial value. As $\Delta y_t \sim \text{iid}(\nu, \omega^2)$ is stationary, the random walk is $I(1)$.

- Occasionally, it is assumed that $\nu = 0$ (that is, **the random walk without drift**).

The above derivations show that for the random walk $\text{Var}(y_t) \rightarrow \infty$ and $\text{Corr}(y_t, y_{t-h}) \rightarrow 1$ for all (also large) $h > 0$ as $t \rightarrow \infty$. The random walk processes are hence strongly autocorrelated and their time paths have trending patterns.

- A simulated figure below depicts a simulated realization of the random walk (left, and its first difference Δy_t on the right) with the assumption $u_t \sim \text{nid}(0, 1)$ and $y_0 = \nu = 0$ (so that the deterministic (linear) trend term $y_0 + \nu t$ is absent).
- As $y_t = \sum_{j=1}^t u_j$ is random, the underlying trend is often called a stochastic trend. If $\nu \neq 0$, this trend wonders about the linear trend $y_0 + \nu t$. All of these features are generally shared by more general $I(1)$ processes.

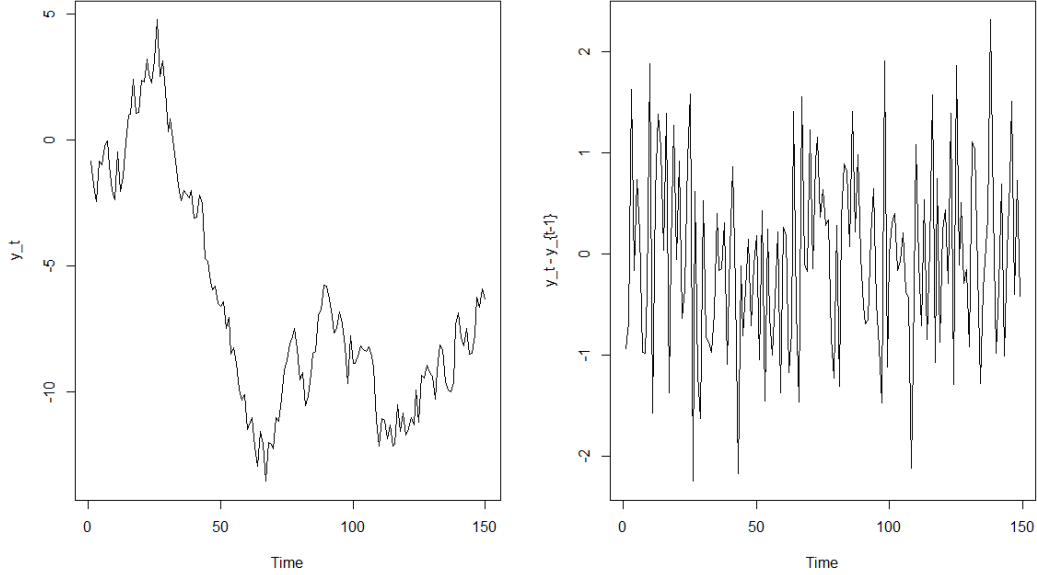


Figure: A simulated realization of the random walk process (left) and its first difference on the right.

5.2 ARIMA process

The random walk is the simplest example of an **ARIMA(p, d, q) process**. Generally, a process y_t is called an ARIMA(p, d, q) process if it becomes a stationary and invertible ARMA(p, q) process after differencing d times, i.e., $\Delta^d y_t \sim \text{ARMA}(p, q)$.

- In practice, the case $d = 1$ is by far the most common.
- When $d = 0$, the special case is an ARMA(p, q) process, and thus also an AR(p) process when $q = 0$.

5.3 Unit root process

Consider the AR(p) process $a(B)x_t = u_t$, where $a(B) = 1 - a_1 B - \dots - a_p B^p$.

- Here and below we apply the notation x_t , when we assume that the underlying process does not contain a constant or other deterministic terms.

It can be shown (see Appendix A) that we can write

$$a(B) = \Delta - \phi B - \phi_1 \Delta B - \dots - \phi_{p-1} \Delta B^{p-1},$$

where $\phi = -a(1)$, and so that the $AR(p)$ process has the representation

$$\Delta x_t = \phi x_{t-1} + \phi_1 \Delta x_{t-1} + \dots + \phi_{p-1} \Delta x_{t-p+1} + u_t, \quad t = 1, 2, \dots,$$

where $u_t \sim \text{iid}(0, \omega^2)$.

Assume now that the polynomial $a(z)$ satisfies the condition

$$\text{If } a(z) = 0, \text{ then } |z| < 1 \text{ or } z = 1$$

This condition entails that the roots of the polynomial $a(z)$ locate outside the (complex) unit circle or at the point $z = 1$.

- If all of the roots are outside the unit circle, then x_t is (at least) asymptotically stationary.

Assume now that there is precisely a single unit root. The fundamental lemma of algebra then implies that $a(z) = (1 - z)b(z)$, where $b(z)$ is a polynomial of order at most $p - 1$ and such that its roots are outside the unit circle (see Appendix A).

- It is clear that this is equivalent to assuming above that $\phi = 0$ and when $\phi = 0$ the above mentioned polynomial $b(z) = 1 - \phi_1 z - \dots - \phi_{p-1} z^{p-1} \equiv \phi(z)$.
- Hence, $\Delta x_t \sim I(0)$ and, as the roots of $b(z)$ are outside the unit circle, we can find initial values $\Delta x_{t-1}, \dots, \Delta x_{t-p+1}$ such that Δx_t is stationary.
- It is then also possible to write the process Δx_t as $\Delta x_t = \phi(B)^{-1} u_t$. Note that $\phi(B)^{-1} u_t$ is effectively an $AR(p - 1)$ written as an infinite order MA presentation.

5.4 Unit root testing

Augmented Dickey-Fuller test (ADF test) is one (most commonly used) unit root test in the econometric literature. It is based on the test regression model (cf. the discussion above on unit root processes)

$$\Delta x_t = \nu + \phi x_{t-1} + \sum_{j=1}^{p-1} \phi_j \Delta x_{t-j} + \eta_t, \quad t = 1, 2, \dots,$$

where $\eta_t \sim \text{nid}(0, \sigma^2)$, $\nu = -\phi\mu$. The null hypothesis is $H_0 : \phi = 0$, which implies the presence of a unit root.

- The test statistic can be the “t-ratio” related to the parameter ϕ (i.e. estimate divided by its standard error), whose asymptotic distribution is non-standard.
- All asymptotic results also hold without the normality assumption (i.e., $\eta_t \sim \text{iid}(0, \sigma^2)$) is sufficient. If the null hypothesis (unit root) is not rejected, according to test regression equation, an $\text{AR}(p - 1)$ model can be based on the differenced time series. In other words, it is in that case necessary to take differences of x_t , which indicates unit root type nonstationarity for the levels (x_t).

Implementing the ADF test requires specifying the number of lags, p . This selection can be made using fixed rules of thumb, information criteria, or sequential testing procedures.

- In practice, it is advisable to consider a range of lag lengths—often determined by the data frequency—rather than relying on a single specification. Conclusions regarding the presence of a unit root should ideally be robust across these different examinations.

Additionally, based on the visual characteristics of the time series and the theoretical nature of the data, the test regression must often be generalized to include deterministic terms. The general model is given by

$$y_t = d_t + x_t, \quad t = 1, 2, \dots,$$

where x_t is as above and the deterministic term $d_t = \nu$ (as now above) or $d_t = \nu_1 + \nu_2 t$.

- It is also possible to add indicator functions to account for (deterministic) seasonal effects.

A critical limitation: The low power of the ADF test. As discussed more detail in the TSE material (see below), an important limitation of the ADF test is its low statistical power. In practical terms, this means: The ADF test often fails to reject the null hypothesis of a unit root, even when the time series is actually stationary.

- In other words, the ADF test often exhibits low statistical power; it may fail to correctly identify a stationary process, particularly when the process is close to non-stationarity (e.g., an $\text{AR}(1)$ process with an autoregressive coefficient near 1).

Consequently, failing to reject the null hypothesis does not strictly prove that the process contains a unit root. Instead, it may simply indicate that the data provides insufficient information to reject the hypothesis—a common issue when the sample size is small or the process is a ‘near-unit root’ process

A key practical takeaway is that a truly stationary time series can frequently yield a p-value of the ADF test that is too high to be considered statistically significant (i.e., it fails to reject the null hypothesis of a unit root). You can observe this phenomenon by running a simulation experiment, say, with an $\text{AR}(1)$ process with varying autoregressive coefficient and compute the corresponding ADF test statistics.

See details: [Link to the TSE material](#)

Example (continued). When analyzing the *dexch* and *rdif* time series, visual inspection alone (by simply looking at the time series plots) is quite revealing: the former is clearly not a unit root process, whereas the interest rate differential might be considered as a unit root process.

For the ADF tests, it seems reasonable in both cases to include a constant term but to exclude a deterministic trend. Below, we report the ADF test results for two fixed lag lengths (6 and 8):

```
dexch

p=6; Test Statistic: -4.026; p-value: 0.002
p=8; Test Statistic: -2.949; p-value: 0.044

====
rdif

p=6; Test Statistic: -1.858; p-value: 0.351
p=8; Test Statistic: -1.939; p-value: 0.314
```

The ADF test results are as expected. We can safely conclude (especially given the typically low power of the ADF test) that the differenced exchange rate is not a unit root process. Regarding the interest rate differential, the unit root hypothesis cannot be rejected at conventional significance levels. However, it is important to remember that here the time series are relatively short and the ADF test is known to have low power (i.e., it often fails to reject the unit root hypothesis even when the process is stationary). The estimation results for *rdif* also indicate strong persistence, though this does not necessarily imply a unit root process.

6 VAR(p) process

6.1 Definition

In the case of a multivariate time series $\mathbf{y}_t = (y_{1t}, \dots, y_{Kt})$, the AR(p) process/model is generalized to the **vector autoregressive process of order p** , that is the **VAR(p) process**

$$\mathbf{y}_t = \boldsymbol{\nu} + \mathbf{A}_1 \mathbf{y}_{t-1} + \dots + \mathbf{A}_p \mathbf{y}_{t-p} + \mathbf{u}_t, \quad \mathbf{u}_t \sim \text{iid}(\mathbf{0}, \boldsymbol{\Sigma}_u),$$

where $\mathbf{A}_1, \dots, \mathbf{A}_p$ are fixed coefficient matrices ($K \times K$) and $\boldsymbol{\nu}_t = (\nu_1, \dots, \nu_K)$ is fixed ($K \times 1$) vector of constant (intercept) terms allowing for the possibility of a nonzero mean $\mathbf{E}(\mathbf{y}_t) \neq \mathbf{0}$. Furthermore, $\mathbf{u}_t = (u_{1t}, \dots, u_{Kt})$ is an unobserved (K -dimensional, $K \times 1$) IID error term (or “shock” or “innovation”). Unless stated otherwise, we assume that the covariance matrix $\mathbf{E}(\mathbf{u}_t \mathbf{u}_t') = \boldsymbol{\Sigma}_u$ is positive definite (i.e., non-singular and invertible matrix), and due to the IID (or weaker white noise) assumption $\mathbf{E}(\mathbf{u}_t \mathbf{u}_s') = \mathbf{0}$ for $s \neq t$.

- In a VAR(p) process/model, or just VAR(p) or a **vector autoregression**, “V” stands for “vector”.
- The idea of the VAR(p) model is to regress each component of $\mathbf{y}_t$ on its own lags and on the lags of the other variables, as in a (multiple-equation) linear regression.
- The model can describe lagged or dynamic dependencies among variables. For example, one may ask how y_{2t} affects the future path of y_{1t} and how y_{1t} affects the future path of y_{2t} .

Example and a special case: Two-variable VAR(1) process. Let us start with considering a bivariate case ($K = 2$). That is we have two time series y_{1t} and y_{2t} . A bivariate (i.e., a two-variable) first-order VAR process (i.e., a VAR(1) process) is given as

$$\begin{aligned} y_{1t} &= \nu_1 + a_{11}y_{1,t-1} + a_{12}y_{2,t-1} + u_{1t} \\ y_{2t} &= \nu_2 + a_{21}y_{1,t-1} + a_{22}y_{2,t-1} + u_{2t}, \end{aligned}$$

where u_{1t} and u_{2t} are two error terms (independent of the history of y_{1t} and y_{2t}) that may be correlated.

In a matrix notation, the bivariate VAR(1) process can be written

$$\begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} = \begin{bmatrix} \nu_1 \\ \nu_2 \end{bmatrix} + \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} u_{1t} \\ u_{2t} \end{bmatrix}.$$

Equivalently,

$$\mathbf{y}_t = \boldsymbol{\nu} + \mathbf{A}_1 \mathbf{y}_{t-1} + \mathbf{u}_t,$$

where now $\mathbf{y}_t = (y_{1t}, y_{2t})$, $\boldsymbol{\nu} = (\nu_1, \nu_2)$, and $\mathbf{u}_t = (u_{1t}, u_{2t})$. Recall that

$$\mathbb{E}(\mathbf{y}_t) = \begin{bmatrix} \mathbb{E}(y_{1t}) \\ \mathbb{E}(y_{2t}) \end{bmatrix} \equiv \boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}.$$

Moreover, the assumption $\mathbf{u}_t \sim \text{iid}(\mathbf{0}, \boldsymbol{\Sigma}_u)$ means here that

$$\mathbb{E}(\mathbf{u}_t) = \begin{bmatrix} \mathbb{E}(u_{1t}) \\ \mathbb{E}(u_{2t}) \end{bmatrix} = \mathbf{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

and

$$\begin{aligned} \boldsymbol{\Sigma}_u &= \mathbb{E}\left((\mathbf{u}_t - \mathbb{E}(\mathbf{u}_t))(\mathbf{u}_t - \mathbb{E}(\mathbf{u}_t))'\right) \\ &= \begin{bmatrix} \text{Var}(u_{1t}) & \text{Cov}(u_{1t}, u_{2t}) \\ \text{Cov}(u_{2t}, u_{1t}) & \text{Var}(u_{2t}) \end{bmatrix}. \end{aligned}$$

Companion form of the VAR(p) process. The VAR(p) process can also be rewritten using so called **companion form**

$$\mathbf{Y}_t = \boldsymbol{\nu} + \mathbf{A} \mathbf{Y}_{t-1} + \mathbf{U}_t,$$

where

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 & \dots & \mathbf{A}_{p-1} & \mathbf{A}_p \\ \mathbf{I}_K & \mathbf{0}_K & \dots & \mathbf{0}_K & \mathbf{0}_K \\ \mathbf{0}_K & \mathbf{I}_K & \dots & \mathbf{0}_K & \mathbf{0}_K \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0}_K & \mathbf{0}_K & \dots & \mathbf{I}_K & \mathbf{0}_K \end{bmatrix}, \quad (Kp \times Kp)$$

where

$$\begin{aligned} \mathbf{Y}_t &= [\mathbf{y}'_t \dots \mathbf{y}'_{t-p+1}]' \quad (Kp \times 1), \quad \boldsymbol{\nu} = [\boldsymbol{\nu}' \dots \mathbf{0}']' \quad (Kp \times 1) \quad \text{and} \\ \mathbf{U}_t &= [\mathbf{u}'_t \mathbf{0}' \dots \mathbf{0}']' \quad (Kp \times 1). \end{aligned}$$

In other words, we have the following presentation

$$\begin{bmatrix} \mathbf{y}_t \\ \mathbf{y}_{t-1} \\ \vdots \\ \mathbf{y}_{t-p+1} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 & \dots & \mathbf{A}_{p-1} & \mathbf{A}_p \\ \mathbf{I}_K & \mathbf{0}_K & \dots & \mathbf{0}_K & \mathbf{0}_K \\ \mathbf{0}_K & \mathbf{I}_K & \dots & \mathbf{0}_K & \mathbf{0}_K \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0}_K & \mathbf{0}_K & \dots & \mathbf{I}_K & \mathbf{0}_K \end{bmatrix} \begin{bmatrix} \mathbf{y}_{t-1} \\ \mathbf{y}_{t-2} \\ \vdots \\ \mathbf{y}_{t-p} \end{bmatrix} + \begin{bmatrix} \mathbf{u}_t \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix},$$

indicating that there holds equalities $\mathbf{y}_t = \boldsymbol{\nu} + \mathbf{A}_1 \mathbf{y}_{t-1} + \dots + \mathbf{A}_p \mathbf{y}_{t-p} + \mathbf{u}_t$, $\mathbf{y}_{t-1} = \mathbf{y}_{t-1}$, and so on to $\mathbf{y}_{t-p+1} = \mathbf{y}_{t-p+1}$. Given assumption $\mathbf{u}_t \sim \text{iid}(\mathbf{0}, \boldsymbol{\Sigma}_u)$, it follows that

$$\mathbf{U}_t \sim \text{iid}(\mathbf{0}, \boldsymbol{\Sigma}_U), \quad \boldsymbol{\Sigma}_U = \text{diag}[\boldsymbol{\Sigma}_u \ \mathbf{0} \ \dots \ \mathbf{0}]$$

where $\boldsymbol{\Sigma}_U = \text{diag}[\boldsymbol{\Sigma}_U \ \dots \ \mathbf{0}]$ is a block diagonal matrix with the given $(K \times K)$ matrices along the diagonal. The covariance matrix $\boldsymbol{\Sigma}_U$ is singular (non-invertible) even if $\boldsymbol{\Sigma}_u$ is positive definite.

6.2 Stationarity of the VAR(p) process

In order the VAR(p) to define a stationary process (or even a stochastic process), the coefficient matrices $\mathbf{A}_1, \dots, \mathbf{A}_p$ must satisfy suitable conditions.

Stationarity (stability) condition of the VAR(p) process. We show that companion form based model equation defines a stationary process, if all of the eigenvalues of the matrix $\mathbf{A}$ are smaller than one in absolute value.

This condition is equivalent to the determinant condition

$$\det(\mathbf{I}_{Kp} - \mathbf{A}z) \neq 0, \quad |z| \leq 1, \quad |z| < 1 \quad (z \in \mathbb{C}),$$

where $|z|$ is the absolute value (or the modulus) of the complex number z .

- If $z \in \mathbb{C}$, that is a complex number, i.e., $z = x + iy$ ($i^2 = -1$), then $|z| = \sqrt{x^2 + y^2}$ is the modulus of z .

Notice that when the process $\mathbf{Y}_t$ (written in terms of the companion form presentation) is (strictly) stationary, then so is the process $\mathbf{y}_t$ as $\mathbf{y}_t$ is a linear transformation of $\mathbf{Y}_t$.

- To illustrate the stationarity assumption further, consider the VAR in a companion form when $t \geq 1$ so that the process $\mathbf{Y}_t$ is well defined provided we specify the initial value $\mathbf{Y}_0$. By iteration, we obtain

$$\begin{aligned} \mathbf{Y}_1 &= \underline{\boldsymbol{\nu}} + \mathbf{A}\mathbf{Y}_0 + \mathbf{U}_1, \\ \mathbf{Y}_2 &= \underline{\boldsymbol{\nu}} + \mathbf{A}\mathbf{Y}_1 + \mathbf{U}_2 = (\mathbf{I}_K + \mathbf{A})\underline{\boldsymbol{\nu}} + \mathbf{A}^2\mathbf{Y}_0 + \mathbf{A}\mathbf{U}_1 + \mathbf{U}_2 \\ &\vdots \\ \mathbf{Y}_t &= \underline{\boldsymbol{\mu}} + \mathbf{A}^t\mathbf{Y}_0 + \sum_{j=0}^{t-1} \mathbf{A}^j \mathbf{U}_{t-j}, \end{aligned}$$

where $\underline{\mu} = [(\sum_{j=0}^{t-1} \mathbf{A}^j)\underline{\nu} \quad (\sum_{j=0}^{t-1} \mathbf{A}^j)\underline{\nu} \quad \dots]$ (that is multiple copies of the same vector-valued element). In order to define a “suitable” initial value we note that when stationarity condition holds, then $\mathbf{A}^j \rightarrow \mathbf{0}$ geometrically, as $j \rightarrow \infty$, implying that $\sum_{j=0}^{\infty} |\mathbf{A}^j|^2 < \infty$ (see Appendix A).

- We can hence define the (companion form-based) linear process (cf. the univariate case, univariate linear process)

$$\mathbf{Y}_t - \underline{\mu} = \mathbf{A}^t \sum_{j=0}^{\infty} \mathbf{A}^j \mathbf{U}_{-j} + \sum_{j=0}^{\infty} \mathbf{A}^j \mathbf{U}_{t-j}, \quad t \geq 1.$$

It follows that $\mathbf{Y}_t, t \geq 1$ is stationary. It is also clear that a process defined in this way yields a solution to the VAR(p) equation can be extended so that, under stability condition of the companion form, the VAR(p) process defines the stationary linear process

$$\mathbf{Y}_t - \underline{\mu} = \sum_{j=0}^{\infty} \mathbf{A}^j \mathbf{U}_{t-j}.$$

Define the matrix $\mathbf{J} = [\mathbf{I}_K \quad \mathbf{0} \cdots \mathbf{0}]$ ($K \times Kp$). We have

$$\mathbf{y}_t = \mathbf{J} \mathbf{Y}_t \quad \text{and} \quad \mathbf{U}_t = \mathbf{J}' \mathbf{u}_t,$$

so that we get the VMA(∞) presentation of the VAR(p) process

$$\mathbf{y}_t - \underline{\mu} = \sum_{j=0}^{\infty} \Psi_j \mathbf{u}_{t-j},$$

where $\Psi_j = \mathbf{J} \mathbf{A}^j \mathbf{J}'$ and hence $\Psi_0 = \mathbf{I}_K$. Given that $\mathbf{A}^j \rightarrow \mathbf{0}$ geometrically, as $j \rightarrow \infty$, the matrices Ψ_j have the same property, especially $\sum_{j=0}^{\infty} \|\Psi_j\|^2 < \infty$.

If we consider $\mathbf{y}_t$ over periods $t \geq 1$ alone and allow nonstationary initial values $(\mathbf{y}_{-p+1}, \dots, \mathbf{y}_0)$, then $\{\mathbf{y}_t, t \geq 1\}$ is nonstationary. The above derivations show that the impact of the initial values disappears when t increases, so that eventually the process becomes stationary. In this type of setting, it is customary to say that the process is **asymptotically stationary** and $\det(\mathbf{I}_{Kp} - \mathbf{A}z) \neq 0, |z| \leq 1, |z| < 1 (z \in \mathbb{C})$ is often called the **stability condition**.

Another Formulation for the stationarity (stability) condition. Using the lag operator $\mathbf{B}$, we can rewrite the VAR(p) process simply as

$$\mathbf{A}(\mathbf{B})\mathbf{y}_t = \underline{\nu} + \mathbf{u}_t, \quad \mathbf{u}_t \sim \text{iid}(\mathbf{0}, \Sigma_u),$$

where $\mathbf{A}(\mathbf{B}) = \mathbf{I}_K - \mathbf{A}_1 \mathbf{B} - \dots - \mathbf{A}_p \mathbf{B}^p$. It can be shown that **the above stationarity (stability) condition is equivalent to the condition**

$$\det \mathbf{A}(z) \neq 0, \quad |z| \leq 1$$

Here $\mathbf{A}(z)$ is a polynomial matrix, a matrix whose elements are polynomials, where the lag operator $\mathbf{B}$ is the argument. Alternatively, $\mathbf{A}(z)$ can be interpreted as a matrix polynomial, whose coefficients are matrices.

6.3 Linear process

By multiplying the $\text{VAR}(p)$ process from the left by the inverse matrix $\mathbf{A}(\mathbf{B})^{-1}$, the above solution must be in short

$$\mathbf{y}_t = \boldsymbol{\mu} + \boldsymbol{\Psi}(\mathbf{B})\mathbf{u}_t,$$

where $\boldsymbol{\Psi}(\mathbf{B}) = \sum_{j=0}^{\infty} \boldsymbol{\Psi}_j \mathbf{B}^j = \mathbf{A}(\mathbf{B})^{-1}$.

- Recall the result $\mathbf{A}(\mathbf{B})^{-1} = [1/\det \mathbf{A}(\mathbf{B})]\mathbf{A}(\mathbf{B})^*$ where $\mathbf{A}(\mathbf{B})^*$ is a finite order matrix polynomial (see Appendix A).

(Multiple-equation and causal) linear process. Linear process for $\mathbf{y}_t = (y_{1t}, \dots, y_{Kt})$ is defined by

$$\mathbf{y}_t = \boldsymbol{\mu} + \sum_{j=0}^{\infty} \boldsymbol{\Psi}_j \mathbf{u}_{t-j}, \quad \mathbf{u}_t \sim \text{iid}(\mathbf{0}, \boldsymbol{\Sigma}_u), \quad \sum_{j=0}^{\infty} \boldsymbol{\Psi}_j^2 < \infty.$$

- The norm $\|\boldsymbol{\Psi}_j\|$ can be defined by stacking the columns of $\boldsymbol{\Psi}_j$ into a single $(K^2 \times 1)$ vector and applying the standard Euclidean vector norm in $\mathbb{R}^{K^2}$ (See Appendix A)
- As introduced in the previous section, a (stationary) $\text{VAR}(p)$ process is an example case of the above linear process.

The (strict) stationarity of the linear process $\{\mathbf{y}_t\}$ follows from the strict stationarity of $\{\mathbf{u}_t\}$, as in the univariate case. Weak stationarity is seen by computing

$$\mathbb{E}(\mathbf{y}_t) = \boldsymbol{\mu} + \sum_{j=0}^{\infty} \mathbb{E}(\boldsymbol{\Psi}_j \mathbf{u}_{t-j}) = \boldsymbol{\mu} + \sum_{j=0}^{\infty} \boldsymbol{\Psi}_j \mathbb{E}(\mathbf{u}_{t-j}) = \boldsymbol{\mu},$$

and by defining a demeaned process $\mathbf{z}_t = \mathbf{y}_t - \boldsymbol{\mu}$, we get

$$\begin{aligned}
\text{Cov}(\mathbf{y}_t, \mathbf{y}_{t+h}) &= \text{Cov}(\mathbf{z}_t, \mathbf{z}'_{t+h}) \\
&= \mathbb{E}(\mathbf{z}_t \mathbf{z}'_{t+h}) \\
&= \mathbb{E}\left(\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \boldsymbol{\Psi}_i \mathbf{u}_{t-i} \mathbf{u}'_{t+h-j} \boldsymbol{\Psi}'_j\right) \\
&= \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \boldsymbol{\Psi}_i \mathbb{E}(\mathbf{u}_{t-i} \mathbf{u}'_{t+h-j}) \boldsymbol{\Psi}'_j \\
&= \sum_{j=0}^{\infty} \boldsymbol{\Psi}_j \boldsymbol{\Sigma}_u \boldsymbol{\Psi}'_{j+h}.
\end{aligned}$$

Here, we have made use of the linearity of expectation, the fact that the innovation series $\mathbf{u}_t$ is independent implying that $\mathbb{E}(\mathbf{u}_{t-s} \mathbf{u}'_{t+h-j}) = \mathbf{0}$ for all $j \neq h + s$, and $\mathbb{E}(\mathbf{u}_t \mathbf{u}'_t) = \boldsymbol{\Sigma}_u$.

- By the properties of matrix norm, $\|\boldsymbol{\Psi}_j \boldsymbol{\Sigma}_u \boldsymbol{\Psi}'_j\| \leq \|\boldsymbol{\Sigma}_u\| \|\boldsymbol{\Psi}_j\|^2$ so that by employing the triangular inequality we can observe that it is finite (see Appendix A). It then follows that $\mathbf{y}_t$ is weakly stationary.
- As an additional remark, the above derivations on $\mathbb{E}(\mathbf{y}_t)$ and $\text{Cov}(\mathbf{y}_t, \mathbf{y}_{t+h})$ also hold, when $\mathbf{u}_t$ is just a serially uncorrelated series (i.e., $\text{Cov}(\mathbf{u}_t, \mathbf{u}_{t+h}) = \mathbf{0}$ for all $h \neq 0$) so that weak stationarity holds in this case too.

Covariance function of VAR(p) process. Given that a stationary VAR(p) process has the linear process representation, its mean (see above) and covariance function can be obtained by steps similar to those above and as a special case. We get

$$\boldsymbol{\Gamma}_h = \sum_{j=0}^{\infty} \boldsymbol{\Psi}_j \boldsymbol{\Sigma}_u \boldsymbol{\Psi}'_{j+h} = \boldsymbol{\Gamma}'_{-h}$$

where $\boldsymbol{\Psi}_j$ are as obtained also above and more specifically to the selected VAR(p).

- Because $\boldsymbol{\Psi}_j \rightarrow \mathbf{0}$ geometrically, when $j \rightarrow \infty$, the covariance function $\boldsymbol{\Gamma}_h$ dampens geometrically as h tends to ∞ . The speed of dampening is slower the closer the absolute value of the largest eigenvalue of $\mathbf{A}$ is to unity.
- Note that in the case $K = p = 1$, the matrix $\mathbf{A}$ is an autoregressive coefficient and $\mathbf{A}^l$, the l :th autocorrelation coefficient).

6.4 VAR(p) and impulse response function

In the context of VMA(∞) presentation, we stated that $\Psi_j = \mathbf{J}\mathbf{A}^j\mathbf{J}'$. An alternative way to express the matrices Ψ_j is obtained from the identity $\Psi(\mathbf{B})\mathbf{A}(\mathbf{B}) = \mathbf{I}_K$, that is,

$$\begin{aligned}\mathbf{I}_K &= (\Psi_0 + \Psi_1\mathbf{B} + \Psi_2\mathbf{B}^2 + \dots)(\mathbf{I}_K - \mathbf{A}_1\mathbf{B} - \dots - \mathbf{A}_p\mathbf{B}^p) \\ &= \Psi_0 + (\Psi_1 - \Psi_0\mathbf{A}_1)\mathbf{B} + (\Psi_2 - \Psi_1\mathbf{A}_1 - \Psi_0\mathbf{A}_2)\mathbf{B}^2 + \dots\end{aligned}$$

It follows that

$$\begin{aligned}\mathbf{I}_K &= \Psi_0 \\ \mathbf{0} &= \Psi_1 - \Psi_0\mathbf{A}_1 \\ \mathbf{0} &= \Psi_2 - \Psi_1\mathbf{A}_1 - \Psi_0\mathbf{A}_2 \\ &\vdots\end{aligned}$$

so that we can compute Ψ_1 recursively by using equations

$$\Psi_0 = \mathbf{I}_K \tag{6.1}$$

$$\Psi_j = \sum_{i=1}^j \Psi_{j-i}\mathbf{A}_i, \quad j = 1, 2, \dots, \tag{6.2}$$

where $\mathbf{A}_i = \mathbf{0}$, for $i > p$.

As noted earlier, a VAR(p) model can be seen as a system of several linear regressions, where regressors are the lags of the response variables. Hence, compared with the usual linear regression, the regressors are selected in a particular way.

- The unobserved error term is largely similar to the error of a conventional regression model. As in a regression, the coefficient matrices $\mathbf{A}_1, \dots, \mathbf{A}_p$ can be interpreted to describe how lagged variables affect the response variables. However, the values of individual elements in these matrices can seldom be clearly interpreted in isolation.

Usually, instead of focusing on individual coefficients, we are interested in dynamic effects.

- Specifically, how does a change in one of the variables, say y_{1t} , affect the future paths of all components of $\mathbf{y}_t$?

It is difficult to describe this type of effect by looking at individual elements in the matrices $\mathbf{A}_1, \dots, \mathbf{A}_p$.

- This can be seen already in the case $K = 2$. Draw a figure where the impacts of y_{1t} and y_{2t} on future values $y_{1,t+j}$ and $y_{2,t+j}$ are marked by arrows. The figure will be rather messy even if there are only two variables!

Another interpretation is based on the idea that the components of the error term $\mathbf{u}_t$ are regarded as impulses or shocks that drive the variables.

- In this case, we are interested in seeing how a unit-sized impulse affects (and spreads to) the components of the process and their future values. We will return to this question in more detail in the **Macroeconometrics course!**

However, from the recursion of Ψ_j , we can quite easily see that elements of the matrices Ψ_j describe how an impulse at time t_0 affects the process $\mathbf{y}_t$ at time $t_0 + j$. For this reason, Ψ_j are at times called impulse responses.

- More specifically, these are often referred to as **forecast error impulse responses**. As mentioned, in macroeconomic applications, we typically need to impose additional (identification) restrictions on the VAR model, leading to a **structural VAR (SVAR) model**. These models will be considered in more detail in the Macroeconometrics course.

6.5 Extra: VARMA(p, q) process

As we have seen earlier in this material, the univariate AR(p) process could be generalized into an ARMA(p, q) process by adding the linear combinations of $\mathbf{u}_{t-1}, \dots, \mathbf{u}_{t-q}$ into the AR(p) model equation. The corresponding multivariate generalization, **VARMA(p, q) process**, is

$$\mathbf{y}_t = \boldsymbol{\nu} + \mathbf{A}_1 \mathbf{y}_{t-1} + \dots + \mathbf{A}_p \mathbf{y}_{t-p} + \mathbf{u}_t - \mathbf{M}_1 \mathbf{u}_{t-1} - \dots - \mathbf{M}_q \mathbf{u}_{t-q},$$

where $\mathbf{u}_t \sim \text{iid}(\mathbf{0}, \boldsymbol{\Sigma}_u)$ and $\mathbf{M}_1, \dots, \mathbf{M}_q$ are coefficient matrices.

- Other authors (like Lütkepohl) define the VARMA process using plus signs:

$$\mathbf{y}_t = \boldsymbol{\nu} + \mathbf{A}_1 \mathbf{y}_{t-1} + \dots + \mathbf{A}_p \mathbf{y}_{t-p} + \mathbf{u}_t + \mathbf{M}_1 \mathbf{u}_{t-1} + \dots + \mathbf{M}_q \mathbf{u}_{t-q}.$$

As in the univariate case, we can note that a sufficient condition for the stationarity of a VARMA(p, q) process is as for the VAR(p) process. Using a lag operator equation, the VARMA process can be written as

$$\mathbf{A}(\mathbf{B})\mathbf{y}_t = \mathbf{M}(\mathbf{B})\mathbf{u}_t$$

where $\mathbf{A}(\mathbf{B}) = \mathbf{I}_K - \mathbf{A}_1 \mathbf{B} - \dots - \mathbf{A}_p \mathbf{B}^p$ and $\mathbf{M}(\mathbf{B}) = \mathbf{I}_K - \mathbf{M}_1 \mathbf{B} - \dots - \mathbf{M}_q \mathbf{B}^q$. When the stationarity condition holds, this equation can be multiplied from the left by the inverse matrix $\mathbf{A}(\mathbf{B})^{-1}$ to get the well-defined linear process

$$\mathbf{y}_t = \boldsymbol{\mu} + \boldsymbol{\Psi}(\mathbf{B})\mathbf{u}_t,$$

where $\boldsymbol{\Psi}(\mathbf{B}) = \sum_{j=0}^{\infty} \boldsymbol{\Psi}_j \mathbf{B}^j = \mathbf{A}(\mathbf{B})^{-1} \mathbf{M}(\mathbf{B})$ and $\boldsymbol{\Psi}_0 = \mathbf{I}_K$. As with the VAR(p) process, $\boldsymbol{\Psi}_j \rightarrow \mathbf{0}$ geometrically as $j \rightarrow \infty$ and as such the covariance function of the VARMA(p, q) process dampens geometrically to zero as the lag grows infinitely.

The theory and application of VARMA(p, q) processes is significantly more challenging than VAR(p) processes.

- This is largely due to the properties of the polynomial matrices, which makes the structural theory and model selection more complicated than with VAR(p) processes.
- This can be seen especially with the uniqueness and identification conditions, which in the univariate case $K = 1$ requires that (now) the polynomials $\mathbf{A}(z)$ and $\mathbf{M}(z)$ do not have common factors and at least one of the higher order coefficients is non-zero.

The next example (Extra below) demonstrates why the uniqueness becomes more complicated in the multivariate case $K > 1$.

Extra: Example of a bivariate VARMA(1,1) process

Let us consider as an example of VARMA processes. Specifically we take a look on a bivariate zero-mean (for simplicity) VARMA(1,1) process

$$\begin{bmatrix} 1 & aB \\ 0 & 1 \end{bmatrix} \mathbf{y}_t = \begin{bmatrix} 1 & mB \\ 0 & 1 \end{bmatrix} \mathbf{u}_t, \quad a \neq m.$$

The first order polynomial matrices $\mathbf{A}(B)$ and $\mathbf{M}(B)$ seem to fulfill a condition similar to the uniqueness condition of the univariate ARMA(1,1) process. By multiplying the model equation above from the left by the inverse matrix

$$\mathbf{A}(B)^{-1} = \begin{bmatrix} 1 & -aB \\ 0 & 1 \end{bmatrix}$$

we get the linear presentation of the process

$$\mathbf{y}_t = \begin{bmatrix} 1 & (m-a)B \\ 0 & 1 \end{bmatrix} \mathbf{u}_t$$

which is a vector MA(1) or VMA(1) process (correspondingly we could talk about a VMA(q) process where $q = \infty$ is also possible). By multiplying model equation from the left by the inverse matrix $\mathbf{M}(B)^{-1}$ we get correspondingly

$$\begin{bmatrix} 1 & (a-m)B \\ 0 & 1 \end{bmatrix} \mathbf{y}_t = \mathbf{u}_t$$

or a VAR(1) process. Hence a VARMA(1,1) process (as above) can be presented as a VAR(1) and a VMA(1) process. If in this kind of a situation we are trying to fit the data with a VARMA(1,1) model, the estimation procedure gets into trouble, as it finds three models that fit the data equally well. The problem gets resolved of course if there is prior knowledge on the structure of the process which enables to choose either a VAR(1) or VMA(1) model

to be estimated. Usually however, this kind of prior knowledge is not at hand and in more complicated cases even recognizing the problem isn't as easy as in this simple example.

A rather complete uniqueness theory has been developed for VARMA(p, q) processes on which model selection can be applied in practice using some model selection criteria.

- A theory for estimating (Gaussian) maximum likelihood has also been developed. These questions will be, however, ignored wholly on this course.
- The statistical methods based on VAR(p) processes which have been described can be justified as approximations, as when the invertibility condition

$$\det(\mathbf{M}(z)) \neq 0, \quad |z| \leq 1,$$

holds, the VARMA(p, q) process can be presented as

$$\mathbf{M}(\mathbf{B}^{-1})\mathbf{A}(\mathbf{B})\mathbf{y}_t = \mathbf{u}_t,$$

where $\mathbf{M}(\mathbf{B})^{-1}\mathbf{A}(\mathbf{B}) = \sum_{j=0}^{\infty} \mathbf{\Pi}_j \mathbf{B}^j$ and the coefficient matrices $\mathbf{\Pi}_j$ converge to zero geometrically as $j \rightarrow \infty$ ($\mathbf{\Pi}_0 = \mathbf{I}_K$).

VARMA(p, q) process can thus be approximated with a VAR(h) process as long a h is “sufficiently large”.

7 VAR(p): Estimation and hypothesis testing

7.1 Parameter estimation

Consider an K -dimensional observed time series that can be described by the VAR(p) model

$$\mathbf{y}_t = \boldsymbol{\nu} + \mathbf{A}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{A}_p \mathbf{y}_{t-p} + \mathbf{u}_t, \quad t = 1, \dots, T,$$

where the initial values $\mathbf{y}_{-p+1}, \dots, \mathbf{y}_0$ are also observed and the coefficient matrices $\mathbf{A}_1, \dots, \mathbf{A}_p$ meet the stationarity/stability condition of the VAR(p) process.

In order to apply maximum likelihood estimation, assume that the error term is normally distributed

$$\mathbf{u}_t \sim \text{nid}(\mathbf{0}, \boldsymbol{\Sigma}_u),$$

where the covariance matrix $\boldsymbol{\Sigma}_u$ is positive definite. The asymptotic results below will hold also under the more general assumption $\mathbf{u}_t \sim \text{iid}(\mathbf{0}, \boldsymbol{\Sigma}_u)$.

Likelihood function. Define the notation

$$\mathbf{x}_t = [1 \quad \mathbf{y}'_{t-1} \quad \cdots \quad \mathbf{y}'_{t-p}]' = [1 \quad \mathbf{Y}'_{t-1}]' \quad ((Kp+1) \times 1)$$

and

$$\boldsymbol{\Pi} = \begin{bmatrix} \boldsymbol{\pi}'_1 \\ \vdots \\ \boldsymbol{\pi}'_K \end{bmatrix} = [\boldsymbol{\nu} : \mathbf{A}_1 : \dots : \mathbf{A}_p]. \quad (K \times (Kp+1))$$

Write the VAR(p) model as

$$\mathbf{y}_t = \boldsymbol{\Pi} \mathbf{x}_t + \mathbf{u}_t, \quad t = 1, \dots, T.$$

This constitutes the equations ($\boldsymbol{\pi}'_i \mathbf{x}_t = \mathbf{x}'_t \boldsymbol{\pi}_i$)

$$y_{kt} = \mathbf{x}'_t \boldsymbol{\pi}_k + u_{kt}, \quad k = 1, \dots, K, \quad t = 1, \dots, T.$$

Hence we can rewrite the model as

$$\mathbf{y}_t = \mathbf{X}'_t \boldsymbol{\pi} + \mathbf{u}_t, \quad t = 1, \dots, T,$$

where $\boldsymbol{\pi} = [\boldsymbol{\pi}'_1 \cdots \boldsymbol{\pi}'_K]'$ ($K(Kp+1) \times 1$), and

$$\mathbf{X}'_t = \text{diag}[\mathbf{x}'_t \cdots \mathbf{x}'_t] = \mathbf{I}_K \otimes \mathbf{x}'_t, \quad (K \times K(Kp+1))$$

where $\otimes$ denotes Kronecker product (see Appendix A).

(Conditional) likelihood function. To derive the likelihood function, we will follow the same principles as with univariate AR and ARMA models. Denote

$$\underline{\mathbf{Y}}_t = [\mathbf{y}'_{-p+1} \cdots \mathbf{y}'_0 \mathbf{y}'_1 \cdots \mathbf{y}'_t]', \quad t = 0, 1, \dots, T.$$

Denote the joint density of $\underline{\mathbf{Y}}_t$ by $f_{\underline{\mathbf{Y}}_t}$. Then the joint density of the whole sample can be derived by using the formula of the conditional density (as in the univariate case!):

$$f_{\underline{\mathbf{Y}}_T} = f_{\mathbf{y}_T|\underline{\mathbf{Y}}_{T-1}} \cdot f_{\underline{\mathbf{Y}}_{T-1}} = f_{\mathbf{y}_T|\underline{\mathbf{Y}}_{T-1}} \cdot f_{\mathbf{y}_{T-1}|\underline{\mathbf{Y}}_{T-2}} \cdot f_{\underline{\mathbf{Y}}_{T-2}} = \cdots = \prod_{t=1}^T f_{\mathbf{y}_t|\underline{\mathbf{Y}}_{t-1}} \cdot f_{\underline{\mathbf{Y}}_0}$$

The linear process representation of the VAR(p) process implies that $\mathbf{y}_{t-j}$ and $\mathbf{u}_t$, $j \geq 1$, are independent and in particular that $\underline{\mathbf{Y}}_{t-1}$ and $\mathbf{u}_t$ are independent.

- This is the result we have already used in univariate models and now we extend it to this multiple-equation case.

This and the VAR(p) model equation together imply that the conditional density of $\mathbf{y}_t$ given $\underline{\mathbf{Y}}_{t-1}$ depends only on the first p subvectors $\mathbf{y}_{t-1}, \dots, \mathbf{y}_{t-p}$, or just $\underline{\mathbf{Y}}_{t-1}$. Therefore, we get the distribution result

$$\mathbf{y}_t|\underline{\mathbf{Y}}_{t-1} \quad \text{or} \quad \mathbf{y}_t|\underline{\mathbf{Y}}_{t-1} \sim \mathbf{N}(\mathbf{X}'_t \boldsymbol{\pi}, \boldsymbol{\Sigma}_u).$$

- Here one can use the following general result that is related to the multivariate normal distribution. If (for some general matrices with suitable dimensions) $\mathbf{Y} = \mathbf{X} + \mathbf{Z}$, where $\mathbf{X}$ and $\mathbf{Z}$ are independent and $\mathbf{Z} \sim \mathbf{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then $\mathbf{Y} | (\mathbf{X} = \mathbf{x}) \sim \mathbf{N}(\boldsymbol{\mu} + \mathbf{x}, \boldsymbol{\Sigma})$.

Given the formula of the density function of a multivariate normal, we have

$$f_{\mathbf{y}_t|\underline{\mathbf{Y}}_{t-1}}(\mathbf{y}_t|\underline{\mathbf{Y}}_{t-1}) = (2\pi)^{-1/2} \det(\boldsymbol{\Sigma}_u)^{-1/2} \exp \left\{ -\frac{1}{2} (\mathbf{y}_t - \mathbf{X}'_t \boldsymbol{\pi})' \boldsymbol{\Sigma}_u^{-1} (\mathbf{y}_t - \mathbf{X}'_t \boldsymbol{\pi}) \right\}.$$

In order to obtain the density ($f_{\underline{\mathbf{Y}}_T}$) of the whole sample ($\underline{\mathbf{Y}}_T$) we must still find the distribution of the initial value vector $\underline{\mathbf{Y}}_0$. In what follows, we interpret $\underline{\mathbf{Y}}_0$ as fixed, or equivalently we condition the underlying joint density on its value. This results in the so called **conditional likelihood function**, which treats the product

$$f_{\mathbf{y}_1|\underline{\mathbf{Y}}_0} \cdots f_{\mathbf{y}_T|\underline{\mathbf{Y}}_{T-1}}$$

as a function of the parameters. Taking the logarithmic terms we obtain the so called log-likelihood function (we ignore the constant part that is independent of the parameter)

$$l(\boldsymbol{\pi}, \boldsymbol{\Sigma}_u) = -\frac{T}{2} \log(\det(\boldsymbol{\Sigma}_u)) - \frac{1}{2} \sum_{t=1}^T (\mathbf{y}_t - \mathbf{X}_t' \boldsymbol{\pi})' \boldsymbol{\Sigma}_u^{-1} (\mathbf{y}_t - \mathbf{X}_t' \boldsymbol{\pi}).$$

It is clear that in large samples the impact of the initial values $\mathbf{y}_{-p+1}, \dots, \mathbf{y}_0$ is negligible and that the initial values have no impact on the asymptotic properties of the likelihood function.

- Unless stated otherwise, hereafter likelihood function refers to this conditional likelihood function.

Maximizing likelihood function. Consider maximizing the likelihood function with respect to the parameters of the model. Take the partial derivative with respect to $\boldsymbol{\pi}$ of the (conditional) log-likelihood function (see Appendix B)

$$\frac{\partial}{\partial \boldsymbol{\pi}} l(\boldsymbol{\pi}, \boldsymbol{\Sigma}_u) = \sum_{t=1}^T \mathbf{X}_t \boldsymbol{\Sigma}_u^{-1} (\mathbf{y}_t - \mathbf{X}_t' \boldsymbol{\pi}).$$

By the definition of the matrix $\mathbf{X}_t$ and using the properties of the Kronecker product (see Appendix A) we obtain

$$\mathbf{X}_t \boldsymbol{\Sigma}_u^{-1} = (\mathbf{I}_K \otimes \mathbf{x}_t) (\boldsymbol{\Sigma}_u^{-1} \otimes \mathbf{1}) = \boldsymbol{\Sigma}_u^{-1} \otimes \mathbf{x}_t = (\boldsymbol{\Sigma}_u^{-1} \otimes \mathbf{I}_{Kp+1}) (\mathbf{I}_K \otimes \mathbf{x}_t),$$

where $(\mathbf{I}_K \otimes \mathbf{x}_t) = \mathbf{X}_t$. Hence, we can write

$$\frac{\partial}{\partial \boldsymbol{\pi}} l(\boldsymbol{\pi}, \boldsymbol{\Sigma}_u) = (\boldsymbol{\Sigma}_u^{-1} \otimes \mathbf{I}_{Kp+1}) \sum_{t=1}^T \mathbf{X}_t (\mathbf{y}_t - \mathbf{X}_t' \boldsymbol{\pi}).$$

Setting $\partial l(\boldsymbol{\pi}, \boldsymbol{\Sigma}_u) / \partial \boldsymbol{\pi} = \mathbf{0}$, we can subtract the (non-singular) factor $(\boldsymbol{\Sigma}_u^{-1} \otimes \mathbf{I}_{Kp+1})$ so that the likelihood equation concerning the parameter $\boldsymbol{\pi}$ is

$$\sum_{t=1}^T \mathbf{X}_t \mathbf{X}_t' \boldsymbol{\pi} = \sum_{t=1}^T \mathbf{X}_t \mathbf{y}_t.$$

This equation yields the unique solution

$$\hat{\boldsymbol{\pi}} = \left(\sum_{t=1}^T \mathbf{X}_t \mathbf{X}_t' \right)^{-1} \sum_{t=1}^T \mathbf{X}_t \mathbf{y}_t,$$

which is the **ordinary least squares (OLS) estimator of the coefficient parameters of the VAR model**.

For further illustration, write the underlying system of equations in matrix form as

$$\begin{bmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_T \end{bmatrix} = \begin{bmatrix} \mathbf{X}'_1 \\ \vdots \\ \mathbf{X}'_T \end{bmatrix} + \begin{bmatrix} \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_T \end{bmatrix},$$

or in short

$$\underline{\mathbf{y}} = \underline{\mathbf{X}}\boldsymbol{\pi} + \underline{\mathbf{u}}.$$

Using straightforward matrix calculus one see that the OLS estimator is obtained by the usual OLS formula

$$\hat{\boldsymbol{\pi}} = (\underline{\mathbf{X}}'\underline{\mathbf{X}})^{-1} \underline{\mathbf{X}}'\underline{\mathbf{y}}.$$

- Note, however, that $\text{Cov}(\underline{\mathbf{u}}) = \mathbf{I}_T \otimes \boldsymbol{\Sigma}_u$, which differs from the standard assumption of a linear model.
- The reason why MLE still yields the OLS estimator has to do with the special structure of the regressor matrix $\underline{\mathbf{X}}$.
- Later we will see that the result does not apply when the structure of $\underline{\mathbf{X}}$ changes. By direct calculation one can see that the subvectors $\hat{\boldsymbol{\pi}}_1, \dots, \hat{\boldsymbol{\pi}}_K$ of $\hat{\boldsymbol{\pi}}$ are the standard OLS estimators based on the individual equations.

It turns out that the maximum likelihood estimator for the covariance matrix is

$$\hat{\boldsymbol{\Sigma}}_u = \frac{1}{T} \sum_{t=1}^T (\mathbf{y}_t - \mathbf{X}'_t \hat{\boldsymbol{\pi}}) (\mathbf{y}_t - \mathbf{X}'_t \hat{\boldsymbol{\pi}})'$$

Details on how to obtain it are provided below as Extra information.

Extra: Covariance matrix estimation

Consider next the maximum likelihood estimation of the covariance matrix $\boldsymbol{\Sigma}_u$. It is handy to first derive MLE for the inverse matrix $\boldsymbol{\Sigma}_u^{-1} \equiv \boldsymbol{\Phi}$ and then (by the invariance property of the MLE) obtain the MLE for $\boldsymbol{\Sigma}_u$ through the inverse $\boldsymbol{\Phi}^{-1} = \boldsymbol{\Sigma}_u$. Applying the parameter matrix $\boldsymbol{\Phi}$ and the short-hand notation $\mathbf{u}_t(\boldsymbol{\pi}) = \mathbf{y}_t - \mathbf{X}'_t \boldsymbol{\pi}$, we can write the log-likelihood as

$$l(\boldsymbol{\pi}, \boldsymbol{\Sigma}_u) = \frac{T}{2} \log(\det(\boldsymbol{\Phi})) - \frac{1}{2} \sum_{t=1}^T \mathbf{u}_t(\boldsymbol{\pi})' \boldsymbol{\Phi} \mathbf{u}_t(\boldsymbol{\pi}).$$

- Note that we have here simplified the derivation by (erroneously) not taking the symmetry of $\boldsymbol{\Phi}$ into account. As noted in Appendix A, when we account for the symmetry correctly, the derivative becomes more complicated, while still the same likelihood equation is obtained for $\boldsymbol{\Phi}$ and hence the same MLE. Intuitively, this is plausible, because the maximization results in a symmetric (and positive definite) solution, even if the symmetry condition is ignored.

Applying the derivation rules of Appendix A, we obtain

$$\frac{\partial}{\partial \Phi} \mathbf{u}_t(\boldsymbol{\pi})' \Phi \mathbf{u}_t(\boldsymbol{\pi}) = \mathbf{u}_t(\boldsymbol{\pi}) \mathbf{u}_t(\boldsymbol{\pi})' \quad \text{and} \quad \frac{\partial}{\partial \Phi} \log(\det(\Phi)) = \Phi^{-1} = \boldsymbol{\Sigma}_u$$

so that

$$\frac{\partial}{\partial \Phi} l(\boldsymbol{\pi}, \boldsymbol{\Sigma}_u) = \frac{T}{2} \boldsymbol{\Sigma}_u - \frac{1}{2} \sum_{t=1}^T \mathbf{u}_t(\boldsymbol{\pi}) \mathbf{u}_t(\boldsymbol{\pi})'$$

Setting the right hand side to zero and replacing $\boldsymbol{\pi}$ in $\mathbf{u}_t(\boldsymbol{\pi})$ by the above derived ML solution $\hat{\boldsymbol{\pi}}$ we obtain the solution

$$\hat{\boldsymbol{\Sigma}}_u = \frac{1}{T} \sum_{t=1}^T (\mathbf{y}_t - \mathbf{X}_t' \hat{\boldsymbol{\pi}}) (\mathbf{y}_t - \mathbf{X}_t' \hat{\boldsymbol{\pi}})'.$$

Given that $\hat{\boldsymbol{\pi}}$ minimizes the log-likelihood function $l(\boldsymbol{\pi}, \boldsymbol{\Sigma}_u)$, we can conclude on the basis of the invariance property of MLE that $\hat{\boldsymbol{\pi}}$ and $\hat{\boldsymbol{\Sigma}}_u$ are (conditional) ML estimators for $\boldsymbol{\pi}$ and $\boldsymbol{\Sigma}_u$.

Asymptotic properties of maximum likelihood estimators. Our starting point is (as is usual in the MLE theory) that standard limit theorems (the law of large numbers and the central limit theorem) apply under the considered VAR(p) model. To simplify notation assume the initial values $\mathbf{y}_{-p+1}, \dots, \mathbf{y}_0$ are stationary.

To obtain the asymptotic distribution of $\hat{\boldsymbol{\pi}}$ compute

$$\frac{\partial^2}{\partial \boldsymbol{\pi} \partial \boldsymbol{\pi}'} l(\boldsymbol{\pi}, \boldsymbol{\Sigma}_u) = -(\boldsymbol{\Sigma}_u^{-1} \otimes \mathbf{I}_{Kp+1}) \sum_{t=1}^T \mathbf{X}_t \mathbf{X}_t' = -\boldsymbol{\Sigma}_u^{-1} \sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t',$$

where the latter equation is implied by $\mathbf{X}_t = \mathbf{I}_K \otimes \mathbf{x}_t$ and the properties of the Kronecker product. As defined above, $\mathbf{x}_t = [1 \quad \mathbf{Y}_{t-1}']'$, the Fisher information matrix for the parameter $\boldsymbol{\pi}$ is

$$\mathbf{I}_{\boldsymbol{\pi}\boldsymbol{\pi}}(\boldsymbol{\pi}, \boldsymbol{\Sigma}_u) \equiv -\mathbb{E} \left[\frac{\partial^2}{\partial \boldsymbol{\pi} \partial \boldsymbol{\pi}'} l(\boldsymbol{\pi}, \boldsymbol{\Sigma}_u) \right] = T(\boldsymbol{\Sigma}_u^{-1} \otimes \boldsymbol{\Gamma}_{\mathbf{x}}),$$

where

$$\boldsymbol{\Gamma}_{\mathbf{x}} = \mathbb{E}(\mathbf{x}_t \mathbf{x}_t') = \begin{bmatrix} 1 & \mathbb{E}(\mathbf{Y}_{t-1}') \\ \mathbb{E}(\mathbf{Y}_{t-1}') & \mathbb{E}(\mathbf{Y}_{t-1} \mathbf{Y}_{t-1}') \end{bmatrix}.$$

Next note that (see also Appendix A)

$$\begin{aligned} \mathbb{E} \left[\frac{\partial^2}{\partial \boldsymbol{\pi} \partial \sigma_{ij}} l(\boldsymbol{\pi}, \boldsymbol{\Sigma}_u) \right] &= \sum_{t=1}^T \mathbb{E} \left[\mathbf{X}_t \left(\frac{\partial}{\partial \sigma_{ij}} \boldsymbol{\Sigma}_u^{-1} \right) (\mathbf{y}_t - \mathbf{X}_t' \boldsymbol{\pi}) \right] \\ &= \sum_{t=1}^T \mathbb{E} \left[\mathbf{X}_t \left(\frac{\partial}{\partial \sigma_{ij}} \boldsymbol{\Sigma}_u^{-1} \right) \right] \mathbb{E}(\mathbf{u}_t) \\ &= \mathbf{0}. \end{aligned}$$

Here the second equation follows from $\mathbf{u}_t = \mathbf{y}_t - \mathbf{X}_t' \boldsymbol{\pi}$ and the fact that $\mathbf{u}_t$ is independent of $\mathbf{y}_{t-1}, \mathbf{y}_{t-2}, \dots$ and hence of $\mathbf{X}_t$ (see the definition of $\mathbf{X}_t$).

The above derivations imply that the Fisher information matrix for $\boldsymbol{\pi}$ and $\boldsymbol{\Sigma}_u$ is block diagonal, i.e., the parameters $\boldsymbol{\pi}$ and $\boldsymbol{\Sigma}_u$ are orthogonal.

- The general maximum likelihood theory then implies that the estimators $\hat{\boldsymbol{\pi}}$ and $\hat{\boldsymbol{\Sigma}}_u$ are asymptotically independent and the asymptotic distribution of $\hat{\boldsymbol{\pi}}$ is as in a situation where $\boldsymbol{\Sigma}_u$ is known.

In other words, $\hat{\boldsymbol{\pi}}$ is asymptotically normal with expectation $\boldsymbol{\pi}$ and the covariance matrix $\mathbf{I}_{\boldsymbol{\pi}\boldsymbol{\pi}}(\boldsymbol{\pi}, \boldsymbol{\Sigma}_u)^{-1}$. As $(\boldsymbol{\Sigma}_u^{-1} \otimes \boldsymbol{\Gamma}_x)^{-1} = \boldsymbol{\Sigma}_u \otimes \boldsymbol{\Gamma}_x^{-1}$, we obtain by the definition of $\mathbf{I}_{\boldsymbol{\pi}\boldsymbol{\pi}}(\boldsymbol{\pi}, \boldsymbol{\Sigma}_u)$ that

$$\hat{\boldsymbol{\pi}} \stackrel{as}{\sim} \mathbf{N}(\boldsymbol{\pi}, T^{-1} (\boldsymbol{\Sigma}_u \otimes \boldsymbol{\Gamma}_x^{-1})).$$

We do not consider the asymptotic distribution of $\hat{\boldsymbol{\Sigma}}_u$. However, we will later invoke the fact that $\hat{\boldsymbol{\Sigma}}_u \xrightarrow{p} \boldsymbol{\Sigma}_u$.

Extra: Some additional notes

The above result holds also without the assumption on stationary initial values provided the matrix $T^{-1} (\boldsymbol{\Sigma}_u \otimes \boldsymbol{\Gamma}_x^{-1})$ is replaced by the general formula of the Fisher information matrix

$$\mathbf{I}_{\boldsymbol{\pi}\boldsymbol{\pi}}(\boldsymbol{\pi}, \boldsymbol{\Sigma}_u) = \boldsymbol{\Sigma}_u^{-1} \otimes \sum_{t=1}^T \mathbf{E}(\mathbf{x}_t \mathbf{x}_t').$$

The same point holds in the following where we consider restricted estimation and hypothesis testing (see below).

- It is, however, useful to note that the accuracy of the asymptotic approximation in finite samples depends on the strength of the autocorrelation of the process.
- When the stability condition of the VAR(p) is “nearly” invalid, the asymptotic approximation may be misleading in small (or even in fairly large) samples.

Restricted estimation of VAR(p). It is also important to note that parameter estimation can be carried out subject to linear constraints to parameters.

- Details on this are available below as Extra material.

In summary, it can be concluded that estimation goes largely in the same (under the restrictions) as above and the resulting estimators have analogous (asymptotic) properties vs. the unrestricted case above. This is also important from the hypothesis testing perspective as introduced in the next Section.

Extra: Estimation subject to linear constraints

Consider the VAR(p) model (from the previous section), where the parameters $\mathbf{A}_1, \dots, \mathbf{A}_p$ (and possibly $\boldsymbol{\nu}$) are subject to linear constraints. Such restrictions can be expressed conveniently so that the parameter $\boldsymbol{\pi}$ is assumed to satisfy

$$\boldsymbol{\pi} = \mathbf{H}\boldsymbol{\delta} + \mathbf{a}, \quad (7.1)$$

where $\mathbf{H}$ is a known $K(Kp+1) \times m$ matrix, $\mathbf{a}$ a known $K(Kp+1)$ vector and $\boldsymbol{\delta}$ is an unknown $m \times 1$ parameter vector. It is also assumed that $\mathbf{H}$ is of full column rank, i.e., $\text{rank}(\mathbf{H}) = m$.

It is quite common to express linear constraints by using the equation

$$\mathbf{R}\boldsymbol{\pi} = \mathbf{b},$$

where $\mathbf{R}$ ($q \times K(Kp+1) \times m$) and $\mathbf{b}$ ($q \times 1$) are known and $\text{rank}(\mathbf{R}) = q$. Using linear algebra one can see that the two ways are equivalent, when $q + m = K(Kp+1)$. The former is useful when we consider estimation (in this section) and the latter is useful when we consider a Wald test (the next section).

Most commonly we are interested in situations where some of the parameters are restricted to be zero. For example, if we choose $\mathbf{H} = [\mathbf{I}_{K^2p} : \mathbf{0}]'$ and $\mathbf{a} = \mathbf{0}$, then the last K components of $\boldsymbol{\pi}$ are assumed to be zero so that $\mathbf{y}_{t-p}$ has no explanatory (or predictive) power for the last component y_{Kt} of $\mathbf{y}_t$. In this case, $\mathbf{R} = [\mathbf{0} : \mathbf{I}_K]$ and $\mathbf{b} = \mathbf{0}$.

The log-likelihood function for a restricted model is obtained by replacing $\boldsymbol{\pi}$ by $\mathbf{H}\boldsymbol{\delta} + \mathbf{a}$. Denote

$$\mathbf{z}_t = \mathbf{y}_t - \mathbf{X}_t' \mathbf{a} \text{ and } \mathbf{W}_t = \mathbf{H}' \mathbf{X}_t.$$

Then, the log-likelihood function

$$l(\boldsymbol{\delta}, \boldsymbol{\Sigma}_u) = -\frac{T}{2} \log(\det(\boldsymbol{\Sigma}_u)) - \frac{1}{2} \sum_{t=1}^T (\mathbf{z}_t - \mathbf{W}_t' \boldsymbol{\delta})' \boldsymbol{\Sigma}_u^{-1} (\mathbf{z}_t - \mathbf{W}_t' \boldsymbol{\delta}).$$

Differentiate this function with respect to the parameters. As in the case of the derivative, we get

$$\frac{\partial}{\partial \boldsymbol{\delta}} l(\boldsymbol{\delta}, \boldsymbol{\Sigma}_u) = \sum_{t=1}^T \mathbf{W}_t \boldsymbol{\Sigma}_u^{-1} (\mathbf{z}_t - \mathbf{W}_t' \boldsymbol{\delta}).$$

Unlike in the case of $\boldsymbol{\pi}$, the matrix $\boldsymbol{\Sigma}_u^{-1}$ cannot be moved to the front of the sum. For this reason the equation $\partial l(\boldsymbol{\delta}, \boldsymbol{\Sigma}_u) \partial \boldsymbol{\delta} = \mathbf{0}$ cannot be solved without knowing $\boldsymbol{\Sigma}$, but instead the solution (usually) reduces to the form

$$\boldsymbol{\delta} = \left(\sum_{t=1}^T \mathbf{W}_t \boldsymbol{\Sigma}_u^{-1} \mathbf{W}_t' \right)^{-1} \sum_{t=1}^T \mathbf{W}_t \boldsymbol{\Sigma}_u^{-1} \mathbf{z}_t.$$

Using the notation $\boldsymbol{\Sigma}_u^{-1} = \boldsymbol{\Phi}$ we can repeat the derivations from the previous section and conclude that the first order condition for $\boldsymbol{\Phi}$, i.e. $\partial l(\boldsymbol{\delta}, \boldsymbol{\Sigma}_u) \partial \boldsymbol{\Phi} = \mathbf{0}$, leads to the equation

$$\boldsymbol{\Sigma}_u = \frac{1}{T} \sum_{t=1}^T (\mathbf{z}_t - \mathbf{W}_t' \boldsymbol{\delta}) (\mathbf{z}_t - \mathbf{W}_t' \boldsymbol{\delta})'.$$

Given that the solution for $\boldsymbol{\delta}$ requires knowing $\boldsymbol{\Sigma}_u$ and the solution for $\boldsymbol{\Sigma}_u$ requires knowing $\boldsymbol{\delta}$, numerical solution methods are required (except special cases). An illustrative method proceeds as follows. Replace the unknown $\boldsymbol{\Sigma}_u$ by its unconstrained estimate $\widehat{\boldsymbol{\Sigma}}_u$ to yield

$$\tilde{\boldsymbol{\delta}}^{(1)} = \left(\sum_{t=1}^T \mathbf{W}_t \widehat{\boldsymbol{\Sigma}}_u^{-1} \mathbf{W}_t' \right)^{-1} \sum_{t=1}^T \mathbf{W}_t^{-1} \widehat{\boldsymbol{\Sigma}}_u^{-1} \mathbf{z}_t.$$

This then yields a new estimate $\widetilde{\boldsymbol{\Sigma}}_u^{(1)}$, which can again be used to replace $\widehat{\boldsymbol{\Sigma}}_u$ in $\tilde{\boldsymbol{\delta}}^{(1)}$. This results in a new estimate $\tilde{\boldsymbol{\delta}}^{(2)}$ for $\boldsymbol{\delta}$. This procedure can be continued until the estimates stop changing. This procedure is just an illustration. In practice, computer programs employ other algorithms. Nonetheless, the simple one step estimator $\tilde{\boldsymbol{\delta}}^{(1)}$ has good asymptotic properties, as it can be shown to be asymptotically equivalent to the true MLE (the reason being the orthogonality of $\boldsymbol{\pi}$ and $\boldsymbol{\Sigma}_u$).

Denote the ML estimators for $\boldsymbol{\delta}$ and $\boldsymbol{\Sigma}_u$ by $\tilde{\boldsymbol{\delta}}$ and $\widetilde{\boldsymbol{\Sigma}}_u$ and derive their asymptotic distribution by relying on the general theory of ML estimation. Differentiate the log likelihood function with respect to $\boldsymbol{\delta}$ once more (see the derivation results in Appendix A) to obtain

$$\frac{\partial^2}{\partial \boldsymbol{\delta} \partial \boldsymbol{\delta}'} l(\boldsymbol{\delta}, \boldsymbol{\Sigma}_u) = - \sum_{t=1}^T \mathbf{W}_t \boldsymbol{\Sigma}_u^{-1} \mathbf{W}_t'$$

so that the Fisher information matrix for $\boldsymbol{\delta}$ is

$$\mathcal{J}_{\boldsymbol{\delta}\boldsymbol{\delta}}(\boldsymbol{\delta}, \boldsymbol{\Sigma}_u) = \sum_{t=1}^T \mathbb{E}(\mathbf{W}_t \boldsymbol{\Sigma}_u^{-1} \mathbf{W}_t'),$$

which in the stationary case equals $T \mathbb{E}(\mathbf{W}_t \boldsymbol{\Sigma}_u^{-1} \mathbf{W}_t')$. Similar calculations to those in the previous section show that $\mathbb{E}(\partial^2 l(\boldsymbol{\delta}, \boldsymbol{\Sigma}_u) \partial \boldsymbol{\delta} \partial \sigma_{ij}) = 0$ so that the parameters $\boldsymbol{\delta}$ and $\boldsymbol{\Sigma}_u$ are orthogonal. Hence, the asymptotic distribution of $\tilde{\boldsymbol{\delta}}$ is

$$\tilde{\boldsymbol{\delta}} \stackrel{as}{\sim} \mathbf{N}(\boldsymbol{\delta}, \mathcal{J}_{\boldsymbol{\delta}\boldsymbol{\delta}}(\boldsymbol{\delta}, \boldsymbol{\Sigma})_u^{-1}).$$

Here we can account for the fact that $\mathbf{W}_t = \mathbf{H}'\mathbf{X}_t$, where $\mathbf{H}$ is known and fixed, but this does not result in any more illustrative formula.

On the part of the estimator $\tilde{\Sigma}_u$ for the constrained covariance matrix Σ_u it suffices to mention that it is consistent, which will be relied on later. Given what was noted above, we can write the estimator $\tilde{\Sigma}_u$ as

$$\tilde{\Sigma}_u = \frac{1}{T} \sum_{t=1}^T (\mathbf{z}_t - \mathbf{W}_t' \tilde{\boldsymbol{\delta}}) (\mathbf{z}_t - \mathbf{W}_t' \tilde{\boldsymbol{\delta}})'.$$

7.2 Hypothesis testing

We derive the Wald test for the restriction

$$H_0 : \mathbf{R}\boldsymbol{\pi} = \mathbf{b},$$

which is assumed to hold in the following (when examining the null hypothesis). In addition, we assume the maximum likelihood estimators $\hat{\boldsymbol{\pi}}$ and $\hat{\Sigma}_u$ satisfy the asymptotic properties considered in unrestricted estimation.. Formally, the asymptotic normality of (assuming stationary initial values) is expressed as

$$\sqrt{T}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}) \xrightarrow{d} \mathbf{N}(\mathbf{0}, \Sigma_u \otimes \Gamma_x^{-1}).$$

As $\mathbf{R}\boldsymbol{\pi} = \mathbf{b}$, it follows from the linear transformation multivariate normal distribution (see Appendix B) that

$$\sqrt{T}(\mathbf{R}\hat{\boldsymbol{\pi}} - \mathbf{b}) = \mathbf{R}\sqrt{T}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}) \xrightarrow{d} \mathbf{Z} \sim \mathbf{N}_q(\mathbf{0}, \mathbf{R}(\Sigma_u \otimes \Gamma_x^{-1})\mathbf{R}'),$$

and further that (see Appendix B)

$$\sqrt{T}(\mathbf{R}\hat{\boldsymbol{\pi}} - \mathbf{b})' [\mathbf{R}(\Sigma_u \otimes \Gamma_x^{-1})\mathbf{R}']^{-1} (\mathbf{R}\hat{\boldsymbol{\pi}} - \mathbf{b}) \xrightarrow{d} \mathbf{Z}' [\mathbf{R}(\Sigma_u \otimes \Gamma_x^{-1})\mathbf{R}']^{-1} \mathbf{Z} \sim \chi_q^2$$

where the last result follow from well known result on multivariate normal and the χ^2 distributions.

- For clarity, here (and elsewhere if needed) the notation $\mathbf{N}_q(\cdot, \cdot)$ signifies a multivariate normal distribution of dimension q .
- The mentioned result on multivariate normal and χ^2 -distribution states that if $\mathbf{Z} \sim \mathbf{N}_k(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then $(\mathbf{Z} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{Z} - \boldsymbol{\mu}) \sim \chi_k^2$ (assuming $\boldsymbol{\Sigma}$ is positive definite).

The square form on the left hand side measures the realism of the restriction, but contains the unknown matrix $\Sigma_u \otimes \Gamma_x^{-1}$, which has to be replaced by a consistent estimator in a test.

- As $\mathbf{\Gamma}_x = \mathbf{E}(\mathbf{x}_t \mathbf{x}_t')$, where $\mathbf{x}_t = [1 \quad \mathbf{Y}'_{t-1}]' = [1 \quad \mathbf{y}'_{t-1} \quad \dots \mathbf{y}'_{t-p}]'$, and given the general consistency results for the first and second order moments of a stationary process together with the consistency of $\widehat{\mathbf{\Sigma}}_u$, allows us to conclude the result

$$[\mathbf{R}(\widehat{\mathbf{\Sigma}}_u \otimes \widehat{\mathbf{\Gamma}}_x^{-1}) \mathbf{R}']^{-1} \xrightarrow{p} [\mathbf{R}(\mathbf{\Sigma}_u \otimes \mathbf{\Gamma}_x^{-1}) \mathbf{R}']^{-1},$$

where

$$\widehat{\mathbf{\Gamma}}_x = \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t'.$$

In conclusion, we then obtain the **Wald test statistic** (see Appendix B)

$$\mathbf{W} = T(\mathbf{R}\widehat{\boldsymbol{\pi}} - \mathbf{b})' [\mathbf{R}(\widehat{\mathbf{\Sigma}}_u \otimes \widehat{\mathbf{\Gamma}}_x^{-1}) \mathbf{R}']^{-1} (\mathbf{R}\widehat{\boldsymbol{\pi}} - \mathbf{b}) \xrightarrow{d} \chi_q^2$$

Large values of the test statistic are critical about the null hypothesis ($H_0 : \mathbf{R}\boldsymbol{\pi} = \mathbf{b}$). In practice, we apply the test by computing the approximate p-value

$$p = \Pr\{\mathbf{W} \geq \mathbf{W}(\mathbf{y})\} \approx \Pr\{\chi_q^2 \geq \mathbf{W}(\mathbf{y})\},$$

where χ_q^2 indicates a χ_q^2 -distributed random variable and $\mathbf{W}(\mathbf{y})$ is the value of the test statistic based on the observed time series data to be analyzed.

Likelihood ratio (LR) test. Instead of the Wald test, one can apply the likelihood ratio (LR) statistic. The general formula is

$$\text{LR} = 2 \left[l(\widehat{\boldsymbol{\pi}}, \widehat{\mathbf{\Sigma}}_u) - l(\widetilde{\boldsymbol{\pi}}, \widetilde{\mathbf{\Sigma}}_u) \right],$$

where $\widetilde{\boldsymbol{\pi}}$ and $\widetilde{\mathbf{\Sigma}}_u$ are the constrained maximum likelihood estimators for $\boldsymbol{\pi}$ and $\mathbf{\Sigma}_u$ (see the previous section). Using the formulae of the log-likelihood functions one can conclude by direct calculation that

$$l(\widehat{\boldsymbol{\pi}}, \widehat{\mathbf{\Sigma}}_u) = -\frac{T}{2} \log \det(\widehat{\mathbf{\Sigma}}_u) - \frac{TK}{2} \quad \text{and} \quad l(\widetilde{\boldsymbol{\pi}}, \widetilde{\mathbf{\Sigma}}_u) = -\frac{T}{2} \log \det(\widetilde{\mathbf{\Sigma}}_u) - \frac{TK}{2}$$

Hence, we get the **LR test statistic**

$$\text{LR} = 2 \left[\log \det(\widetilde{\mathbf{\Sigma}}_u) - \log \det(\widehat{\mathbf{\Sigma}}_u) \right] \xrightarrow{d} \chi_q^2$$

where the asymptotic distribution can be concluded by the arguments as in the case of the Wald test. Large values of the test statistic are critical to the null hypothesis and approximate p-values are obtained as in the Wald test.

Extra: Properties of the constrained ML estimator

The above test statistics can be generalized to cover the parameter $\boldsymbol{\delta}$ of the restricted VAR(p) model (see the restriction of interest). The constrained ML estimator was seen to obey the asymptotic normal distribution as

$$\sqrt{T}(\tilde{\boldsymbol{\delta}} - \boldsymbol{\delta}) \xrightarrow{d} \mathbf{N} \left(\mathbf{0}, \left(\sum_{t=1}^T \mathbf{E}(\mathbf{W}_t \boldsymbol{\Sigma}_u^{-1} \mathbf{W}_t') \right)^{-1} \right)$$

In order to apply the Wald test we need a valid empirical counterpart for the matrix $\sum_{t=1}^T \mathbf{E}(\mathbf{W}_t \boldsymbol{\Sigma}_u^{-1} \mathbf{W}_t')$. One obvious alternative is $T^{-1} \sum_{t=1}^T \mathbf{W}_t \tilde{\boldsymbol{\Sigma}}_u^{-1} \mathbf{W}_t'$, which allows one to derive the Wald test as above in the case of unconstrained ML estimates. The square roots of the diagonal elements of $\sum_{t=1}^T \mathbf{W}_t \tilde{\boldsymbol{\Sigma}}_u^{-1} \mathbf{W}_t'$ yield approximates for standard errors for the components of $\tilde{\boldsymbol{\delta}}$. Let $s.e.(\tilde{\boldsymbol{\delta}}_i)$ denote the standard error of the i :th component of $\tilde{\boldsymbol{\delta}}$. Then

$$\tilde{\boldsymbol{\delta}}_i / s.e.(\tilde{\boldsymbol{\delta}}_i) \stackrel{as}{\sim} \mathbf{N}(\boldsymbol{\delta}_i, 1)$$

so that the approximate 95% confidence interval for $\boldsymbol{\delta}_i$ is $\tilde{\boldsymbol{\delta}}_i \pm 1.96 s.e.(\tilde{\boldsymbol{\delta}}_i)$.

In the same way, one can form confidence intervals for the elements of the unconstrained parameter $\boldsymbol{\pi}$.

8 VAR(p): Model selection and diagnostic checking

8.1 Lag length selection

Using statistical tests in model selection. It is common to start the modeling from an unrestricted VAR model, because there is usually no sufficiently strong a prior information what the correct and specific parameter restrictions are or one wants to test some theory-based restrictions by Wald or LR tests.

In other words, the first task is to select the unknown lag length p of the VAR(p) model.

If one has reasons to choose a specific lag length, one can examine the appropriateness of the choice by testing VAR(p) against a higher order VAR. If the alternative model is the VAR($p + s$) model

$$\mathbf{y}_t = \boldsymbol{\nu} + \sum_{j=1}^{p+s} \mathbf{A}_j \mathbf{y}_{t-j} + \mathbf{u}_t, \quad \mathbf{u}_t \sim \text{nid}(\mathbf{0}, \boldsymbol{\Sigma}_u),$$

then one conducts a test for the hypothesis

$$H_0 : \mathbf{A}_{p+1} = \dots = \mathbf{A}_s = \mathbf{0}.$$

Let

$$\widehat{\boldsymbol{\Sigma}}(p^*) = T^{-1} \sum_{t=1}^T (\mathbf{y}_t - \mathbf{X}_t' \widehat{\boldsymbol{\pi}}) (\mathbf{y}_t - \mathbf{X}_t' \widehat{\boldsymbol{\pi}})'$$

be the MLE for the error covariance matrix of the unrestricted VAR(p^*) model. Then, the above hypothesis can be tested by the LR test statistic

$$\text{LR} = 2 \left[\log \det(\widehat{\boldsymbol{\Sigma}}_u(p)) - \log \det(\widehat{\boldsymbol{\Sigma}}_u(p+s)) \right] \underset{as}{\sim} \chi_{K^2 s}^2$$

Note that in this test statistic (as in the ones below) the estimators $\widehat{\boldsymbol{\Sigma}}_u(p)$ and $\widehat{\boldsymbol{\Sigma}}_u(p+s)$ apply the same number of observations. That is, the same sample period is used (this is not always automatically the case in different softwares.)

Sequential testing. As in the lag length selection of the univariate AR(p) model, also in the VAR context it is at times common to conduct a series of tests in a row by starting from a “sufficiently” long lag length P .

- If the hypothesis $\mathbf{A}_P = \mathbf{0}$ is not rejected, one turns to testing the hypothesis $\mathbf{A}_{P-1} = \mathbf{0}$.
- If this hypothesis is not rejected one keeps on testing until the first rejection occurs
- If none (hypotheses) is rejected, then one concludes that the order is zero.

This method is also similar to using the partial autocorrelation function to select the order of a univariate AR(p) model. In that case, if the first rejection occurs with an order that seems too large, one can proceed in the same manner without strict formal testing as is the case when a partial autocorrelation is used.

When the order of the VAR model is selected, one can apply Wald and LR tests to test for more detailed specification restrictions. This is natural when the underlying background theory implies interesting hypotheses.

8.2 Model selection criteria

There are various statistical criteria for selecting the order of an unrestricted VAR model, each based on different theoretical principles. When faced with a large number of alternative models, or when practical constraints, such as computationally intensive forecasting experiments, are present, the model selection criteria discussed below offer an effective strategy for model specification

Let again P be a predetermined sufficiently large lag order and $\widehat{\Sigma}(p^*)$ as above. Model selection criteria or information criteria are typically of the form

$$C(p^*) = \log \det(\widehat{\Sigma}_u(p^*)) + \frac{f(T)}{T}d, \quad p^* = 1, \dots, P,$$

where p^* is the order of the model, d is the number of corresponding unrestricted autoregressive and intercept parameters ($d = K(Kp^* + 1)$ for an unrestricted VAR(k) model)) and f is the so called *penalty function* that satisfies $f(T)/T \rightarrow 0$ as $T \rightarrow \infty$.

- The idea of the penalty function is to prevent one from choosing too large a model.
- The first part of the criterion measures the goodness of fit and if a larger p^* does not increase it much enough then the larger model is not favored. This becomes clear by noticing that when comparing the values of the criterion function the first term could be the maximum of the $-K - (2/T) \times \log$ likelihood function.
- **For good (preferable) models $C(p^*)$ takes on small values.**

As for univariate models, some familiar penalty functions are

$$\text{AIC} : f(T) = 2 \quad (\text{Akaike, 1974})$$

$$\text{HQ} : f(T) = 2 \log(\log T) \quad (\text{Hannan and Quinn, 1979})$$

$$\text{BIC} : f(T) = \log T \quad (\text{Schwarz, 1978; Rissanen, 1978})$$

AIC is the most liberal and (favors larger models), while BIC is the most stringent (favors smaller models).

- There is a version of HQ, where the constant term (2) is replaced by another constant.
- HQ and BIC result in consistent estimators for the lag order, when the lag order choice is based on minimization of the criterion.
- In practice it is recommended that one uses model selection criteria as an additional device and that the model choice is not made mechanically.

As was noted earlier, model selection criteria can also be used to compare models subject to constraints. In such cases, the maximum likelihood estimator on the right hand side of information criterion is formed from a restricted VAR model.

- This procedure is applied especially when it is difficult to formalize relevant hypotheses and there are several alternative models.
- One possibility is to estimate all possible model variants and compute the criterion for each model (as in the case of unrestricted models). This tends to lead to rather tedious computations. Hence various step-wise procedures have been suggested. In the case of zero restrictions one possibility is to drop one explanatory variable (that is, a component from the lag $\mathbf{y}_{t-j}$, $j = 1, \dots, p$) at the time and compute the criterion for the corresponding models. In the next step one starts from the model whose criterion value in the previous step had the largest drop compared to the initial model. This is repeated so that one of the explanatory variables is dropped until the criterion value no longer drops compared to the previous step.

Example (continued), model selection and estimation outcome. Let's apply the previously presented model selection procedure to the exchange rate/interest rate data. When the maximum lag is chosen as $P = 6$, we get the following results when using the LR test (denoted by $\text{LR}(p^*)$) between the VAR models with lag lengths p^* and $p^* - 1$:

Test	LR test stat.	p-value
LR(2)	17.154	0.002
LR(3)	2.099	0.717
LR(4)	4.602	0.331
LR(5)	3.153	0.535
LR(6)	1.671	0.796

The results point clearly to $p = 2$ (given the maximum lag length was sufficiently long to begin with), which results to the following estimated results:

$$\begin{bmatrix} rdiff_t \\ dexch_t \end{bmatrix} = \begin{bmatrix} 0.003 \\ (0.014) \\ -0.002 \\ (0.002) \end{bmatrix} + \begin{bmatrix} 1.139 & -0.223 \\ (0.090) & (0.661) \\ 0.024 & 0.405 \\ (0.012) & (0.089) \end{bmatrix} \begin{bmatrix} rdiff_{t-1} \\ dexch_{t-1} \end{bmatrix} + \begin{bmatrix} -0.176 & 0.635 \\ (0.090) & (0.647) \\ -0.028 & -0.181 \\ (0.012) & (0.087) \end{bmatrix} \begin{bmatrix} rdiff_{t-2} \\ dexch_{t-2} \end{bmatrix} + \begin{bmatrix} \hat{u}_{1t} \\ \hat{u}_{2t} \end{bmatrix}. \quad (8.1)$$

The approximate standard error are in brackets under the estimates. The correlation coefficient of the residuals is -0.272 so the errors of the model are slightly correlated. The estimated model fulfills the stationarity condition since the absolute value of the highest eigenvalue of the ML estimate for the companion form matrix $\mathbf{A}$ is 0.951.

- Some estimates are small in relation to their standard errors, which indicates that further restrictions on the model could be inserted in.
- We will get back to this later (also as an example of constrained VAR estimation).

8.3 Diagnostic checks

Residuals are also useful for investigating model adequacy. For an unconstrained model the residuals are defined by

$$\hat{\mathbf{u}}_t = \mathbf{y}_t - \hat{\boldsymbol{\nu}} - \hat{\mathbf{A}}_1 \mathbf{y}_{t-1} - \dots - \hat{\mathbf{A}}_p \mathbf{y}_{t-p}, \quad t = 1, \dots, T.$$

The residuals should have similar properties to those of the theoretical errors $\mathbf{u}_t$. To investigate this it is useful

- to examine residual plots,
- check whether there are individual or groups of “outliers”,
- whether there are signs of changes in variance, or other clearly systematic features.

It is also often recommended to examine whether the residuals are (multi)normally distributed, although the asymptotic estimation and test theory does not entail normality of the error terms (iid assumption suffices). Empirical auto- and cross-correlation functions for the residuals may also reveal potential serial dependencies in the model errors. If $\mathbf{u}_t \sim \text{iid}(\mathbf{0}, \boldsymbol{\Sigma}_u)$, then the residual auto- and cross-correlations are asymptotically normal with mean zero. Unlike in the case of theoretical model errors, these are not asymptotically independent and the $N(0, 1/T)$ approximation (for the one variable) does not hold.

- The true asymptotic distribution can be derived, but is not considered in the course.

- Some econometric programs (e.g., R) may print residual auto- and cross-correlations along with their true asymptotic standard errors or corresponding critical bounds that allow one to examine whether the residuals are serially dependent.

Portmanteau test. It is also possible to test for the presence of residual serial correlation. The so called Portmanteau test is directly based on residual auto- and cross-correlation functions (a multivariate extension of the Ljung-Box test). One version of it is

$$Q_H^* = T^2 \sum_{h=1}^H \frac{1}{T-h} \text{tr}(\mathbf{S}_h' \mathbf{S}_0^{-1} \mathbf{S}_h \mathbf{S}_0^{-1}), \quad \mathbf{S}_h = \sum_{t=1}^{T-h} \hat{\mathbf{u}}_t \hat{\mathbf{u}}_{t+h}',$$

where the null hypothesis is that there is no autocorrelation (here in the residuals) remaining. If the chosen unrestricted VAR(p) model error $\mathbf{u}_t \sim \text{iid}(\mathbf{0}, \boldsymbol{\Sigma}_u)$, then Q_H^* is asymptotically χ^2 -distributed with degrees of freedom equal to $K^2(H-p)$. If the model is restricted, the value of the degree of freedom of the χ^2 -distribution is K^2H minus the number of unrestricted autoregressive parameters. The asymptotic result entails that H is “large”.

Although there may be no reason to suspect residual (auto)correlation, the residuals might not be independent, as zero correlation is only related to linear independency.

- In order to examine possible nonlinear dependencies (in a restricted sense) one can use auto- and cross-correlations based on squared residuals.
- In this case, the usual standard errors and test statistics are not valid without appropriate adjustments.

Example (continued). Below, we report the fitted values of the VAR(2) model for *rdif* and *dexch*. The plots also depict the residual time series along with their sample auto- and cross-correlation coefficients. The variation in the second residual series (*dexch*) appears to be slightly higher at the beginning, while the first residual series (*rdif*) exhibits some notable spikes. Despite these features, the residuals appear to be “reasonably random”. The auto- and cross-correlations of the residuals are generally small, with the exception of a few isolated spikes at higher-order lags.

Diagram of fit and residuals for rdif

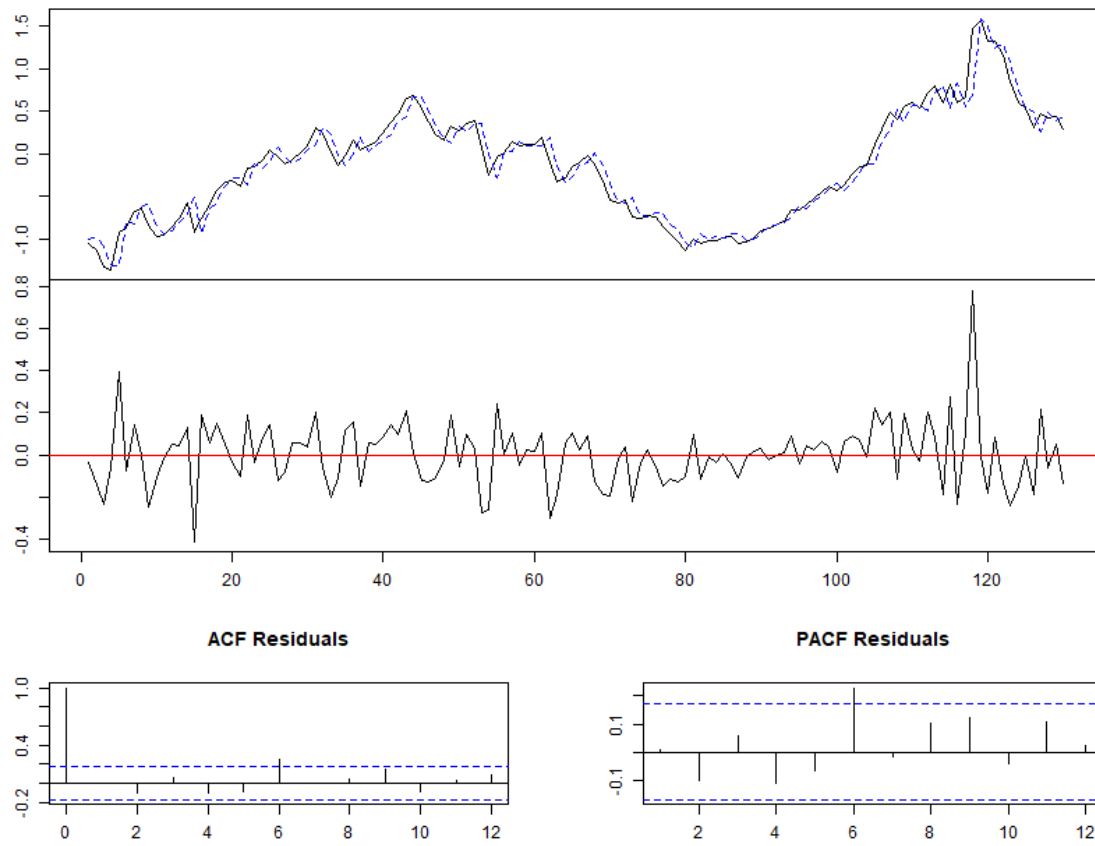
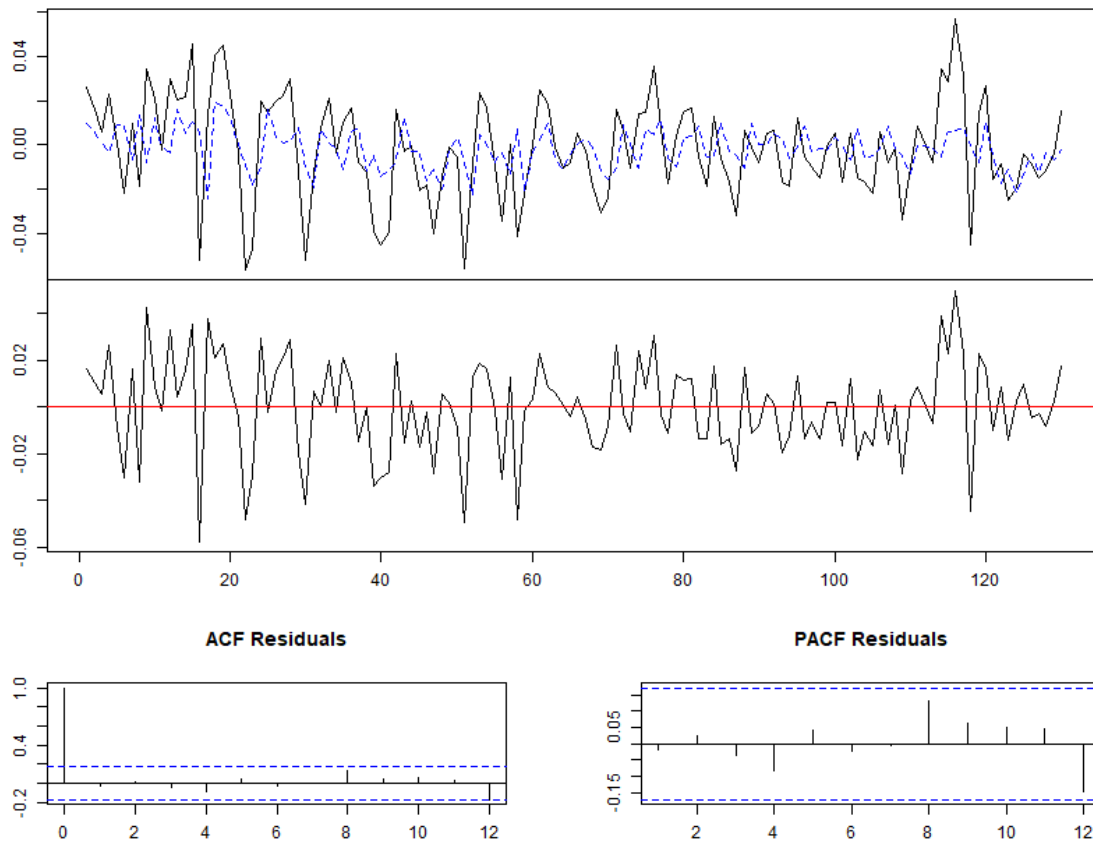
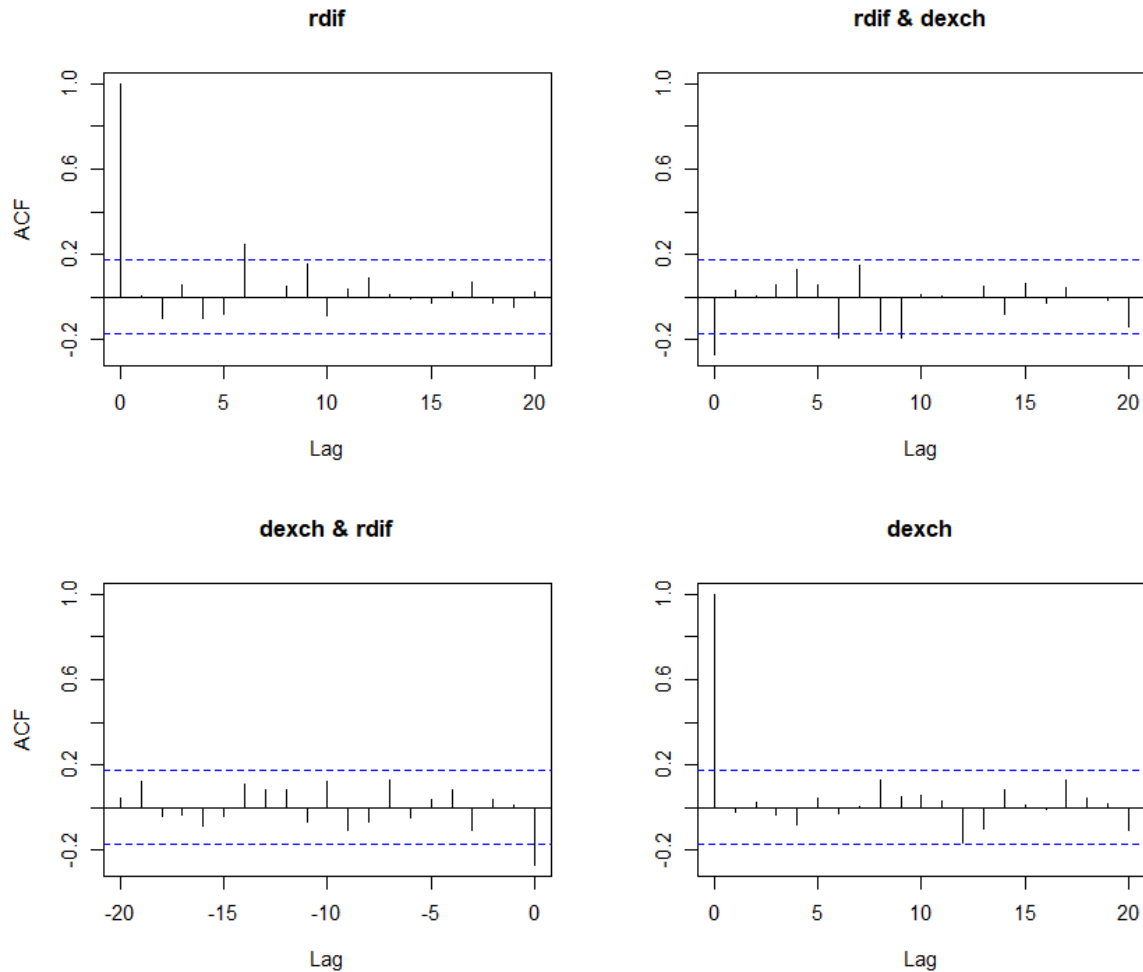


Diagram of fit and residuals for dexch





We employ the Portmanteau test (as discussed above) to check for remaining autocorrelation in the residuals. The results reported below show that the p-values exceed standard significance levels, indicating that the model specification is adequate (i.e., we fail to reject the null hypothesis of no autocorrelation at the conventional statistical significance levels).

In addition to checking for autocorrelation, we examine the distributional properties of the residuals. Plotting the histogram of the (standardized) residuals reveals a distribution with heavier tails than the normal distribution (leptokurtosis)—a feature hinted at by earlier graphical examinations (see the attached R code for plots not explicitly depicted here).

- Furthermore, the slight correlation observed in the squared residuals at certain lags points toward conditional heteroskedasticity (ARCH effects) and a consequent deviation from Gaussianity.

- Overall, despite these distributional nuances, the model can be regarded as a reasonably successful specification.

Portmanteau Test (adjusted):

H=12: Chi-squared = 51.062, df = 40, p-value = 0.1129

H=6: Chi-squared = 21.248, df = 16, p-value = 0.1692

H=15: Chi-squared = 58.683, df = 52, p-value = 0.2438

9 Forecasting with VAR model

Consider forecasting a $\text{VAR}(p)$ process in a theoretical situation, where the parameters of the process are assumed to be known.

- In practice, one must use estimated parameters when computing forecasts.

As discussed in univariate time series modelling, we know from probability theory that the optimal prediction of a random variable Y given another variable X is the conditional expectation (conditional mean) $E(Y|X)$.

- Recall the properties **CEV1–CEV4** (see also Appendix A).
- Here optimality refers to a prediction that has a minimum mean square error (MSE).

When $\mathbf{Y}$ is a vector, we obtain the same result, when our goal is to minimize the MSE matrix. An illustration follows. Let $\tilde{\mathbf{Y}} = g(\mathbf{X})$ be any prediction for $\mathbf{Y}$ given $\mathbf{X}$ (a function of $\mathbf{X}$). Then

$$\text{MSE}(\tilde{\mathbf{Y}}) = E[(\mathbf{Y} - \tilde{\mathbf{Y}})(\mathbf{Y} - \tilde{\mathbf{Y}})']$$

is the mean square error matrix of the forecast $\tilde{\mathbf{Y}}$. Furthermore, we get

$$\mathbf{a}'\text{MSE}(\tilde{\mathbf{Y}})\mathbf{a} \geq \mathbf{a}'\text{MSE}(E(\mathbf{Y}|\mathbf{X}))\mathbf{a}$$

for all vectors $\mathbf{a}$ (with comfortable dimension).

- In other words, the matrix $\text{MSE}(\tilde{\mathbf{Y}}) - \text{MSE}(E(\mathbf{Y}|\mathbf{X}))$ is positive semi-definite, written as $\text{MSE}(\tilde{\mathbf{Y}}) - \text{MSE}(E(\mathbf{Y}|\mathbf{X})) \geq 0$.
- By choosing $\mathbf{a} = (0, \dots, 0, 1, 0, \dots, 0)$, we observe that the mean square error of any component of $\tilde{\mathbf{Y}}$ is at least as large as the one of $E(\mathbf{Y}|\mathbf{X})$.
- We have $\text{MSE}(\tilde{Y}) = E(Y - \tilde{Y})^2$, when Y is a scalar-valued.

We apply the above result to a $\text{VAR}(p)$ process. First, we form the optimal prediction of $\mathbf{y}_{t+1}$ given that the whole history $\mathbf{y}_t, \mathbf{y}_{t-1}, \dots$ is known.

Given the $\text{VAR}(p)$ process, we have

$$E_t(\mathbf{y}_{t+1}) = E(\mathbf{y}_{t+1}|\mathbf{y}_{t-j}, j \geq 0) = E(\mathbf{y}_{t+1}|\mathbf{y}_t, \dots, \mathbf{y}_{t-p+1}) \equiv E(\mathbf{y}_{t+1}|\mathcal{F}_t) \equiv \mathbf{y}_t(1),$$

where $\mathcal{F}_t = \{\mathbf{y}_t, \dots, \mathbf{y}_{t-p+1}\}$ denotes **information set** containing the predictive information. By the independence of the error, we find that $\mathbf{u}_t$ and $\mathbf{y}_{t-j}$ are independent for $j > 0$. Therefore, based on CEV2, we get (also in this vector-valued case) $\mathbf{E}_t(\mathbf{u}_{t+j}) = \mathbf{E}(\mathbf{u}_{t+j}) = \mathbf{0}$ ($j > 0$), and

$$\begin{aligned}\mathbf{y}_t(1) &= \boldsymbol{\nu} + \mathbf{A}_1 \mathbf{E}_t(\mathbf{y}_t) + \dots + \mathbf{A}_p \mathbf{E}_t(\mathbf{y}_{t+1-p}) + \mathbf{E}_t(\mathbf{u}_{t+1}) \\ &= \boldsymbol{\nu} + \mathbf{A}_1 \mathbf{y}_t + \dots + \mathbf{A}_p \mathbf{y}_{t+1-p},\end{aligned}$$

where $\boldsymbol{\nu} = \mathbf{A}(1)\boldsymbol{\mu}$. Generally, for the forecast horizons $h \geq 1$,

$$\mathbf{E}_t(\mathbf{y}_{t+h}) = \boldsymbol{\nu} + \mathbf{A}_1 \mathbf{E}_t(\mathbf{y}_{t+h-1}) + \dots + \mathbf{A}_p \mathbf{E}_t(\mathbf{y}_{t+h-p})$$

or

$$\mathbf{y}_t(h) = \boldsymbol{\nu} + \mathbf{A}_1 \mathbf{y}_t(h-1) + \dots + \mathbf{A}_p \mathbf{y}_t(h-p),$$

where $\mathbf{y}_t(h-j) = \mathbf{E}_t(\mathbf{y}_{t+h-j})$ and $\mathbf{y}_t(h-j) = \mathbf{y}_{t+h-j}$ when $h \leq j$.

- In other words, $\mathbf{y}_t(s) = \mathbf{E}_t(\mathbf{y}_{t+s})$ for $s > 0$ and $\mathbf{y}_t(s) = \mathbf{y}_{t+s}$ for $s \leq 0$.

By using the linear VMA(∞) representation of the VAR process (cf. the analogous results for the univariate case), we can present the forecast as

$$\mathbf{y}_t(h) = \mathbf{E}_t(\mathbf{y}_{t+h}) = \boldsymbol{\mu} + \sum_{j=0}^{\infty} \boldsymbol{\Psi}_j \mathbf{E}_t(\mathbf{u}_{t+h-j}) = \boldsymbol{\mu} + \sum_{j=h}^{\infty} \boldsymbol{\Psi}_j \mathbf{u}_{t+h-j}$$

from which we can see that the forecast error can be written as

$$\mathbf{y}_{t+h} - \mathbf{y}_t(h) = \sum_{j=0}^{h-1} \boldsymbol{\Psi}_j \mathbf{u}_{t+h-j} \quad (\boldsymbol{\Psi}_0 = \mathbf{I}_K).$$

This expression shows that the expectation of the forecast error is zero so that the forecast is unbiased. We also find that the covariance matrix of the error is

$$\sum_{j=0}^{h-1} \boldsymbol{\Psi}_j \boldsymbol{\Sigma}_u \boldsymbol{\Psi}_j'.$$

In the case of a normal errors, $\mathbf{u}_t \sim \text{nid}(\mathbf{0}, \boldsymbol{\Sigma}_u)$, the forecast error is also normally distributed

$$\mathbf{y}_{t+h} - \mathbf{y}_t(h) \sim \text{N} \left(\mathbf{0}, \sum_{j=0}^{h-1} \boldsymbol{\Psi}_j \boldsymbol{\Sigma}_u \boldsymbol{\Psi}_j' \right).$$

Using this results we can form confidence intervals for the components of $\mathbf{y}_{t+h}$ that we seek to forecast. Let $\sigma_i^2(h)$ denote the i :th diagonal element of $\sum_{j=0}^{h-1} \boldsymbol{\Psi}_j \boldsymbol{\Sigma}_u \boldsymbol{\Psi}_j'$. Then (using obvious notation)

$$\frac{y_{i,t+h} - y_{i,t}(h)}{\sigma_i(h)} \sim \text{N}(0, 1),$$

and

$$\begin{aligned} 0.95 &= \mathbf{P} \left\{ -1.96 \leq \frac{y_{i,t+h} - y_{i,t}(h)}{\sigma_i(h)} \leq 1.96 \right\} \\ &= \mathbf{P} \left\{ y_{i,t}(h) - 1.96\sigma_i(h) \leq y_{i,t+h} \leq y_{i,t}(h) + 1.96\sigma_i(h) \right\}. \end{aligned}$$

That is, the 95% confidence interval for the i :th component in $\mathbf{y}_{t+h}$ is $y_{i,t}(h) \pm 1.96\sigma_i(h)$.

Finally, in practice and as mentioned at the beginning of this section, unknown parameters need to be replaced by their estimates in forecast computation so that the above formulae hold only approximately.

- Note also that the above confidence intervals were presented for fixed i and h .
- When we consider several confidence intervals simultaneously we face the usual problems of constructing joint confidence intervals.

10 Granger and instantaneous causality

10.1 General idea of Granger causality

Consider two real-valued random variables x and z . How should one define causality from the former to the latter variable? Without delving deeper to the philosophical implications, one can think that a necessary condition for causality is that the “cause” does not occur after the “consequence”. If x is a cause for z , then one expects that x has predictive content for z .

Let us formalize this intuition. Let x_t and z_t be real-valued stochastic processes and consider predicting the future values of z_t at time t .

- Let $\mathcal{F}_t$ denote the **information set** (as already described in forecasting sections above) hypothetical variables set, which includes x_t and all other variables that are observed at time t and could be used to predict the future values z_{t+h} , $h \geq 1$.
- The optimal (minimum mean square error) prediction of z_{t+h} given $\mathcal{F}_t$ is denoted $z_t(h|\mathcal{F}_t)$.
- Correspondingly, let $z_t(h|\mathcal{F}_t \setminus \mathcal{F}_t^x)$ be the optimal prediction of z_{t+h} when the history of x , denoted by $\mathcal{F}_t^x = \{x_s, s \leq t\}$, is removed from the information set.

Let $\Sigma_z(h|\mathcal{F}_t)$ and $\Sigma_z(h|\mathcal{F}_t \setminus \mathcal{F}_t^x)$ be the corresponding mean square errors (MSE). We say that x_t **Granger causes** z_t if

$$\Sigma_z(h|\mathcal{F}_t) < \Sigma_z(h|\mathcal{F}_t \setminus \mathcal{F}_t^x)$$

for at least one t and $h \in \{1, 2, \dots\}$. That is, **under Granger causality, we can obtain more accurate predictions for z_{t+h}** (for some $h > 0$) by exploiting information on the process x_t .

If $\mathbf{x}_t$ and $\mathbf{z}_t$ are vector-valued processes, the mean square errors become matrices. In this case, we say that $\mathbf{x}_t$ Granger causes $\mathbf{z}_t$ if

$$\Sigma_{\mathbf{z}}(h|\mathcal{F}_t \setminus \mathcal{F}_t^{\mathbf{x}}) \neq \Sigma_{\mathbf{z}}(h|\mathcal{F}_t)$$

for at least for one $h > 0$. The interpretation remains essentially similar: the history of $\mathbf{x}_t$ improves the prediction of $\mathbf{z}_t$.

10.2 Granger causality in VAR models

The information set $\mathcal{F}_t$ applied in the general definition above is often infeasible in practice. In applied work, the information set is restricted to selected observed variables. Granger causality is most often studied in the context of **VAR models**, where the relevant information set consists of the components of the vector process and their lags.

Bivariate VAR model ($K = 2$): Granger causality can be easily tested within a VAR model. Consider first, for simplicity, a bivariate VAR(p) process

$$\begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} = \begin{bmatrix} \nu_1 \\ \nu_2 \end{bmatrix} + \sum_{j=1}^p \begin{bmatrix} a_{11,j} & a_{12,j} \\ a_{21,j} & a_{22,j} \end{bmatrix} \begin{bmatrix} y_{1,t-j} \\ y_{2,t-j} \end{bmatrix} + \begin{bmatrix} u_{1t} \\ u_{2t} \end{bmatrix}.$$

If

$$a_{12,j} = 0, j = 1, 2, \dots, p,$$

then the lags of y_{2t} do not help to predict y_{1t} and **there is no Granger causality from y_{2t} to y_{1t}** (that is, y_{2t} is not Granger-causal for y_{1t}).

- That is, the restriction $a_{12,1} = a_{12,2} = \dots = a_{12,p} = 0$ implies the absence of such a Granger causality relationship.

When working with an estimated VAR model (for the empirical data), this hypothesis can be tested by the Wald or likelihood ratio (LR) test, following a χ_p^2 -distribution under the null hypothesis.

- At times, such as in the **vars** package in R, the F -test statistic is used instead of the Wald test statistic, but the logic is the same.

Larger VAR models ($K > 2$) and indirect effects: When the VAR model contains more than two variables, Granger causality between two specific variables cannot always be tested by testing zero-restrictions in a simple pairwise manner. Consider the following three-variable VAR process:

$$\begin{bmatrix} y_{1t} \\ y_{2t} \\ y_{3t} \end{bmatrix} = \begin{bmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{bmatrix} + \sum_{j=1}^p \begin{bmatrix} a_{11,j} & a_{12,j} & a_{13,j} \\ a_{21,j} & a_{22,j} & a_{23,j} \\ a_{31,j} & a_{32,j} & a_{33,j} \end{bmatrix} \begin{bmatrix} y_{1,t-j} \\ y_{2,t-j} \\ y_{3,t-j} \end{bmatrix} + \begin{bmatrix} u_{1t} \\ u_{2t} \\ u_{3t} \end{bmatrix}.$$

If, for instance, $a_{12,1} = \dots = a_{12,p} = 0$, the lags of y_{2t} do not help forecasting y_{1t} one period ahead. However, because the lags of y_{2t} may affect y_{3t} , which in turn affects y_{1t} , the lags of y_{2t} may help forecasting $y_{1,t+1}$, $y_{1,t+2}$ etc. In other words, there may be Granger causality through **indirect effects**.

General condition. Generally, we can consider Granger causality from one **group of variables** to another. Consider a stationary VAR(p) process and partition $\mathbf{y}_t = [\mathbf{y}'_{1t} \ \mathbf{y}'_{2t}]'$, where $\mathbf{y}_{1t}$ is $K_1 \times 1$ and $\mathbf{y}_{2t}$ is $K_2 \times 1$, ($K_1 + K_2 = K$), and write the VAR(p) process

$$\begin{bmatrix} \mathbf{y}_{1t} \\ \mathbf{y}_{2t} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\nu}_1 \\ \boldsymbol{\nu}_2 \end{bmatrix} + \sum_{j=1}^p \begin{bmatrix} \mathbf{A}_{11,j} & \mathbf{A}_{12,j} \\ \mathbf{A}_{21,j} & \mathbf{A}_{22,j} \end{bmatrix} \begin{bmatrix} \mathbf{y}_{1,t-j} \\ \mathbf{y}_{2,t-j} \end{bmatrix} + \begin{bmatrix} \boldsymbol{\zeta}_{1t} \\ \boldsymbol{\zeta}_{2t} \end{bmatrix}.$$

Here $\mathbf{A}_{\cdot,j}$ are matrices. It intuitively seems that Granger causality from $\mathbf{y}_{2t}$ to $\mathbf{y}_{1t}$ relates to whether $\mathbf{A}_{12,j} \neq \mathbf{0}$ for some j . In other words, variables in $\mathbf{y}_{2t}$ **does not** Granger cause $\mathbf{y}_{1t}$, if and only if $\mathbf{A}_{12,j} = \mathbf{0}$ for all $j = 1, \dots, p$.

Extra: More details for Granger causality

For simplicity assume $p = 1$ and denote $\mathbf{A}_{ij,1} = \mathbf{A}_{ij}$. The general version follows similar logic.

Assume $\mathbf{A}_{12} = \mathbf{0}$. Then the equation above implies a stationary VAR(1) presentation for $\mathbf{z}_t$:

$$\mathbf{z}_t = \mathbf{A}_{11}\mathbf{z}_{t-1} + \boldsymbol{\zeta}_t.$$

It follows that the conditional expectation depends only on the history of $\mathbf{z}$:

$$\mathbb{E}(\mathbf{z}_{t+h} | \mathbf{z}_s, s \leq t) = \mathbf{A}_{11}^h \mathbb{E}(\mathbf{z}_{t+h-1} | \mathbf{z}_s, s \leq t), \quad h \geq 1.$$

On the other hand, when $\mathbf{A}_{12} = \mathbf{0}$, the full model implies:

$$\mathbb{E}(\mathbf{z}_{t+h} | \mathbf{y}_s, s \leq t) = \mathbf{A}_{11}^h \mathbb{E}(\mathbf{z}_{t+h-1} | \mathbf{y}_s, s \leq t), \quad h \geq 1.$$

Comparing these two recursions (starting from $\mathbb{E}(\mathbf{z}_{t+1} | \cdot) = \mathbf{A}_{11}\mathbf{z}_t$), one observes by induction that the predictions are identical for all h and t . Hence, there is no Granger causality.

Now, assume $\mathbf{A}_{12} \neq \mathbf{0}$. As $\mathbf{u}_{t+1}$ and $\mathbf{y}_s, s \leq t$, are independent, we have $\mathbb{E}(\mathbf{u}_{t+1} | \mathbf{y}_s, s \leq t) = \mathbf{0}$ and the model yields:

$$\begin{aligned} \mathbf{z}_{t+1} - \mathbb{E}(\mathbf{z}_{t+1} | \mathbf{y}_s, s \leq t) &= \mathbf{z}_{t+1} - \mathbf{A}_{11}\mathbf{z}_t - \mathbf{A}_{12}\mathbf{x}_t \\ &= \boldsymbol{\zeta}_{t+1}. \end{aligned}$$

However, projecting only on the history of $\mathbf{z}$:

$$\begin{aligned} \mathbf{z}_{t+1} - \mathbb{E}(\mathbf{z}_{t+1} | \mathbf{z}_s, s \leq t) &= \mathbf{z}_{t+1} - \mathbf{A}_{11}\mathbf{z}_t - \mathbf{A}_{12}\mathbb{E}(\mathbf{x}_t | \mathbf{z}_s, s \leq t) \\ &= \boldsymbol{\zeta}_{t+1} + \mathbf{A}_{12}[\mathbf{x}_t - \mathbb{E}(\mathbf{x}_t | \mathbf{z}_s, s \leq t)]. \end{aligned}$$

The last term is non-zero because $\mathbf{A}_{12}\mathbf{x}_t$ is not a function of the history of $\mathbf{z}$ alone. Thus, the mean square error (covariance matrix of the prediction error) differs between the two information sets. It follows that $\mathbf{x}$ Granger causes $\mathbf{z}$. $\square$

The “no Granger causality” hypothesis corresponds to $K_1 K_2 p$ zero restrictions:

$$H_0 : \mathbf{A}_{12,1} = \mathbf{A}_{12,2} = \cdots = \mathbf{A}_{12,p} = \mathbf{0}.$$

This hypothesis can be tested using Wald or Likelihood Ratio (LR) tests.

- Under the null hypothesis, these test statistics are asymptotically $\chi^2_{K_1 K_2 p}$ -distributed.
- Large values of the test statistic lead to rejection of the null hypothesis (i.e., finding evidence of Granger causality).
- As mentioned, the F -test is also commonly used in finite samples (e.g., in R/RStudio).

Example: Three-variable process. Consider the VAR(1) process:

$$\mathbf{y}_t = \begin{bmatrix} y_{1t} \\ y_{2t} \\ y_{3t} \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.5 & 0 & 0 \\ 0.1 & 0.1 & 0.3 \\ 0 & 0.2 & 0.3 \end{bmatrix} \mathbf{y}_{t-1} + \mathbf{u}_t.$$

Partitioning $\mathbf{y}_t = [y_{1t} \quad \mathbf{y}'_{2t}]'$ so that $\mathbf{y}_{2t} = [y_{2t} \quad y_{3t}]'$.

- Here, the sub-matrix $\mathbf{A}_{12}$ corresponds to the top-right block $\begin{bmatrix} 0 & 0 \end{bmatrix}$. Since this is zero, $\mathbf{y}_{2t}$ (variables y_{2t} and y_{3t}) **does not Granger cause** y_{1t} .
- On the other hand, looking at the coefficients affecting y_{2t} (the second row), the coefficient for $y_{1,t-1}$ is $0.1 \neq 0$. Thus, y_{1t} **Granger causes** $\mathbf{y}_{2t}$.

Granger causality when involving $I(1)$ variables. Standard Granger causality tests (such as using F or Wald tests in a standard VAR), as above, rely on the assumption that variables are stationary ($I(0)$).

- If you apply standard tests to non-stationary $I(1)$ variables, the test statistics will **not** follow standard asymptotic distributions (like χ^2 or F distributions).
- This often leads to “spurious causality”—finding relationships where none exist.

Different scenarios:

- Scenario A: Variables are $I(1)$ and cointegrated. If variables share a long-run equilibrium (cointegration), you must use a Vector Error Correction Model (VECM). Cointegration and VECM will be introduced in the next section.

- Scenario B: Variables are $I(1)$ and not cointegrated. If variables are non-stationary and have no long-run relationship, you must difference them. Using differenced time series, the limitation is that tests test short-run causality only. Long-run information is removed by differencing (but this is acceptable since no long-run relationship exists).
- Scenario C: The robust and preferred solution is the Toda-Yamamoto (TY) procedure, which is valid regardless of whether the variables are $I(0)$, $I(1)$, or cointegrated. It avoids the potential biases associated with pre-testing for cointegration. The core idea is to estimate a VAR in levels (not differences) but add extra lags to “fix” the asymptotic distribution of the Wald test statistics. Steps:
 1. **Select lag length (p):** Determine the optimal lag length p for the VAR using, for example, information criteria (AIC, SC, or HQ).
 2. **Determine integration order (d_{max}):** Identify the maximum order of integration for the variables in the system (usually $d_{max} = 1$ for (economic) time series).
 3. **Estimate augmented VAR:** Estimate a VAR in levels with a total lag length of $p + d_{max}$.
 4. **Perform Wald Test:** Test the Granger causality restrictions **only on the first p lags**. The coefficients for the extra d_{max} lags are ignored; they function essentially as nuisance parameters to ensure the test statistic follows a standard χ^2 distribution. Therefore, the test regression for a bivariate system where we test if $\mathbf{x}$ Granger-causes $\mathbf{y}$:

$$\mathbf{y}_t = \boldsymbol{\nu} + \sum_{i=1}^p \mathbf{A}_i \mathbf{y}_{t-i} + \underbrace{\sum_{j=p+1}^{p+d_{max}} \mathbf{A}_j \mathbf{y}_{t-j}}_{\text{ignored lags}} + \sum_{i=1}^p \mathbf{B}_i \mathbf{x}_{t-i} + \underbrace{\sum_{j=p+1}^{p+d_{max}} \mathbf{B}_j \mathbf{x}_{t-j}}_{\text{ignored lags}} + u_t$$

and test the hypothesis

$$H_0 : \mathbf{B}_1 = \mathbf{B}_2 = \dots = \mathbf{B}_p = \mathbf{0}.$$

10.3 Instantaneous (contemporaneous) causality

The definition of Granger causality discussed above strictly concerns **predictive** causality: whether past values of a variable help predict future values of another. It does not address **instantaneous** (or **contemporaneous**) causality—that is, whether knowing the value of x_{t+1} helps to more accurately forecast z_{t+1} given the history $\mathcal{F}_t$.

Formally, there exists instantaneous Granger causality between processes x_t and z_t if:

$$\Sigma_z(1|\mathcal{F}_t) \neq \Sigma_z(1|\mathcal{F}_t \cup \{x_{t+1}\}).$$

Here, the variable set $\mathcal{F}_t \cup \{x_{t+1}\}$ includes the future realization x_{t+1} in addition to the history $\mathcal{F}_t$. In other words, adding the contemporaneous value of x to the information set improves the forecast of z .

Symmetry and interpretation. A crucial property of instantaneous causality is **symmetry**. It can be shown that if x_t is instantaneously causal for z_t , then z_t is also instantaneously causal for x_t .

- Consequently, this statistical definition does not determine the “direction” of causality.
- The direction must be inferred from other sources, such as economic theory. For instance, if theory suggests causality runs only from x to z , a significant instantaneous relationship can be interpreted in favor of that causal link.

Instantaneous causality in VAR models. In the context of VAR models, instantaneous causality is directly linked to the correlation of the error terms.

- **Bivariate case** ($K = 2$): In a bivariate VAR, there is **no instantaneous causality** between y_{1t} and y_{2t} if and only if the errors u_{1t} and u_{2t} are **uncorrelated**. If the errors are correlated, knowing the realization of y_{1t} (and thus u_{1t}) provides information about u_{2t} , thereby improving the forecast of y_{2t} .
- **General case** ($K > 2$): In systems with more than two variables, the absence of instantaneous causality corresponds to the structure of the error covariance matrix Σ_u . There is no instantaneous causality between two groups of variables if Σ_u is **block diagonal** with respect to those groups.

Example: Consider a VAR process for $\mathbf{y}_t = (y_{1t}, y_{2t}, y_{3t})$ with the following error covariance matrix:

$$\Sigma_u = \begin{bmatrix} 2.0 & 0 & 0 \\ 0 & 1.5 & 0.4 \\ 0 & 0.4 & 1.0 \end{bmatrix}.$$

Here, the covariance between the error of y_{1t} and the errors of the other variables is zero (the top row and left column off-diagonal elements are 0). However, the errors for y_{2t} and y_{3t} are correlated (0.4). Conclusion: There is no instantaneous causality between y_{1t} and the group $\mathbf{y}_{2t} = (y_{2t}, y_{3t})$.

Testing: Instantaneous causality can be tested statistically using a Wald-type test for zero restrictions on the corresponding off-diagonal elements of the error covariance matrix Σ_u .

Example (continued). The estimates from the VAR(2) model indicate clearly, that the data does not support Granger causality in direction $dexch_t \rightarrow rdif_t$ (compare the relevant estimates to their standard errors). By using a formal likelihood ratio (LR) test we get the value of test statistic of 1.08 and a (approximate) p-value of 0.583. As a part of the LR test, we need and get the following estimates for the constrained model (that is, an example of constrained/restricted estimation result)

$$\begin{bmatrix} rdif_t \\ dexch_t \end{bmatrix} = \begin{bmatrix} 0.002 \\ (0.014) \\ -0.002 \\ (0.002) \end{bmatrix} + \begin{bmatrix} 1.144 & 0 \\ (0.087) & \\ 0.024 & 0.405 \\ (0.012) & (0.089) \end{bmatrix} \begin{bmatrix} rdif_{t-1} \\ dexch_{t-1} \end{bmatrix} + \begin{bmatrix} -0.183 & 0 \\ (0.087) & \\ -0.028 & -0.181 \\ (0.012) & (0.087) \end{bmatrix} \begin{bmatrix} rdif_{t-2} \\ dexch_{t-2} \end{bmatrix} + \begin{bmatrix} \tilde{u}_{1t} \\ \tilde{u}_{2t} \end{bmatrix}.$$

Compared to the unconstrained model and its estimation result, the estimation accuracy has improved slightly (as one could expect), as seen by comparing the standard errors.

Granger causality in direction $rdif_t \rightarrow dexch_t$ seems to be realistic based on the estimates.

- This requires the estimation of another constrained model,
- The resulting p-value of the LR test statistic of 6.54 is 0.038.

Furthermore, when using R's causality-function, leading to the F-test statistics (instead of asymptotic Wald test) to the corresponding null hypotheses as above (with the LR test), we obtain the p-values of 0.618 and 0.052. These lead, as expected, similar conclusions to the above LR test-based results. That is, in line with most (theoretical) economic arguments, we can conclude stronger predictive relation from interest rate differential to changes in the exchange rate than other way round.

11 Cointegration

11.1 Introduction to cointegration

Instead of stationary variables, let us now assume that we have two variables, y_{1t} and y_{2t} , where either or both are nonstationary—specifically **integrated of order one**, denoted as $I(1)$.

Cointegration in the bivariate case. An important special case of linear regression between nonstationary $I(1)$ variables arises when they share a **common stochastic trend**. Suppose there exists a linear relationship such that for some parameter δ , the linear combination

$$z_t = y_{1t} - \delta y_{2t} = [1 \quad -\delta] \mathbf{y}_t$$

is stationary, $I(0)$, even though y_{1t} and y_{2t} are individually nonstationary $I(1)$. In such a case, y_{1t} and y_{2t} are said to be **cointegrated**.

- The vector $[1 \quad -\delta]'$ is called the **cointegration vector**.
- **Long-run equilibrium:** Cointegration is often interpreted as a long-run relationship. If the equilibrium is defined by $y_{1t} = \tilde{\delta} y_{2t}$, then z_t represents the “equilibrium error”—the extent to which y_t deviates from its equilibrium value.
- Because $z_t \sim I(0)$, this error fluctuates around zero with constant variance. On average, the system remains in equilibrium.

Why not just difference the data? An alternative way to analyze variables that are nonstationary (or integrated of order one) is to use their first differences, Δx_t and Δy_t , which are stationary. However, if a cointegration relationship exists, differencing is not an optimal strategy because it discards the long-run level relationship (the “common trend”) between the series.

The problem of overdifferencing. To understand why simply differencing all $I(1)$ variables can be problematic, consider a two-dimensional example where $y_{1t} \sim I(1)$ and $y_{2t} \sim I(0)$.

- The vector $\mathbf{y}_t$ is effectively $I(1)$, but differencing the entire vector is unnecessary for the second component.

Assume $(\Delta y_{1t}, y_{2t})$ follows a stable VAR(1) process:

$$\begin{bmatrix} 1 - a_{11}\mathbf{B} & 0 \\ 0 & 1 - a_{22}\mathbf{B} \end{bmatrix} \begin{bmatrix} \Delta y_{1t} \\ y_{2t} \end{bmatrix} = \begin{bmatrix} \nu_1 \\ \nu_2 \end{bmatrix} + \begin{bmatrix} u_{1t} \\ u_{2t} \end{bmatrix},$$

where $\mathbf{u}_t$ is white noise. If we base our analysis on the fully differenced series $\Delta \mathbf{y}_t$, the second equation becomes:

$$(1 - a_{22}\mathbf{B})\Delta y_{2t} = (1 - \mathbf{B})u_{2t}.$$

Here, the error term has an MA(1) component $(1 - \mathbf{B})u_{2t}$ with a unit root. This means the model for the differenced series becomes a VARMA(1,1) where the MA part is **non-invertible**. This misuse of differencing (“overdifferencing”) leads to misspecified models that are difficult to estimate.

General definition of cointegration. The situation described above can be formalized for $\mathbf{y}_t = (y_{1t}, \dots, y_{Kt})$. If $\mathbf{y}_t \sim I(1)$, but there is a linear combination $\beta' \mathbf{y}_t$, which is $I(0)$. then an $I(1)$ process $\{\mathbf{y}_t, t = 1, 2, \dots\}$ ($K \times 1$) is said to be cointegrated if there exists a nonzero parameter vector β ($K \times 1$) such that:

$$\beta' \mathbf{y}_t \sim I(0).$$

The parameter β is called the **cointegrating (cointegration) vector** (CI-vector).

- Note that this definition allows for trivial cases (like the overdifferencing example above) where $\beta = (0, \dots, 1, \dots, 0)$, meaning one variable is already stationary.
- There can be several linearly independent CI-vectors. If there are r such vectors, they form the **cointegrating matrix** $\beta = [\beta_1 \ \dots \ \beta_r]$, and r is the **cointegration rank**.

Bivariate example and intuition. To illustrate, consider the process:

$$\begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} + \begin{bmatrix} b \\ 1 \end{bmatrix} \tau_t + \begin{bmatrix} \xi_{1t} \\ \xi_{2t} \end{bmatrix},$$

where $\xi_t = (\xi_{1t}, \xi_{2t})$ is stationary, such as $\xi_t = (\xi_{1t}, \xi_{2t}) \sim \text{iid}(\mathbf{0}, \Sigma_\xi)$, $\tau_t = \sum_{j=1}^t \eta_j$ is a random walk process (containing the stochastic trend, with a fixed initial value τ_0) where $\eta_t \sim \text{iid}(0, \sigma^2)$.

- Individually, y_{1t} and y_{2t} are $I(1)$ because they both contain τ_t . Therefore, the time series they generate wandering behavior over time with increasing variance.

- The non-stationarity of the processes y_{1t} and y_{2t} is driven by a one-dimensional random walk τ_t . However, the linear combination:

$$y_{1t} - by_{2t} = (\mu_1 - b\mu_2) + (\xi_{1t} - b\xi_{2t})$$

eliminates the random walk τ_t . The result is stationary (even iid), and it does not behave like y_{1t} and y_{2t} . Instead, it is fluctuating around the constant mean $\mu_1 - b\mu_2$. Thus, the process is cointegrated with the cointegration vector $(1, -b)$.

- This implies that the linear combination $y_{1t} - by_{2t}$ (where $b \neq 0$) establishes a very strong linear dependence between the processes y_{1t} and y_{2t} , forcing them to move ‘hand in hand’. This is perhaps most evident when $b = 1$ and $\mu_1 = \mu_2$, in which case the linear combination $y_{1t} - y_{2t}$ is stationary with an expected value of zero.
- Although the processes y_{1t} and y_{2t} wander wildly over time, cointegration prevents them from drifting too far apart.

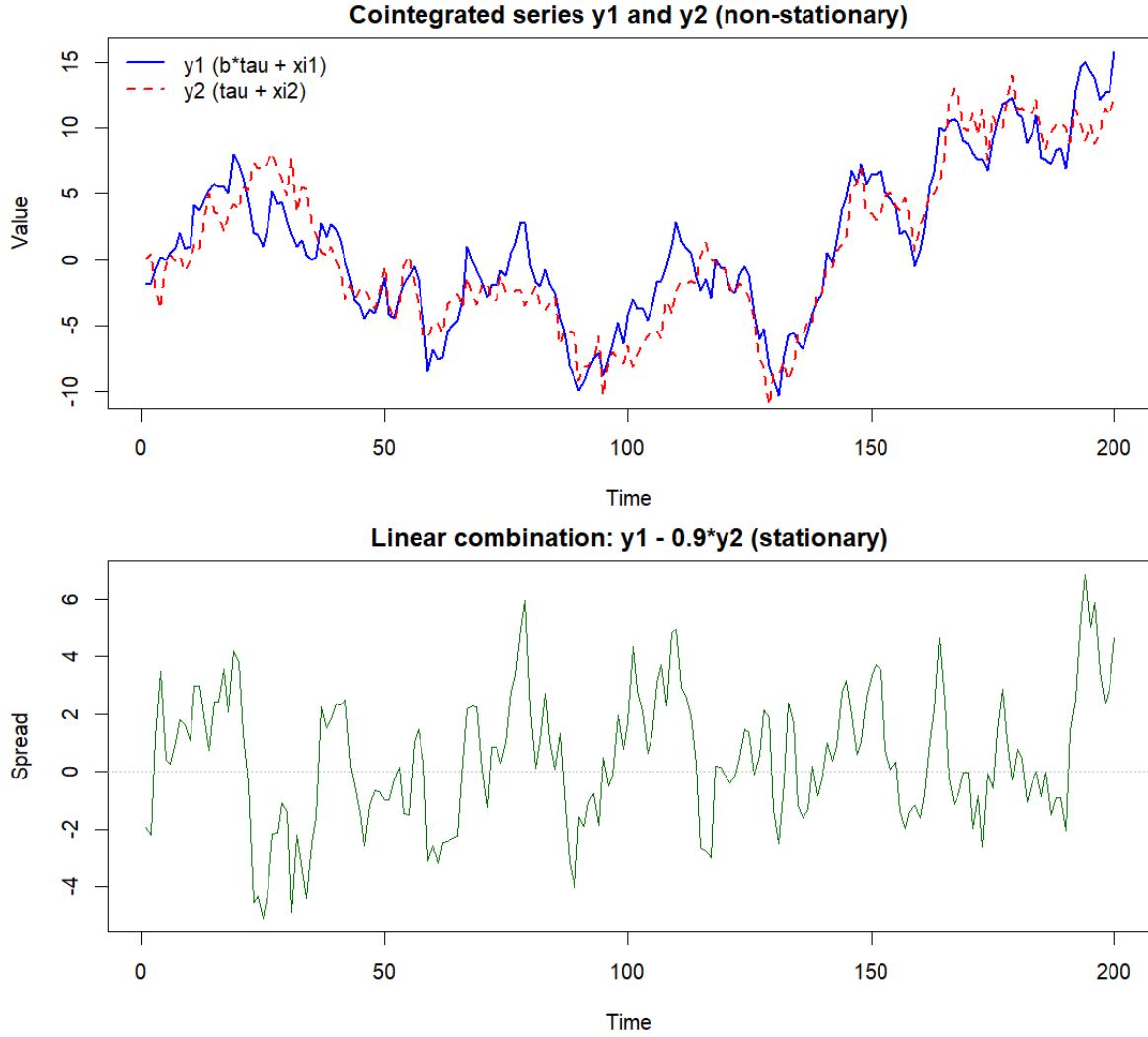


Figure: A simulated cointegrated process (as above) where $T = 200$, $\mu_1 = \mu_2 = 0$, $b = 0.9$, $\sigma^2 = 1$, the (independent) AR(1) processes ξ_{1t} and ξ_{2t} with AR(1)-coefficients of 0.9 and 0.6, respectively.

Empirical example and interpretation. Consider the aggregate income (Y_t) and consumption (C_t) of an economy. Empirically, both series are typically non-stationary. However, the Permanent Income Hypothesis suggests a long-run equilibrium relationship where a stable fraction of income is consumed, implying $C_t = aY_t$, $0 < a < 1$. In logarithmic terms, this linearizes to:

$$\log(C_t) - \log(Y_t) = \log(a)$$

This theoretical equilibrium provides the foundation for a cointegration model. While the individual series $\log(C_t)$ and $\log(Y_t)$ may wander indefinitely (stochastic trends), the linear combination $u_t = \log(C_t) - \log(Y_t) - \log(a)$ is expected to be stationary. Here, the residual u_t represents the transient deviation from the long-run equilibrium. If consumption drifts too far from income, economic forces will eventually pull it back.

Non-uniqueness of CI-vectors. In the absence of restrictions, cointegrating (cointegration) vectors are not unique.

- As an example, in the bivariate example above, if $(1, -b)$ is a CI-vector, then any multiple $(c, -cb)$ is also a CI-vector, as the linear combination remains stationary.
- Why this matters? Because of this non-uniqueness, software (like R or EViews) usually reports “normalized” cointegrating vectors. They typically set the first coefficient to 1 (just as we did with above for y_{1t}) to make the results interpretable.

While the vectors themselves are not unique, the **cointegrating space** (the subspace spanned by these vectors) is unique.

11.2 Cointegrated VAR and VECM model

Consider the K -dimensional VAR(p) process

$$\mathbf{x}_t = \mathbf{A}_1 \mathbf{x}_{t-1} + \cdots + \mathbf{A}_p \mathbf{x}_{t-p} + \mathbf{u}_t, \quad t = 1, 2, \dots$$

where $\mathbf{u}_t \sim \text{iid}(\mathbf{0}, \Sigma_u)$.

First rewrite the model using the lag polynomial $\mathbf{A}(B) = \mathbf{I}_K - \mathbf{A}_1 B - \cdots - \mathbf{A}_p B^p$. Using the algebraic result on lag polynomials (see Appendix A) similar to the univariate unit root case, we can decompose it as:

$$\mathbf{A}(B) = \mathbf{A}(1) B + \left(\mathbf{I}_K - \sum_{j=1}^{p-1} \mathbf{\Lambda}_j B^j \right) \Delta,$$

where $\mathbf{\Lambda}_j = -\mathbf{A}_{j+1} - \cdots - \mathbf{A}_p$. Substituting this back allows us to write the VAR(p) as a **Vector Error Correction Model (VECM) presentation**:

$$\Delta \mathbf{x}_t = \mathbf{\Theta} \mathbf{x}_{t-1} + \sum_{j=1}^{p-1} \mathbf{\Lambda}_j \Delta \mathbf{x}_{t-j} + \mathbf{u}_t, \quad t = 1, 2, \dots$$

where $\mathbf{\Theta} = -\mathbf{A}(1)$.

The reduced rank condition. Now, assume that the standard stationarity condition ($\det(\mathbf{A}(z)) \neq 0$ for all $|z| \leq 1$) **does not hold**. Instead, assume the polynomial $\det(\mathbf{A}(z))$ has roots at $z = 1$ (unit roots).

- Consequently, $\det(\mathbf{A}(1)) = 0$, which implies that the matrix Θ is singular (reduced rank). Let the rank of Θ be $r < K$, that is, $\text{rank}(\Theta) = r$.

Using a standard result from matrix algebra, any $(K \times K)$ matrix of rank r can be decomposed into the product of two $K \times r$ matrices of rank r :

$$\Theta = \alpha\beta',$$

where α and β are $(K \times r)$ matrices with $\text{rank}(\alpha) = \text{rank}(\beta) = r$. Plugging this into the above model equation yields the **VECM model**:

$$\Delta \mathbf{x}_t = \alpha\beta' \mathbf{x}_{t-1} + \sum_{j=1}^{p-1} \Lambda_j \Delta \mathbf{x}_{t-j} + \mathbf{u}_t, \quad t = 1, 2, \dots$$

Interpretation: If we assume that $\mathbf{x}_t \sim I(1)$, and $E(\mathbf{x}_t) = \mathbf{0}$, then the term $\Delta \mathbf{x}_t$ is stationary ($I(0)$). Since the error term $\mathbf{u}_t$ is also $I(0)$, for the equation to balance, the term $\beta' \mathbf{x}_{t-1}$ must also be stationary ($I(0)$). This implies that the process $\mathbf{x}_t$ is **cointegrated**.

Cointegrating vectors. The columns of the matrix β are the cointegrating vectors. The linear combinations $\beta' \mathbf{x}_{t-1}$ represent deviations from the long-run equilibrium relationships between the components of $\mathbf{x}_t$.

- These deviations are called “equilibrium errors”.
- Since equilibrium errors are stationary, they fluctuate randomly around zero with constant variance. Once the system deviates from equilibrium, it tends to return to it.

Adjustment matrix. The parameter α is often called the **adjustment matrix** (or loading/weight matrix).

- It describes how the errors $\beta' \mathbf{x}_{t-1}$ feed back into the system to affect the changes $\Delta \mathbf{x}_t$.
- Essentially, α determines the speed and direction of the correction mechanism that forces the non-stationary variables back toward their equilibrium relationship.

Example case: A simple VECM presentation of the VAR(2) model. The bivariate cointegrated VAR(2) model with one cointegrating vector ($r = 1$) has the following VECM representation

$$\begin{aligned} \begin{bmatrix} \Delta x_{1t} \\ \Delta x_{2t} \end{bmatrix} &= \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} \begin{bmatrix} \beta_1 & \beta_2 \end{bmatrix} \begin{bmatrix} x_{1,t-1} \\ x_{2,t-1} \end{bmatrix} + \begin{bmatrix} \lambda_{11,1} & \lambda_{12,1} \\ \lambda_{21,1} & \lambda_{22,1} \end{bmatrix} \begin{bmatrix} \Delta x_{1,t-1} \\ \Delta x_{2,t-1} \end{bmatrix} + \begin{bmatrix} u_{1t} \\ u_{2t} \end{bmatrix} \\ &= \begin{bmatrix} \alpha_1(\beta_1 x_{1,t-1} + \beta_2 x_{2,t-1}) \\ \alpha_2(\beta_1 x_{1,t-1} + \beta_2 x_{2,t-1}) \end{bmatrix} + \begin{bmatrix} \lambda_{11,1} & \lambda_{12,1} \\ \lambda_{21,1} & \lambda_{22,1} \end{bmatrix} \begin{bmatrix} \Delta x_{1,t-1} \\ \Delta x_{2,t-1} \end{bmatrix} + \begin{bmatrix} u_{1t} \\ u_{2t} \end{bmatrix}. \end{aligned}$$

The Granger Representation Theorem. In the derivation above, we assumed $\mathbf{x}_t \sim I(1)$. The connection between the unit root assumption in $\mathbf{A}(1)$ and the VECM representation is formalized by the Granger Representation Theorem.

- Details are beyond the scope of this course.

Impulse responses in cointegrated systems. In the context of a stationary VAR process, impulse response functions are based on the coefficients Ψ_j of the VMA(∞) representation. While an $I(1)$ process does not have a standard VMA(∞) representation, we can still define the impulse responses using the same recursive logic.

For a stationary VAR, the coefficient matrices are computed as:

$$\Psi_j = \sum_{i=1}^j \Psi_{j-i} \mathbf{A}_i, \quad \Psi_0 = \mathbf{I}_K, \quad j = 1, 2, \dots,$$

where $\mathbf{A}_i = \mathbf{0}$ when $i > p$. This formula remains well-defined even if the starting point is a cointegrated VAR(p) process $\mathbf{x}_t$. However, there is a fundamental difference in behavior:

- **Stationary case:** Impulse responses converge to zero as $j \rightarrow \infty$.
- **Integrated ($I(1)$) case:** The impulse responses **do not** converge to zero but instead usually converge towards a nonzero constant. This reflects the “permanent” nature of shocks in non-stationary cases. Consequently, the “total impact” (sum of matrix coefficients Ψ_j) often defined for stationary VARs cannot be defined for the $I(1)$ case, but the impulse response analysis itself remains a useful tool.

Adding Deterministic Terms. In the derivations above, we assumed the stationary components had zero mean, which is rarely realistic. We now extend the VECM to include deterministic terms like constants and trends.

Case 1: Nonzero mean. Consider the process $\mathbf{y}_t = \boldsymbol{\mu} + \mathbf{x}_t$, where $\mathbf{x}_t$ follows a VAR(p) and satisfies the Granger Representation Theorem. The resulting VECM is:

$$\Delta \mathbf{y}_t = \boldsymbol{\alpha} (\boldsymbol{\beta}' \mathbf{y}_{t-1} - \boldsymbol{\nu}) + \sum_{j=1}^{p-1} \boldsymbol{\Lambda}_j \Delta \mathbf{y}_{t-j} + \mathbf{u}_t,$$

where $\boldsymbol{\nu} = \boldsymbol{\beta}' \boldsymbol{\mu}$.

- **Interpretation:** The term $\boldsymbol{\nu}$ represents the expectation of the stationary linear combinations. The system corrects towards the equilibrium $\boldsymbol{\beta}' \mathbf{y}_{t-1} = \boldsymbol{\nu}$.

Case 2: Linear trend. A more general model includes a linear trend: $\mathbf{y}_t = \boldsymbol{\mu}_0 + \boldsymbol{\mu}_1 t + \mathbf{x}_t$. The corresponding VECM is:

$$\Delta \mathbf{y}_t = \boldsymbol{\nu}_0 + \boldsymbol{\alpha} (\boldsymbol{\beta}' \mathbf{y}_{t-1} - \boldsymbol{\nu}_1(t-1)) + \sum_{j=1}^{p-1} \boldsymbol{\Lambda}_j \Delta \mathbf{y}_{t-j} + \mathbf{u}_t,$$

where $\boldsymbol{\nu}_1 = \boldsymbol{\beta}' \boldsymbol{\mu}_1$.

- In this general specification, the linear combinations $\boldsymbol{\beta}' \mathbf{y}_t$ are **not** stationary because their expectation contains a linear trend component.

Restricted trend: An important special case arises if $\boldsymbol{\nu}_1 = \mathbf{0}$, simplifying the model to:

$$\Delta \mathbf{y}_t = \boldsymbol{\nu}_0 + \boldsymbol{\alpha} \boldsymbol{\beta}' \mathbf{y}_{t-1} + \sum_{j=1}^{p-1} \boldsymbol{\Lambda}_j \Delta \mathbf{y}_{t-j} + \mathbf{u}_t.$$

Here, the cointegrating relations $\boldsymbol{\beta}' \mathbf{y}_{t-1}$ are stationary around a constant, even though the variables $\mathbf{y}_t$ themselves drift with a linear trend.

11.3 Cointegration relations and identification

As discussed, cointegrating vectors are not unique: if the $r \times K$ matrix $\boldsymbol{\beta}'$ satisfies $\boldsymbol{\beta}' \mathbf{y}_t \sim I(0)$, then $\mathbf{C} \boldsymbol{\beta}' \mathbf{y}_t$ is also $I(0)$ for any non-singular $r \times r$ matrix $\mathbf{C}$. To interpret the model, we must impose **restrictions** or **normalization** to achieve identification.

Because the cointegrating matrix $\boldsymbol{\beta}$ is not uniquely determined, we must impose restrictions to identify the parameters. A standard approach is the **normalization strategy**.

We begin by partitioning the K -dimensional vector of variables $\mathbf{y}_t$ into two groups:

$$\mathbf{y}_t = \begin{bmatrix} \mathbf{z}_t \\ \mathbf{q}_t \end{bmatrix},$$

where $\mathbf{z}_t$ is $r \times 1$ and $\mathbf{q}_t$ is $(K - r) \times 1$.

A common identification assumption is to normalize β such that the first r rows form an identity matrix:

$$\beta' = [\mathbf{I}_r \quad : \quad \mathbf{A}],$$

where $\mathbf{I}_r$ is an $r \times r$ identity matrix and $\mathbf{A}$ is an $r \times (K - r)$ matrix of coefficients to be estimated. Key statistical implications:

- **Linear independence:** This normalization is valid if and only if the r variables chosen for $\mathbf{z}_t$ are linearly independent in the cointegration space. Mathematically, the $r \times r$ sub-matrix of β corresponding to the variables in $\mathbf{z}_t$ must be non-singular (invertible).
- **Common stochastic trends:** In a system with K variables and r cointegrating equations, there are exactly $K - r$ independent sources of non-stationarity. These are known as the common stochastic trends. By partitioning the model this way, we treat the sub-process $\mathbf{q}_t$ as the “engine” of the system; it contains the $K - r$ trends that drive the long-run path of the entire vector.
- **Economic interpretation:** This structure often aligns with economic theories where r variables (in $\mathbf{z}_t$) adjust to an equilibrium determined by the remaining $K - r$ variables (in $\mathbf{q}_t$).

– For example, if $r = 1$, we can express the long-run relation as:

$$z_{1,t} = -a_{11}q_{1,t} - a_{12}q_{2,t} \cdots + \text{error}_t$$

Here, $z_{1,t}$ “follows” the stochastic trends generated by the $\mathbf{q}_t$ variables. If the “drivers” in $\mathbf{q}_t$ shift, $z_{1,t}$ must eventually adjust to maintain the equilibrium.

Estimation and inference. The asymptotic theory for $I(1)$ processes is non-standard. However, standard procedures exist for VECM estimation.

- Assuming the cointegrating rank r is known, the VECM parameters can be estimated using maximum likelihood (Johansen’s Reduced Rank Regression). This method can also handle models with specific structural restrictions on the cointegrating vectors.
- **Lag length selection:** Typically, one first selects the lag length p for the unrestricted VAR (in levels) using information criteria (AIC, BIC, HQ) or sequential testing, and then estimates the VECM with $p - 1$ lags in differences.

Hypothesis testing: Once the model is estimated and identified (e.g., using the normalization $\beta' = [\mathbf{I}_r \quad : \quad \mathbf{A}]$):

- **t-tests:** The standard t -test can be used to evaluate the significance of individual coefficients in α or the non-restricted part of β ($\mathbf{A}$), as these estimates are asymptotically normal.

- **Wald tests:** Hypotheses restricting multiple parameters simultaneously can be tested using the Wald test statistic, which follows an asymptotic χ^2 -distribution with degrees of freedom equal to the number of restrictions.

Diagnostic checks: Standard diagnostic checks used for stationary VAR models (e.g., Portmanteau tests for residual autocorrelation, ARCH-LM tests for heteroskedasticity) are also applicable to VECMs. Additionally, plotting the estimated cointegration relations—the residuals $\hat{\beta}'\mathbf{y}_t$ —provides visual verification that the relations are indeed stationary.

11.4 Testing cointegration

Determining the number of cointegration relationships is of particular interest in different applications.

Engle-Granger approach. At times the cointegration vector is known based on some (economic) theory.

- See, e.g., the previous examples on the consumption-income relationship.

In this case, construct the cointegration relationship $z_t = \beta'\mathbf{y}_t$ and test whether the null hypothesis of no cointegration will be rejected.

If it is known that there exists at most one cointegrating vector, a simple approach to test for the existence of cointegration is the Engle-Granger approach.

- Run a (linear) regression of y_{1t} (the first element of $\mathbf{y}_t$) on the other $K - 1$ variables $y_{2t}, \dots, y_{Kt}$ and testing the unit root in the residuals.
- This is based on the idea of the normalization introduced above.
- Unit root testing can be based on the ADF test (critical values are, however, different!). If the unit root hypothesis is rejected, the hypothesis of no-cointegration is also rejected.

The above static regression gives consistent estimates of the cointegrating vector.

- In a second stage, the error-correction model can be estimated using the estimated cointegrating vector from the first stage.

There are, however, problems with this Engle-Granger approach:

- The results of the tests are sensitive to the left-hand side variable of the regression, that is, to the normalization applied to the cointegrating vector.
- If the cointegrating vector happens not to involve y_{1t} but only $y_{2t}, \dots, y_{Kt}$, the above test is not appropriate.

- The residual-based test tends to lack power because it does not exploit all the available information about the dynamic interactions of the variables.
- More than one cointegrating relationship may exist.

Johansen approach. An alternative approach (that does not suffer from the above drawbacks) was proposed by Johansen (1988), who developed a maximum likelihood estimation procedure.

- This also allows one to test for the number of cointegrating relations.
- The details of the Johansen procedure are complex and we shall only focus on the main aspects (for empirical analysis).

The first step in the Johansen approach involves testing hypotheses about the rank of the long-run matrix Θ , or equivalently the number of columns in β ($K \times r$).

- Imposing the restriction $\alpha\beta'$ and for a given r , it can be shown that the maximum likelihood estimate for β in the VECM equals the matrix containing the r eigenvectors corresponding to the r largest (estimated) eigenvalues of a $K \times r$ matrix.

Let us denote the (theoretical) eigenvalues of this matrix in decreasing order as $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_K$.

- If there are r cointegrating relationships (and Θ has rank r), it must be the case that $\log(1 - \lambda_j) = 0$ for the smallest $K - r$ eigenvalues, that is, for $j = r + 1, r + 2, \dots, K$.

(Johansen approach): trace test

We can use the (estimated) eigenvalues $\hat{\lambda}_1 > \hat{\lambda}_2 > \dots > \hat{\lambda}_K$ to test hypotheses about the rank of Θ .

For example, the hypothesis $H_0 : r \leq r_0$ against $H_1 : r_0 < r \leq K$ can be tested using the likelihood ratio (LR) test statistic

$$\text{LR} = -T \sum_{j=r_0+1}^K \log(1 - \hat{\lambda}_j).$$

Trace test checks whether the smallest $K - r_0$ eigenvalues are significantly different from zero.

- The trace (and the following maximum eigenvalue) test statistic has generally a non-standard asymptotic null distribution (do not have the usual χ^2 -distribution).

(Johansen approach): maximum eigenvalue test

We can test $H_0 : r \leq r_0$ against $H_1 : r = r_0 + 1$ using the maximum eigenvalue test statistic

$$\text{LR}_{max} = -T \log(1 - \hat{\lambda}_{r_0+1}).$$

It is based on the estimated $(r_0 + 1)$ th largest eigenvalue.

As with the unit root tests, the null distributions of both the trace and maximum eigenvalue tests depend on the fact whether a constant (and a time trend) are included in.

Determining the cointegrating rank. The cointegration rank may be specified by means of a sequential testing procedure, starting from a test of no cointegration ($r = 0$ or $\text{rank}(\Theta) = 0$) up to a test of full stationarity ($r = K$ or $\text{rank}(\Theta) = K$).

The full sequence of hypotheses is

$$\begin{aligned} H_0(0) &: \text{rank}(\Theta) = 0 \text{ vs } H_1 : \text{rank}(\Theta) > 0 \\ H_0(1) &: \text{rank}(\Theta) = 1 \text{ vs } H_1 : \text{rank}(\Theta) > 1 \\ &\vdots \\ H_0(K-1) &: \text{rank}(\Theta) = K-1 \text{ vs } H_1 : \text{rank}(\Theta) = K. \end{aligned}$$

Each hypothesis can be tested by the trace or the maximum eigenvalue test. The testing sequence terminates, and the corresponding cointegrating rank is selected, when the null hypothesis cannot be rejected for the first time.

- If the first hypothesis, $H_0(0)$, cannot be rejected, a VAR model for the differenced series is considered.
- Rejection of all the hypotheses implies a VAR model for the levels.

11.5 Empirical example

Let us consider the four variable Canada dataset available in R (vars package).

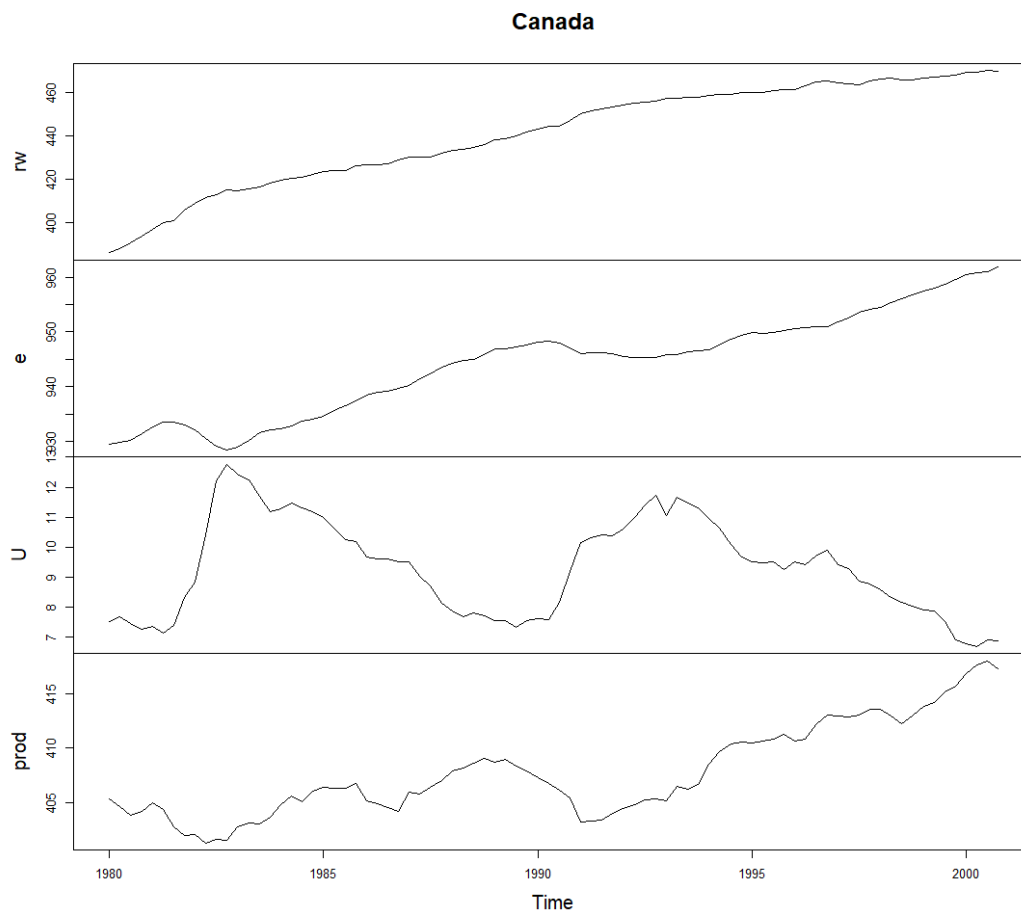


Figure: The four variables are the log of employment (e), labour productivity ($prod$), log real wage (rw) and unemployment rate (U). Sample period: 1980:Q1–2000:Q4.

After preliminary data inspection (summary statistics and time series plots), it is meaningful to conduct unit root tests (here ADF tests).

ADF unit root tests with different deterministic components (we assume, for simplicity, and throughout four lags in the ADF test regression):

Variable	Determ. terms	Test stat.	p-value
prod	constant, trend	-2.21	0.478
\Delta prod	constant	-4.73	0.000
e	constant, trend	-2.84	0.189

\Delta e	constant	-4.42	0.000
U	constant	-2.32	0.170
\Delta U	constant	-3.84	0.003
rw	constant, trend	-2.12	0.525
\Delta rw	constant	-3.04	0.035

It can be concluded that all the four time series can be treated as integrated of order one (i.e. $I(1)$ processes) based on the visual inspection and the results of ADF tests.

The lag length selections p in the $\text{VAR}(p)$ when including the constant and deterministic trend: AIC=3, HQ=2 and BIC=1.

Testing cointegration. Let us consider an example of cointegration analysis. We proceed with lag length specifications of $p = 2$ and $p = 3$, including a deterministic trend. We test the cointegration rank using the Trace test

Test statistics Critical values					
H_0	p=2	p=3	10%	5%	1%
r= 0	86.12	84.92	59.14	62.99	70.05
r<=1	37.33	36.42	39.06	42.44	48.45
r<=2	15.65	18.72	22.76	25.32	30.45
r<=3	4.10	3.85	10.49	12.25	16.26

These results indicate the presence of one cointegration relationship ($r = 1$). In the sequential testing procedure, the null hypothesis $H_0 : \text{rank}(\Theta) = 0$ is clearly rejected. However, the subsequent null hypothesis, $H_0 : \text{rank}(\Theta) \leq 1$, cannot be rejected against the alternative of higher-order cointegration.

- At times the “critical values” above are labeled to the corresponding percentage points (90, 95 and 99%) of the (non-standard) asymptotic distribution (the same numbers).

Estimation results. The VECM estimation yields the following results for the cointegrating vector and the adjustment coefficients. (See also the attached R code. All the details are not reported here)

1. The long-run relationship: The estimated cointegration relationship, normalized on the variable rw (real wages), is given by:

$$ec_{t-1} = rw_{t-1} + \underset{(0.61)}{0.545}, prod_{t-1} - \underset{(0.68)}{0.013}, e_{t-1} + \underset{(1.45)}{1.727}, U_{t-1} - \underset{(0.28)}{0.709}(t-1) \quad (11.1)$$

Standard errors are reported in parentheses underneath the coefficients. The t -values for the long-run coefficients lead the following statistical conclusions:

- Only the deterministic trend coefficient is significantly different from zero at the 5% level ($|t| > 1.96$).
 - The coefficients for productivity ($prod$), employment (e), and unemployment (U) are not statistically significant in this specific specification, suggesting that the long-run equilibrium is driven largely by the trend component relative to real wages.
2. The adjustment coefficients (α): The estimated loading matrix $\hat{\alpha}$ describes how each variable adjusts to the disequilibrium ec_{t-1} . The estimates and their corresponding t -values:

$$\hat{\alpha} = \begin{pmatrix} -0.085 \\ -0.012 \\ -0.016 \\ -0.009 \end{pmatrix} \begin{matrix} (t = -5.71) \\ (t = -0.92) \\ (t = -2.16) \\ (t = -1.49) \end{matrix} \begin{matrix} \leftarrow \Delta rw_t \\ \leftarrow \Delta prod_t \\ \leftarrow \Delta e_t \\ \leftarrow \Delta U_t \end{matrix}$$

Interpretation:

- Significant adjustment (rw and e): The adjustment coefficient for real wages (Δrw) is -0.085 with a highly significant t -value of -5.71 . This implies that real wages actively adjust to correct deviations from the long-run path. Employment (Δe) also shows significant adjustment ($t = -2.16$) at the 5% level.
 - $prod$ and U): The adjustment coefficients for productivity ($\Delta prod$, $t = -0.92$) and unemployment (ΔU , $t = -1.49$) are not statistically different from zero. This suggests that these variables are weakly exogenous with respect to the long-run parameters. In other words, shocks to productivity and unemployment drive the system's long-run movements, but these variables do not correct themselves to restore equilibrium.
3. Short-run dynamics: The regression output summary (not fully detailed here for brevity) shows the short-run dynamics via the lagged differenced terms (Λ_j). Notably:
- The first lag of employment (Δe_{t-1}) is a significant predictor for current unemployment changes (ΔU_t), with a strong negative coefficient (-0.65 , $t = -5.09$).
 - Productivity growth ($\Delta prod_t$) exhibits some autoregressive behavior, as indicated by the significance of its own lags.

12 Big Dependent Data

12.1 Background

So far, we have examined standard VAR model (and its VECM counterpart). In practice, the number of variables in these models must be kept relatively small to maintain estimation accuracy and efficiency. Models with more than, say, six variables are rare. However, especially in many economic applications, this dimensionality constraint presents significant challenges.

Omitted variable bias. In applications where the underlying dynamics are driven by a large information set, restricting the model to a handful of variables creates a substantial risk of omitted variable bias.

- For example, central banks monitor hundreds of variables when conducting monetary policy. Excluding this information from a model can lead to distorted results.
- A classic example is the “price puzzle” observed in early VAR-based monetary policy literature, where a contractionary monetary policy shock appeared to increase the price level. This counter-intuitive result is widely attributed to omitted variable bias: if the model does not include forward-looking inflation expectations (which the central bank observes), it misinterprets a policy rate hike—enacted in anticipation of future inflation—as the cause of the subsequent price rise.

Unobserved variables. Economic theory or other background information (or equivalent results in other application fields) often features variables that lack precise empirical counterparts or are measured with error.

- For instance, “inflation” is a theoretical concept representing the decline in the value of money. In practice, we observe proxies like the Consumer Price Index (CPI) or core inflation (excluding volatile energy and food prices).
- It is often unclear which specific proxy best represents the underlying theoretical variable. A **factor model approach** allows us to extract the common signal (the underlying inflation) from a large panel of different price indicators.

Modelling specific dynamics. Standard VARs focus on varying a few key aggregates. However, researchers are often interested in the dynamic reaction of specific, disaggregated variables (e.g., sector-specific prices or regional housing starts). Including all these variables in a standard VAR is computationally infeasible.

The FAVAR solution. One solution to these dimensionality problems, within the VAR context, is the **Factor Augmented Vector Autoregression (FAVAR)** model. The core idea is to utilize a large dataset (where even the number of variables M may exceed the number of observations T) to construct a small number of **common factors** via **Principal Component Analysis (PCA)**, and then include these factors in the VAR model.

The FAVAR approach offers distinct advantages:

1. **Dimension reduction:** It condenses the information from hundreds of variables into a few factors, keeping the parameter space manageable.
2. **Proxies for unobservables:** Factors can act as better proxies for theoretical concepts (like “economic activity” or “global financial cycle”) than any single observed variable.
3. **Broader analysis:** It allows for the analysis of shocks (in structural VARs) on a vast set of variables that are not explicitly in the VAR system but are part of the large dataset used to extract factors.

In the following sections, we will focus first on extracting factors from large datasets and second on utilizing them in the VAR analysis. This discussion draws heavily on Chapter 16 of Kilian and Lütkepohl (2017).

Note: The notation in this chapter differs slightly from previous sections. Specifically, matrices and dimensions here relate to large-scale factor methods and should not be confused, for example, with notation in the previous cointegration and VECM section.

12.2 Factor models

Static factor model

Let $\mathbf{x}_t$ be an $M \times 1$ vector of observed, demeaned variables. We assume these variables are driven by a small number of unobserved common factors $\mathbf{f}_t$ ($r \times 1$) and idiosyncratic errors. The **static factor model** is defined as:

$$\mathbf{x}_t = \mathbf{\Lambda} \mathbf{f}_t + \mathbf{v}_t,$$

where

- $\mathbf{\Lambda}$ is the $(M \times r)$ matrix of **factor loadings**.
- $\mathbf{v}_t \sim \text{iid}(\mathbf{0}, \mathbf{\Sigma}_v)$ is the vector of **idiosyncratic errors**.

We assume that the common factors and idiosyncratic errors are uncorrelated, i.e., $E(\mathbf{v}_s \mathbf{f}_t') = \mathbf{0}$ for all s, t . Typically, we assume $M \gg r$. The covariance matrix of $\mathbf{x}_t$ implies:

$$\Sigma_x = \Lambda \Sigma_f \Lambda' + \Sigma_v.$$

This model is not uniquely identified without further restrictions. Common identification assumptions include:

1. Orthogonal loadings: $\Lambda' \Lambda = \mathbf{I}_r$.
2. Uncorrelated, standardized factors: $\mathbf{f}_t \sim (\mathbf{0}, \mathbf{I}_r)$, implying Σ_f is a diagonal matrix.

Estimation via PCA

If Λ were known, we could estimate the factors easily. In practice, we estimate the factors by maximizing the variance of $\mathbf{x}_t$ explained by the common component (equivalently, minimizing the variance of the idiosyncratic errors). The objective function is:

$$\min_{\Lambda, \mathbf{f}_1, \dots, \mathbf{f}_T} \frac{1}{T} \sum_{t=1}^T (\mathbf{x}_t - \Lambda \mathbf{f}_t)' (\mathbf{x}_t - \Lambda \mathbf{f}_t).$$

The solution is found via **Principal Component Analysis (PCA)**:

1. Find the r largest eigenvalues $\lambda_1 > \dots > \lambda_r$ of the sample covariance matrix $\mathbf{S}_x = \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t'$ and their corresponding (orthonormal) eigenvectors $\bar{\lambda}_1, \dots, \bar{\lambda}_r$.
2. Set the estimated loading matrix is $\widehat{\Lambda} = [\bar{\lambda}_1, \dots, \bar{\lambda}_r]$.
3. The estimated factors are $\widehat{\mathbf{f}}_t = \widehat{\Lambda}' \mathbf{x}_t$.

The eigenvalues λ_i represent the empirical variance explained by the i -th factor, ordered from largest to smallest.

Dynamic factor model

The static model assumes factors and errors are serially uncorrelated. The **Dynamic Factor Model (DFM)** generalizes this by allowing for autoregressive dynamics:

$$\mathbf{x}_t = \Lambda^f \mathbf{f}_t + \mathbf{v}_t,$$

where the factors and errors follow time-series processes:

$$\begin{aligned} \mathbf{f}_t &= \Gamma_1 \mathbf{f}_{t-1} + \dots + \Gamma_s \mathbf{f}_{t-s} + \boldsymbol{\eta}_t, \\ \mathbf{v}_t &= \mathbf{A}_1 \mathbf{v}_{t-1} + \dots + \mathbf{A}_p \mathbf{v}_{t-p} + \mathbf{u}_t. \end{aligned}$$

Here $\mathbf{A}_i$ are typically diagonal matrices, and $\mathbf{u}_t$ is white noise (or iid) error term.

Note: We assume $\mathbf{x}_t$ is stationary. While factor models exist for integrated processes, the methods presented here assume stationarity.

Dynamic form. If $\mathbf{x}_t$ depends directly on lagged values of the factors, we obtain the **dynamic form** of the model:

$$\mathbf{x}_t = \Lambda_0^f \mathbf{f}_t + \Lambda_1^f \mathbf{f}_{t-1} + \cdots + \Lambda_{q^*}^f \mathbf{f}_{t-q^*} + \mathbf{v}_t.$$

Assuming the same DGP for $\mathbf{f}_t$ and $\mathbf{v}_t$ as in the static form (as above) of the DFM, this model can be written in lag operator notation as

$$\mathbf{x}_t = \Lambda^f(\mathbf{B}) \mathbf{f}_t + \mathbf{v}_t, \quad \Gamma(\mathbf{B}) \mathbf{f}_t = \boldsymbol{\eta}_t, \quad \mathbf{A}(\mathbf{B}) \mathbf{v}_t = \mathbf{u}_t$$

where

$$\begin{aligned} \Lambda^f(\mathbf{B}) &= \Lambda_0^f + \Lambda_1^f \mathbf{B} + \cdots + \Lambda_{q^*}^f \mathbf{B}^{q^*}, \\ \Gamma(\mathbf{B}) &= \mathbf{I}_r - \Gamma_1 \mathbf{B} - \cdots - \Gamma_s \mathbf{B}^s, \\ \mathbf{A}(\mathbf{B}) &= \text{diag}[a_1(\mathbf{B}) \cdots a_M(\mathbf{B})], \end{aligned}$$

where $\mathbf{v}_t$ is the vector of idiosyncratic components, and $\boldsymbol{\eta}_t$ is white noise or iid error term such that $\mathbf{E}(\mathbf{u}_t \boldsymbol{\eta}_s') = \mathbf{0} \forall t, s$.

Interestingly, this dynamic form can be rewritten in a **static representation** by stacking the current and lagged factors into a larger vector $\mathbf{F}_t = (\mathbf{f}_t', \dots, \mathbf{f}_{t-q^*}')'$. The observation equation becomes

$$\mathbf{x}_t = \Lambda^F \mathbf{F}_t + \mathbf{v}_t,$$

where $\Lambda^F = \begin{bmatrix} \Lambda_0^f & \Lambda_1^f & \cdots & \Lambda_{q^*}^f \end{bmatrix}$. This allows us to estimate the dynamic model using the PCA approach described for the static model.

Estimation strategy. Recall that the model can be rewritten as

$$\mathbf{A}(\mathbf{B}) \mathbf{x}_t = \mathbf{A}(\mathbf{B}) \Lambda^f(\mathbf{B}) \mathbf{f}_t + \mathbf{u}_t.$$

A recommended iterative procedure for estimating the DFM (see Stock and Watson, 2005):

1. Initialize: Estimate univariate AR models for each component of $\mathbf{x}_t$ to get initial estimates $\tilde{\mathbf{A}}(\mathbf{B})$.
2. PCA step: Estimate factors $\hat{\mathbf{F}}_t$ from the filtered model $\tilde{\mathbf{A}}(\mathbf{B}) \mathbf{x}_t$ using PCA.
3. Regression step: Estimate the factor loadings and error dynamics by regressing $\mathbf{x}_t$ on the estimated factors.

4. Iterate: Repeat steps 2 and 3 until convergence.

Choosing the number of factors. Since PCA does not rely on the time structure, a common heuristic for selecting r (the number of factors) is the **scree plot** approach or a variance threshold. One specifies a target fraction of total variance (e.g., 50% or 90%) and selects the minimum number of principal components required to explain that cumulative variance.

Asymptotic convergence. Standard theory suggests estimation improves as $M \rightarrow \infty$. However, Boivin and Ng (2006) highlight that simply adding more data is not always beneficial if the new variables have correlated errors or weak factor loadings. Quality matters more than quantity; including noisy or irrelevant variables can degrade the precision of factor estimates.

12.3 FAVAR Model

We can now formally define the FAVAR model to address the omitted variable bias. Assume the “true” process (such as economy) is driven by both observable variables $\mathbf{y}_t$ ($K \times 1$) and unobserved factors $\mathbf{f}_t$ ($r \times 1$). The joint dynamics are given by a demeaned(!) VAR:

$$\mathbf{z}_t = \begin{pmatrix} \mathbf{f}_t \\ \mathbf{y}_t \end{pmatrix} = \sum_{j=1}^p \mathbf{A}_j^* \begin{pmatrix} \mathbf{f}_{t-j} \\ \mathbf{y}_{t-j} \end{pmatrix} + \tilde{\mathbf{u}}_t,$$

where the coefficient matrix is partitioned as:

$$\mathbf{A}_j^* = \begin{pmatrix} \mathbf{A}_{j,r \times r}^* & \mathbf{A}_{j,r \times K}^* \\ \mathbf{A}_{j,K \times r}^* & \mathbf{A}_{j,K \times K}^* \end{pmatrix}.$$

If we were to estimate a standard VAR using only $\mathbf{y}_t$, we would effectively be restricting $\mathbf{A}_{j,K \times r}^* = \mathbf{0}$. If the factors indeed drive $\mathbf{y}_t$, this restriction is invalid and leads to the omitted variable bias.

Estimation and inference. The FAVAR approach proceeds in two steps:

1. **Factor extraction:** Estimate the factors $\hat{\mathbf{f}}_t$ from a large dataset $\mathbf{x}_t$ using, for example, the PCA methods described above.
2. **VAR estimation:** Treat the estimated factors as observed variables and estimate the augmented VAR:

$$\begin{pmatrix} \hat{\mathbf{f}}_t \\ \mathbf{y}_t \end{pmatrix} = \sum_{j=1}^p \mathbf{A}_j \begin{pmatrix} \hat{\mathbf{f}}_{t-j} \\ \mathbf{y}_{t-j} \end{pmatrix} + \mathbf{u}_t.$$

Notes on inference: Because $\hat{\mathbf{f}}_t$ are estimated regressors (“generated regressors”) rather than true observations, standard asymptotic standard errors for impulse responses are incorrect. In practice, bootstrap methods are used to compute valid confidence intervals for FAVAR analysis.

13 Predictive regressions with time series variables

In this section, we focus on predicting (univariate) y_{t+h} at the time t (forecast origin) given the collection of K predictive variables in $\mathbf{x}_t = (x_{1t}, \dots, x_{Kt})$, which may also contain some lagged values of y_t . The general predictive regression model is given by:

$$y_{t+h} = \mathbf{x}_t' \boldsymbol{\beta} + u_{t+h} = \sum_{i=1}^K x_{it} \beta_i + u_{t+h}, \quad t = 1, \dots, T-h.$$

where the forecast horizon h is given and fixed, $\boldsymbol{\beta}$ is the vector of parameters and u_{t+h} is the error term with typical assumptions (such as an iid error term).

- Notice that in this section K is the number of predictive variables to predict univariate time series y_t . This holds also for the introductory machine learning section (next subsection).
- In other words, we do not consider VAR-type of cases in this section.

Standard assumption: Stationarity. **Unless otherwise specified**, in the general predictive regression framework, we assume that the variables included in $\mathbf{x}_t$ and the dependent variable y_t are **stationary**.

- The variables can be (relatively) strongly persistent (i.e., even relatively strong serial correlation), but they are assumed not to be integrated processes (i.e., not $I(1)$ processes).
- Under these conditions, standard asymptotic theory generally applies.

We consider two major deviations from this standard framework below

- **Section 13.1** considers the usual low-dimensional case where some variables are integrated ($I(1)$).
- **Section 13.2** considers the case of “Big Dependent Data” where the number of predictors K is large, but the dependent variable is still univariate.

13.1 Linear regressions containing I(1) variables

Consider the linear regression

$$y_t = \mathbf{x}_t' \boldsymbol{\beta} + u_t,$$

and the two assumptions (i)–(ii):

- (i) The error term u_t is serially uncorrelated and uncorrelated with the regressors included in $\mathbf{x}_t$.
- (ii) All the regressors in $\mathbf{x}_t$ are either deterministic or stationary random variables.

Assume now that only (i) holds, that is the error term u_t is serially uncorrelated and uncorrelated with the regressors included in $\mathbf{x}_t$.

- Even if the model cannot be written in this way, the OLS estimator of the coefficients of the $I(1)$ regressors $\mathbf{x}_t$ is consistent. However, its asymptotic distribution is, in general, nonstandard such that usual inference does not apply.
- If the model can be written such that all the parameters of interest are coefficients of mean zero stationary variables, their OLS estimator is consistent and asymptotically normal.

To examine these points in more detail, let us consider again the following relatively simple regression model

$$y_t = \beta_0 + \beta_1 x_{t-1} + \beta_2 x_{t-2} + u_t,$$

where $x_t \sim I(1)$. The model can be rewritten as

$$y_t = \beta_0 + (\beta_1 + \beta_2)x_{t-1} - \beta_2(x_{t-1} - x_{t-2}) + u_t,$$

or

$$y_t = \beta_0 + \beta_1(x_{t-1} - x_{t-2}) + (\beta_1 + \beta_2)x_{t-2} + u_t.$$

As $(x_{t-1} - x_{t-2}) \sim I(0)$, and hence, standard inference on β_2 (or β_1) holds. Therefore, both β_1 and β_2 cannot simultaneously be written as coefficients of $I(0)$ variables (unless higher lags are included in the model).

- The OLS estimator of β_1 and β_2 is not, in general, jointly asymptotically normal.
- The test statistic on a hypothesis concerning both coefficients (for instance $H_0 : \beta_1 = \beta_2$), in general, does not have the usual asymptotic χ^2 -distribution.

Let us continue with the above linear regression where now also the assumption (i) does not hold and the dependent variable is $I(1)$.

- If the dependent variable is not cointegrated with any of the regressors, the OLS estimator of the coefficients of the $I(1)$ regressors is inconsistent.

- If the dependent variable is cointegrated with at least one of the regressors, the OLS estimator of the parameters of interest
 - that can be written as coefficients of stationary variables, is inconsistent.
 - that cannot be written as coefficients of stationary variables, is consistent, but not asymptotically normal.

For instance, consider the following regression model:

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + u_t,$$

where all variables are $I(1)$, and y_t and x_{2t} are cointegrated and assumption (i) does not hold.

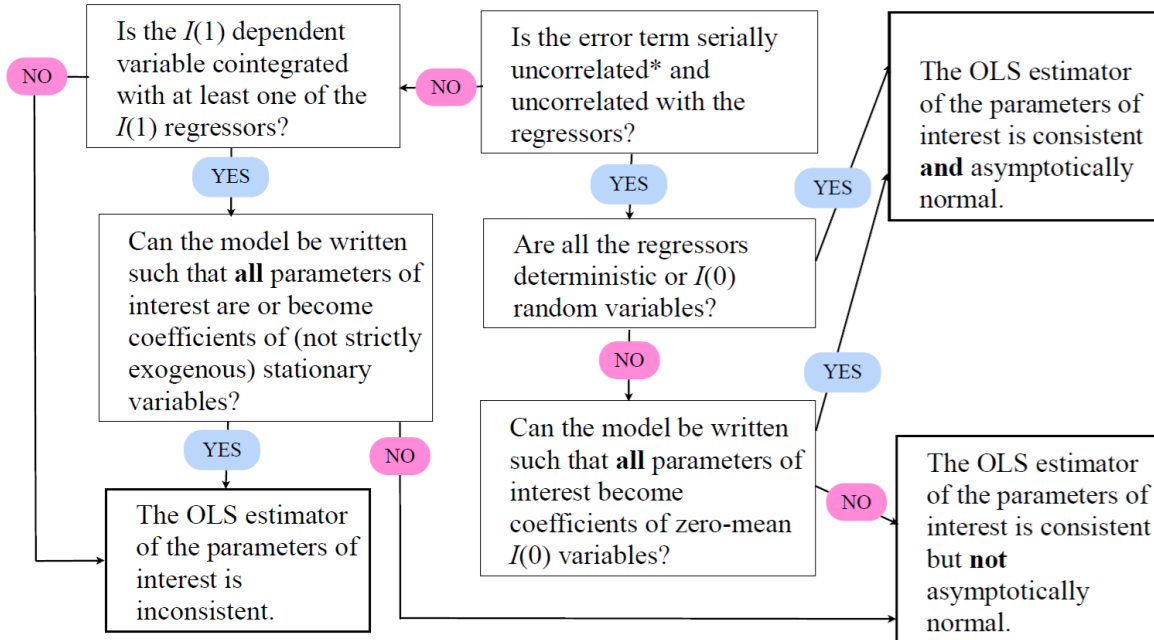
- If x_{1t} and x_{2t} are cointegrated such that $(x_{1t} - \gamma x_{2t}) \sim I(0)$, the model can be written as

$$y_t = \beta_0 + \beta_1(x_{1t} - \gamma x_{2t}) + (\beta_1\gamma + \beta_2)x_{2t} + u_t,$$

and the OLS estimator of β_1 is inconsistent.

- If x_{1t} is not cointegrated with x_{2t} , β_1 cannot be written as a coefficient of an $I(0)$ variable, and hence, its OLS estimator is consistent, but not asymptotically normal.

The complete *cheat sheet* on linear regression with $I(0)$ and $I(1)$ variables is summarized hereby:



* Not autocorrelated after $H-1$ lags.

Figure: Complete cheat sheet on inference in OLS regression models with $I(1)$ variables.

A spurious regression emerges when a statistically significant relationship between two $I(1)$ variables y_t and x_t is found although those are completely unrelated. Assume that a researcher estimates a linear regression

$$y_t = \beta_0 + \beta_1 x_t + u_t,$$

but assuming wrongly that the variables y_t and x_t are stationary. Instead, they are generated by two independent random walks (i.e., y_t and x_t are $I(1)$ variables)

$$\begin{aligned} y_t &= y_{t-1} + u_{1t}, & u_{1t} &\sim \text{iid}(0, \sigma_1^2), \\ x_t &= x_{t-1} + u_{2t}, & u_{2t} &\sim \text{iid}(0, \sigma_2^2). \end{aligned}$$

This typically leads to the estimation result of the above linear regression characterized by a fairly high R^2 (the coefficient of determination), highly autocorrelated residuals $\hat{u}_t$ and a (highly) statistically significant estimate of β_1 . This result is clearly spurious given that the variables are completely independent!

- Estimation results like this should not be taken seriously.
- The reason is that with y_t and x_t being $I(1)$ variables, the error term u_t will also be $I(1)$, not stationary ($I(0)$).

Solution to this problem is to include the lags of y_t and x_t as predictors in the model. In other words, we end up a model

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 y_{t-1} + \beta_3 x_{t-1} + u_t.$$

In this case the OLS estimator is consistent. Thus, in general, including lagged values in the regression is sufficient to solve many of the problems associated with possibly spurious regressions.

13.2 Forecasting with big dependent data

In this section, let us focus on predicting (univariate) y_{t+h} at time t (forecast origin) given the collection of K predictive variables in $\mathbf{x}_t = (x_{1t}, \dots, x_{Kt})$, which may also contain some lagged values of y_t (as discussed above).

- This section strongly follows (notation and presentation) Chapter 7 of Peña and Tsay (2021).

In this section, we allow the possibility that the vector $\mathbf{x}_t$ can be high-dimensional, that is we have a large set of potential predictive variables. This means deviations from the traditional econometric methods as we focus on possibly **big dependent data** emphasizing the case where K , the number of predictors, is large, even larger than the sample length T .

- In this type of data-rich environment, several interesting insights have been developed in recent times.
- In this course, we concentrate only on a brief introduction to **regularized estimators**, specifically **ridge**, **lasso** and **elastic net**, for such predictive regressions where the predictor space can be (very) large.

It is important to note that most modern statistical and machine learning methods introduced in this section were developed for independent (cross-sectional) data. Consequently, they may not be directly or fully applicable to time series without appropriate handling, even though they are frequently used in this manner.

- This research area, focusing on “big dependent data,” is still very much evolving.
- We will discuss the justifications for and modifications to these methods for time series applications where relevant.
- In short, overlooking serial dependence and high multicollinearity can lead to erroneous inferences. For data with short-range dependence, where serial correlation decays quickly (e.g., exponentially) and the series is not highly persistent, tools designed for independent data can often be used with minimal adjustment. However, in the presence of strong serial dependence (highly persistent time series), more significant complications should be expected

We concentrate on the regularized estimation of the following predictive regression model introduced at the beginning of this chapter:

$$y_{t+h} = \mathbf{x}'_t \boldsymbol{\beta} + e_{t+h} = \sum_{i=1}^K x_{it} \beta_i + e_{t+h}, \quad t = 1, \dots, T-h,$$

with a given and fixed forecast horizon h , $\boldsymbol{\beta}$ is the vector of parameters and e_t is the error term with typical assumptions (such as white noise or iid error term)).

- For simplicity, and importantly, in notation it is understood that $\mathbf{x}_t$ may contain the constant term. In the forthcoming shrinkage-based estimators the handling of constant is somewhat different than in the conventional models.
- In these regularized estimations and their applications (that is, in this section), each predictor x_{it} is mean-adjusted and standardized to have unit variance. This assumption is quite natural for independent data, but for time series may encounter some additional difficulties.

The above predictive regression can be rewritten using vectors and matrices as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{E},$$

where the vectors $\mathbf{Y}$ and $\mathbf{E}$ and the matrix $\mathbf{X}$ have suitable dimensions

$$\mathbf{Y} = \begin{bmatrix} y_{1+h} \\ y_{2+h} \\ \vdots \\ y_T \end{bmatrix} \quad (T-h \times 1), \quad \mathbf{X} = \begin{bmatrix} \mathbf{x}'_1 \\ \mathbf{x}'_2 \\ \vdots \\ \mathbf{x}'_{T-h} \end{bmatrix} \quad (T-h \times K) \quad \text{and} \quad \mathbf{E} = \begin{bmatrix} e_{1+h} \\ e_{2+h} \\ \vdots \\ e_T \end{bmatrix} \quad (T-h \times 1).$$

As discussed above, under traditional linear regression analysis, ordinary least squares (OLS) can be used to estimate the parameters when K is sufficiently (much) smaller than the number of observations T :

$$\hat{\boldsymbol{\beta}}_{OLS} = \left(\sum_{t=1}^{T-h} \mathbf{x}_t \mathbf{x}'_t \right)^{-1} \sum_{t=1}^{T-h} \mathbf{x}_t y_{t+h}.$$

- The sum may also run from 1 to T if the necessary h initial values with negative time indices are available

However, if $K \geq T-h$, then the OLS estimator is not unique and will overfit the data. This emphasizes the need for **regularization** and **shrinkage-based estimators**.

One possible approach and algorithm to obtain a regularized estimator is to build upon **lasso estimator**

$$\hat{\boldsymbol{\beta}}_L(\lambda) = \arg \min_{\boldsymbol{\beta}} \left(\frac{\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2^2}{T-h} + \lambda \|\boldsymbol{\beta}\|_1 \right),$$

where $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 = \sum_{t=1}^{T-h} (y_{t+h} - \mathbf{x}'_t \boldsymbol{\beta})^2$ and $\|\boldsymbol{\beta}\|_1 = \sum_{i=1}^K |\beta_i|$.

- Here $\|\mathbf{z}\|_2$ refers to Euclidean norm l_2 , that is for the general vector $\mathbf{z} = (z_1, \dots, z_K)$ is $\|\mathbf{z}\|_2 = \sqrt{\sum_{i=1}^K z_i^2}$. Furthermore, l_1 -norm is defined as $\|\mathbf{z}\|_1 = \sum_{i=1}^K |z_i|$.
- Note that, for convenience, in the Lasso estimator (in general in lasso literature) the predictors are often standardized so that $\sum_{t=1}^{T-h} x_{it} = 0$ and $\sum_{t=1}^{T-h} x_{it}^2 = 1$ for $i = 1, \dots, K$. The dependent variable is mean-adjusted, i.e. $\frac{1}{T-h} \sum_{t=1}^{T-h} y_{t+h} = 0$.

Lasso uses l_1 -penalty (i.e. absolute values of the coefficients).

- It can be shown that this **enables variable selection**. That is, some resulting estimates $\hat{\beta}_i$ will be exactly zero and hence excluded from the final model.
- The fact that only a few coefficients β_i are nonzero implies the **sparsity** of the resulting model (model is **sparse** in the sense that some predictors are excluded).

An alternative regularization-based estimator is the well-known **ridge estimator** (ridge regression). Ridge estimator is obtained when l_2 -penalty is used

$$\widehat{\beta}_R(\lambda) = \arg \min_{\beta} \left(\frac{\|\mathbf{Y} - \mathbf{X}\beta\|_2^2}{T - h} + \lambda \|\beta\|_2^2 \right).$$

In other words, the difference to the lasso estimator is the penalty term. The use of l_2 -penalty means that the ridge estimator provides resulting estimates which are **dense** that is there won't be exactly zero estimates in the same way as with lasso estimator.

An important part of using regularized estimators is the determination of the value of the penalty term via parameter λ . A small penalty might result in overfitting, whereas a large penalty would lead stronger shrinkage in coefficients β_i towards 0.

- An ultimate case is of course $\lambda = \infty$, which would yield (or closeby in the ridge estimator) $\beta_i = 0$ for all i .
- In contrast, if $\lambda = 0$ implies no penalty at all, which may result in severe multicollinearity problems as the resulting ordinary least squares estimator is not necessarily even applicable in such circumstances.
- An interesting case is also obtained when λ is close to zero but nonzero. This leads so called **overparametrized models**.

For independent (cross-sectional) data, **cross-validation (CV)** is typically used to determine the value of penalty parameter λ .

- For time series data, it is somewhat questionable to use CV due to its construction (observations are not independent as in cross-sectional datasets) as seen below.
- Despite this it seems that CV is also quite often used for time series and research in this respect is still ongoing.

One and typical implementation of CV, as introduced in Peña and Tsay (2021, Chapter 7.1), is the following.

- Divide the dataset $S = \{(y_{t+h}, \mathbf{x}_t) | t = 1, \dots, T - h\}$ randomly into, say, 10 subsamples, say $S_1, \dots, S_{10}$.
- Let $S_{(i)}$ be all the data but the subsample S_i . That is $S_{(i)} = \cup_{j \neq i}^{10} S_j$ (i.e. the subsample S_i is excluded). Let $\{\lambda_j | j = 1, \dots, m\}$ be the set of possible values of λ and let $\text{SSE}(\lambda_j)$ be the sum of squared prediction errors defined below for parameter λ_j .

The CV procedure (as in Peña and Tsay (2021, Chapter 7.1) is carried out as follows:

1. For each λ_j ($j = 1, \dots, m$)
 - (a) Initialize $\text{SSE}(\lambda_j) = 0$
 - (b) For $i = 1, \dots, 10$
 - Fit the Lasso, or Ridge, model with penalty λ_j using data $S_{(i)}$
 - Predict the response using the fitted parameters and the predictors in S_i
 - Compute the prediction errors for the subsample S_i
 - Update $\text{SSE}(\lambda_j)$ by adding the sum of squared prediction errors of subsample S_i .
2. Select λ by $\lambda = \arg \min_{\lambda_j} \text{SSE}(\lambda_j)$

This type of 10-fold CV procedure is commonly used in most software packages. However, as already discussed, the validity of such CV is questionable when the data are serially dependent. The above random division to subsamples does not respect serial correlation between observations and can invalidate results for time series data.

- Following, e.g., Peña and Tsay (2021), one alternative approach to CV is to select λ using some information criteria such as BIC or AIC.

Finally, the use of a combination of the l_1 and l_2 penalties leads to **elastic net** (EN) estimator

$$\widehat{\boldsymbol{\beta}}_{EL}(\alpha, \lambda_1, \lambda_2) = \arg \min_{\boldsymbol{\beta}} \left(\frac{\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2^2}{T - h} + \lambda_1 \alpha \|\boldsymbol{\beta}\|_1 + \lambda_2 (1 - \alpha) \|\boldsymbol{\beta}\|_2^2 \right),$$

where $0 \leq \alpha \leq 1$. The elastic net estimator reduces to the lasso estimator when $\alpha = 1$ and to the ridge estimator if $\alpha = 0$. In this formulation of the elastic net, two penalties λ_1 and λ_2 are used, resulting in the need to select two tuning parameters. If we do not prespecify α , one possible way forward is to consider a few possible values of α and compare their predictive performance.

Appendix

Appendix A: Matrix calculus

Matrix Norm

For a matrix $\mathbf{A} = [a_{ij}]$ of dimension $n \times m$ the Euclidean norm is

$$\|\mathbf{A}\| = \sqrt{\sum_{i=1}^n \sum_{j=1}^m a_{ij}^2}$$

The same norm can be expressed by applying the standard Euclidean norm to the $nm \times 1$ vector whose elements constitute the elements of $\mathbf{A}$. Define the vectorization operator $\text{Vec}(\cdot)$ as follows. Partition $\mathbf{A} = [\mathbf{a}_1 \cdots \mathbf{a}_m]$ by columns so that $\mathbf{a}_i$ ($n \times 1$) is the i :th column of $\mathbf{A}$ and define the vector (the vectorized $\mathbf{A}$) as

$$\text{Vec}(\mathbf{A}) = \begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_m \end{bmatrix} \quad (nm \times 1)$$

Then it is clear that $\|\mathbf{A}\|$ is the norm of $\text{Vec}(\mathbf{A})$, that is,

$$\|\mathbf{A}\| = \sqrt{\text{vec}(\mathbf{A})' \text{vec}(\mathbf{A})}$$

The latter definition and known properties of a vector norm enable us to see that the matrix norm has the same properties as the vector norm

- $\|\mathbf{A}\| \geq 0$ and $\|\mathbf{A}\| = 0$ if and only if $\mathbf{A} = \mathbf{0}$
- $\|c\mathbf{A}\| = |c| \|\mathbf{A}\|$ for all $c \in \mathbb{R}$
- $\|\mathbf{A} + \mathbf{B}\| \leq \|\mathbf{A}\| + \|\mathbf{B}\|$ for all $m \times n$ matrices $\mathbf{B}$ (triangular inequality)
- $\|\mathbf{AB}\| \leq \|\mathbf{A}\| \|\mathbf{B}\|$ (submultiplicativity)

A variety of other matrix norms are applied that satisfy the above four conditions (sometimes only the first three ones), but these norms are not needed in this course.

Sometimes it is useful to express the Euclidean norm by applying the trace of a square matrix, i.e., the sum of diagonal elements. If $\mathbf{B} = [b_{ij}]$ is $n \times n$ matrix, then its trace $\text{tr}(\mathbf{B})$ is defined by

$$\text{tr}(\mathbf{B}) = \sum_{i=1}^n b_{ii}$$

It is easy to see from the definition that

$$\text{tr}(\mathbf{B} + \mathbf{C}) = \text{tr}(\mathbf{B}) + \text{tr}(\mathbf{C}) \text{ and } \text{tr}(\mathbf{BD}) = \text{tr}(\mathbf{DB})$$

where $\mathbf{C}$ is $n \times n$ matrix, while in the latter condition $\mathbf{B}$ can be $n \times m$ and $\mathbf{D}$ can be $m \times n$.

The norm of the matrix $\mathbf{A}$ ($n \times m$) can be written

$$\|\mathbf{A}\| = \sqrt{\text{tr}(\mathbf{A}'\mathbf{A})}$$

Appendix ??Matrixque) shows how the matrix norm can be used to analyze the convergence of matrix sequences and series.

Matrix Eigenvalues and Jordan Decomposition

The eigenvalues $\lambda_1, \dots, \lambda_n$ of a square matrix $\mathbf{A}$ ($n \times n$) satisfy (are solutions to) the determinant equation

$$\det(\mathbf{A} - \lambda \mathbf{I}_n) = 0$$

The corresponding eigenvectors $\mathbf{u}_1, \dots, \mathbf{u}_n$ ($\mathbf{u}_i \neq \mathbf{0}$) are obtained from the equations

$$(\mathbf{A} - \lambda_i \mathbf{I}_n) \mathbf{u}_i = \mathbf{0} \iff \mathbf{A} \mathbf{u}_i = \lambda_i \mathbf{u}_i, \quad i = 1, \dots, n$$

If $\mathbf{A}$ ($n \times n$) and $\lambda_1, \dots, \lambda_n$ are as above, there exists a non-singular matrix $\mathbf{P}$ ($n \times n$) such that

$$\mathbf{A} = \mathbf{P} \mathbf{\Lambda} \mathbf{P}^{-1}$$

Here $\mathbf{\Lambda}$ ($n \times n$) is the block diagonal matrix

$$\mathbf{\Lambda} = \begin{bmatrix} \mathbf{\Lambda}_{n_1} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{\Lambda}_{n_2} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{\Lambda}_{n_k} \end{bmatrix} \equiv \text{diag}[\mathbf{\Lambda}_{n_1} \quad \dots \quad \mathbf{\Lambda}_{n_k}]$$

where $n_1 + \dots + n_k = n$ and

$$\mathbf{\Lambda}_{n_i} = \begin{bmatrix} \lambda_i & 1 & 0 & \dots & \dots & 0 \\ 0 & \lambda_i & 1 & & & 0 \\ \vdots & & \ddots & & & \vdots \\ \vdots & & & \ddots & & 0 \\ \vdots & & & & 0 & \lambda_i & 1 \\ 0 & \dots & \dots & \dots & 0 & \lambda_i \end{bmatrix}$$

The cells on the main diagonal of $\mathbf{\Lambda}_{n_i}$ contain the eigenvalue λ_i , the cells of the diagonal right above the main diagonal contain 1 and all other cells contain 0 (the same eigenvalue

may appear in several blocks of $\mathbf{\Lambda}$). This representation of matrix $\mathbf{A}$ is called the *Jordan canonical form* or the *Jordan decomposition*. If the eigenvalues of $\mathbf{A}$ are distinct, the matrix $\mathbf{\Lambda}$ is diagonal and its diagonal elements are the eigenvalues of $\mathbf{A}$, while the columns of $\mathbf{P}$ (that are linearly independent) are the eigenvectors of $\mathbf{A}$.

By computing we can show the following useful result (where $\binom{k}{j} = 0$, if $j > k$)

$$\mathbf{\Lambda}_{n_i}^k = \begin{bmatrix} \lambda_i^k & \binom{k}{1}\lambda_i^{k-1} & \cdots & \binom{k}{j}\lambda_i^{k-j} & \cdots & \binom{k}{n_i-1}\lambda_i^{k-n_i+1} \\ 0 & \ddots & \ddots & & \ddots & \\ \vdots & 0 & \ddots & \ddots & & \ddots \\ \vdots & & 0 & \ddots & \ddots & \\ \vdots & & & 0 & \ddots & \ddots \\ 0 & \cdots & \cdots & \cdots & 0 & \ddots \end{bmatrix},$$

where each diagonal carries the same element (and all elements below the main diagonal are zeros).

Matrix Sequences and Series

The convergence of a matrix sequence $\mathbf{A}_N = [a_{ij,N}]$, $N = 1, 2, \dots$ to the matrix $\mathbf{A} = [a_{ij}]$ (of the same dimension) may be defined to require that $a_{ij,N} \rightarrow a_{ij}$ for all i, j , as $N \rightarrow \infty$. This is denoted by $\lim_{N \rightarrow \infty} \mathbf{A}_N = \mathbf{A}$ or $\mathbf{A}_N \rightarrow \mathbf{A}$ (as $N \rightarrow \infty$).

The convergence of a matrix series $\mathbf{S} = \sum_{i=1}^{\infty} \mathbf{A}_i$ is defined as the limit of the sequence of partial sums $\mathbf{S}_N = \sum_{i=1}^N \mathbf{A}_i$, that is, there exists a matrix $\mathbf{S}$ such that $\mathbf{S}_N \rightarrow \mathbf{S}$, which implies that $\mathbf{S} = \sum_{i=1}^{\infty} \mathbf{A}_i$.

If $\mathbf{A} = [a_{ij}]$ is $n \times m$ matrix, then the definition of the matrix norm implies $\max\{|a_{ij}|\} \leq \|\mathbf{A}\| \leq \sqrt{nm} \max\{|a_{ij}|\}$, where maximization is over $i = 1, \dots, n$ and $j = 1, \dots, m$. Using this together with the definition of convergence of real sequences we observe that for the matrix sequence $\mathbf{A}_N$ it holds that $\mathbf{A}_N \rightarrow \mathbf{A}$ if and only if $\|\mathbf{A}_N - \mathbf{A}\| \rightarrow 0$. An analogous alternative definition can also be presented for matrix series. Given the above notes and well known results on the convergence of real valued series it is clear that a sufficient condition for a matrix series to converge is that the real valued series $\sum_{i=1}^{\infty} \|\mathbf{A}_i\|$ converges (absolute convergence).

Let $\mathbf{A}$ be an $n \times n$ square matrix and $\lambda_1, \dots, \lambda_n$ its eigenvalues. Assume $\rho \equiv \max_{1 \leq i \leq n} |\lambda_i| < 1$. By applying the Jordan decomposition, we obtain

$$\mathbf{A}^N = (\mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1})^N = \mathbf{P}\mathbf{\Lambda}^N\mathbf{P}^{-1}$$

Because $N^k \rho^N \rightarrow 0$, as $N \rightarrow \infty$, for all $k \geq 0$, we obtain using results from Section A.2 that $\mathbf{\Lambda}^N \rightarrow \mathbf{0}$ and hence, given the above equation, we have

$$\mathbf{A}^N \rightarrow \mathbf{0}, \text{ as } N \rightarrow \infty$$

Choose $0 < \bar{\lambda} < 1$ such that $\rho < \bar{\lambda} < 1$ (we can always find such $\bar{\lambda}$). Then, $N^k \rho^N = N^k (\rho/\bar{\lambda})^N \bar{\lambda}^N$, where $N^k (\rho/\bar{\lambda})^N \rightarrow 0$. Hence $N^k \rho^N < \bar{\lambda}^N$ for all sufficiently large N . It follows that the convergence $\mathbf{A}^N \rightarrow \mathbf{0}$ is geometric, that is, there exists a constant $C < \infty$ and a constant $r < 1$ such that for the matrix elements $[\mathbf{A}^N]_{ij}$ it holds that $[\mathbf{A}^N]_{ij} \leq Cr^N$. The same applies to the elements of the matrix $\mathbf{A}^N = \mathbf{P} \mathbf{A}^N \mathbf{P}^{-1}$. By the submultiplicativity of a matrix it also follows that

$$\|\mathbf{A}^N\| = \|\mathbf{P} \mathbf{A}^N \mathbf{P}^{-1}\| \leq \|\mathbf{P}\| \|\mathbf{P}^{-1}\| \|\mathbf{A}^N\|$$

so that $\|\mathbf{A}^N\| \rightarrow 0$ geometrically (If the eigenvalues of $\mathbf{A}$ are distinct, $\mathbf{A}$ is diagonal with diagonal elements among the eigenvalues of $\mathbf{A}$ that are smaller than one in absolute value. Hence the argument behind the result becomes simpler). Altogether we can conclude that $\sum_{i=1}^{\infty} \mathbf{A}_i$ converges ($\mathbf{A}^0 = \mathbf{I}_n$) and it is easily seen that

$$\sum_{i=1}^{\infty} \mathbf{A}_i = (\mathbf{I}_n - \mathbf{A})^{-1}$$

Polynomials, Polynomial Matrices and Power Series

Consider the m :th degree (or order) polynomial function

$$p(z) = \sum_{k=0}^m a_k z^k$$

where a_k and z are real or complex numbers and $a_m \neq 0$. When $z \in \mathbb{C}$, the fundamental theorem of algebra implies that we can write

$$p(z) = a_m (z - \zeta_1) \cdots (z - \zeta_m),$$

where $\zeta_1, \dots, \zeta_m$ are the roots of $p(z)$ such that $p(\zeta_k) = 0$. The case $\zeta_k = \zeta_l$ ($k \neq l$) is possible and when the coefficients a_k are real ($a_k \in \mathbb{R}$ for all k) possible complex roots appear as conjugate pairs so that if $\zeta_k = x_k + iy_k$ ($i^2 = -1$) then for some $l \neq k$, $\zeta_l = x_l - iy_l \equiv \bar{\zeta}_k$.

A *polynomial matrix* is a matrix whose elements are polynomials

$$\mathbf{P}(z) = [p_{ij}(z)], \quad p_{ij}(z) = \sum_{k=0}^{m_{ij}} a_{ij,k} z^k$$

Setting $m = \max \{m_{ij}\}$ we can write

$$\mathbf{P}(z) = \sum_{k=0}^m \mathbf{A}_k z^k,$$

where $\mathbf{A}_k = [a_{ij,k}]$ and $a_{ij,k} = 0$, when $k > m_{ij}$. Given this presentation it is customary to talk about *matrix polynomials*, that is, polynomials whose coefficients are matrices.

Calculations such as $\mathbf{A}(z) + \mathbf{B}(z)$, $\mathbf{A}(z)\mathbf{B}(z)$ and the determinant $\det(\mathbf{A}(z))$ are defined as in the case of usual matrices. For a square matrix polynomial $\mathbf{A}(z)$ ($n \times n$) the inverse matrix $\mathbf{A}(z)^{-1}$ ($n \times n$) satisfies $\mathbf{A}(z)\mathbf{A}(z)^{-1} = \mathbf{I}_n$. As in the case of matrix whose elements are real numbers, we can compute the inverse matrix $\mathbf{A}(z)^{-1}$ by the formula

$$\mathbf{A}(z)^{-1} = \frac{1}{\det(\mathbf{A}(z))} \mathbf{A}(z)^* \quad (\det(\mathbf{A}) \neq 0)$$

where $\mathbf{A}(z)^* = [a_{ij}^*(z)]$ is the adjunct matrix of $\mathbf{A}(z)$ such that $a_{ij}^*(z) = (-1)^{i+j} \times \det(\mathbf{M}_{ij})$, where $\mathbf{M}_{ij}$ is the $(n-1) \times (n-1)$ matrix that results from deleting row i and column j from $\mathbf{A}$ (the same definition applies to the usual matrix). Given that a determinant can be formed by summing products of elements of the matrix, both $\det(\mathbf{A}(z))$ and $\mathbf{A}(z)^*$ are polynomials and the degree of $\det(\mathbf{A}(z))$ is at most mn . Hence, the elements of the inverse matrix $\mathbf{A}(z)^{-1}$ are rational functions of the form $\sum_{j=0}^q a_j z^j / \sum_{j=0}^r b_j z^j$ ($q, r < \infty$).

Return to the usual polynomial $p(z) = \sum_{k=0}^m a_k z^k$. It is easy to see that we can write

$$p(z) = p(1) + (1-z)q(z) \tag{13.1}$$

where

$$q(z) = \sum_{k=0}^{m-1} b_k z^k, \quad b_k = - \sum_{j=k+1}^m a_j, \quad k = 0, \dots, m-1$$

Another similar representation is

$$p(z) = p(1)z + (1-z)r(z) \tag{13.2}$$

where

$$r(z) = \sum_{k=0}^{m-1} c_k z^k, \quad c_0 = a_0, \quad c_k = - \sum_{j=k+1}^m a_j, \quad k = 1, \dots, m-1$$

These representations can be generalized to the case of power series

$$p(z) = \sum_{k=0}^{\infty} a_k z^k \quad (|z| \leq 1)$$

If we assume

$$\sum_{k=0}^{\infty} k |a_k| < \infty$$

it can be easily proved that equations (13.1) and (13.2) hold when $|z| \leq 1$ and m is replaced by ∞ . Furthermore, it holds that $\sum_{k=0}^{\infty} |b_k| < \infty$ and $\sum_{k=0}^{\infty} |c_k| < \infty$.

The above stated results generalize to matrix polynomials. Let

$$\mathbf{P}(z) = [p_{ij}(z)] = \sum_{k=0}^m \mathbf{A}_k z^k \quad (|z| \leq 1)$$

where $\mathbf{A}_k = [a_{ij,k}]$ satisfies the condition

$$\sum_{k=0}^{\infty} k \|\mathbf{A}_k\| < \infty$$

By applying the above derivations to the polynomials $p_{ij}(z)$ we can show the following generalization of (13.1)

$$\mathbf{P}(z) = \mathbf{P}(1)z + (1 - z)\mathbf{Q}(z) \quad (13.3)$$

where

$$\mathbf{Q}(z) = \sum_{k=0}^{\infty} \mathbf{B}_k z^k, \quad \mathbf{B}_k = - \sum_{j=k+1}^{\infty} \mathbf{A}_j \quad \text{and} \quad \sum_{k=1}^{\infty} \|\mathbf{B}_k\| < \infty$$

Similarly we can generalize (13.2) to

$$\mathbf{P}(z) = \mathbf{P}(1)z + (1 - z)\mathbf{R}(z) \quad (13.4)$$

where

$$\mathbf{R}(z) = \sum_{k=0}^{\infty} \mathbf{C}_k z^k, \quad \mathbf{C}_0 = \mathbf{A}_0, \quad \mathbf{C}_k = - \sum_{j=k+1}^{\infty} \mathbf{A}_j, \quad k \geq 1 \quad \text{and} \quad \sum_{k=1}^{\infty} \|\mathbf{C}_k\| < \infty$$

If $\mathbf{P}(z)$ is degree m matrix polynomial (i.e., $\mathbf{A}_k = \mathbf{0}$ for all $k > m$), then $\mathbf{Q}(z)$ and $\mathbf{R}(z)$ are degree $m - 1$ matrix polynomials.

Partitioned Matrices and the Kronecker Product

A partitioned matrix is a matrix whose elements are matrices

$$\mathbf{A} = [\mathbf{A}_{ij}] = \begin{bmatrix} \mathbf{A}_{11} & \cdots & \mathbf{A}_{1m} \\ \vdots & & \vdots \\ \mathbf{A}_{n1} & \cdots & \mathbf{A}_{nm} \end{bmatrix}$$

If $\mathbf{A} = [\mathbf{A}_{ij}]$ and $\mathbf{B} = [\mathbf{B}_{ij}]$, then (assuming that the dimensions agree with the definitions of the calculations)

$$\mathbf{A} + \mathbf{B} = [\mathbf{A}_{ij} + \mathbf{B}_{ij}] \quad \text{and} \quad \mathbf{AB} = \left[\sum_k \mathbf{A}_{ik} \mathbf{B}_{kj} \right]$$

These results follow from standard matrix addition and multiplication operations. In addition, when $\det(\mathbf{A}) \neq 0$ and $\det(\mathbf{A}_{ii}) \neq 0$ ($i = 1, 2$) we have

$$\begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{A}_{1.2}^{-1} & -\mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{A}_{2.1}^{-1} \\ -\mathbf{A}_{22}^{-1} \mathbf{A}_{21} \mathbf{A}_{1.2}^{-1} & \mathbf{A}_{2.1}^{-1} \end{bmatrix},$$

where $\mathbf{A}_{i \cdot j} = \mathbf{A}_{ii} - \mathbf{A}_{ij}\mathbf{A}_{jj}^{-1}\mathbf{A}_{ji}$. It also holds that (a) $\det(\mathbf{A}) = \det(\mathbf{A}_{11})\det(\mathbf{A}_{22})$, when $\mathbf{A}_{12} = \mathbf{0}$ or $\mathbf{A}_{21} = \mathbf{0}$, (b) $\det(\mathbf{A}) = \det(\mathbf{A}_{11})\det(\mathbf{A}_{2 \cdot 1})$, when $\det(\mathbf{A}_{11}) \neq 0$, and (c) $\det(\mathbf{A}) = \det(\mathbf{A}_{22})\det(\mathbf{A}_{1 \cdot 2})$, when $\det(\mathbf{A}_{22}) \neq 0$.

The Kronecker product of $\mathbf{A}$ ($n \times m$) and $\mathbf{B}$ ($p \times q$) is the $np \times mq$ matrix

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & \cdots & a_{1m}\mathbf{B} \\ \vdots & & \vdots \\ a_{n1}\mathbf{B} & \cdots & a_{nm}\mathbf{B} \end{bmatrix} = [a_{ij}\mathbf{B}]$$

The Kronecker product has the following properties

- $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = (\mathbf{AC} \otimes \mathbf{BD})$
- $(\mathbf{A} \otimes \mathbf{B})' = \mathbf{A}' \otimes \mathbf{B}'$
- $(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}$
- $\mathbf{A} \otimes (\mathbf{B} + \mathbf{C}) = \mathbf{A} \otimes \mathbf{B} + \mathbf{A} \otimes \mathbf{C}$
- If $\mathbf{A}$ ($n \times n$) has eigenvalues $\lambda_1, \dots, \lambda_n$ and $\mathbf{B}$ ($m \times m$) has eigenvalues $\mu_1, \dots, \mu_m$, then $\mathbf{A} \otimes \mathbf{B}$ has eigenvalues $\lambda_i\mu_j$, ($i = 1, \dots, n, j = 1, \dots, m$).
- $\text{vec}(\mathbf{ABC}) = (\mathbf{C}' \otimes \mathbf{A})\text{vec}(\mathbf{B})$

Properties of Conditional Expectation

Consider the prediction problem where we seek to predict the value of a random (scalar) variable Y given the value $\mathbf{x}$ of a random vector $\mathbf{X}$. Probability theory tells us that the optimal in the sense of mean square error prediction is obtained by the conditional expectation (conditional mean) of Y given $\mathbf{X} = \mathbf{x}$, denoted $E(Y|\mathbf{X} = \mathbf{x})$. That is, for any function $g(\mathbf{X})$ it holds that

$$E[(Y - E(Y|\mathbf{X} = \mathbf{x}))^2] \leq E[(Y - g(\mathbf{X}))^2]$$

When $\mathbf{X}$ and Y are jointly continuous, the conditional mean is

$$E(Y|\mathbf{X} = \mathbf{x}) = \int_{-\infty}^{\infty} y f_{Y|\mathbf{X}}(y; \mathbf{x}) dy = \int_{-\infty}^{\infty} y \frac{f_{Y, \mathbf{X}}(y, \mathbf{x})}{f_{\mathbf{X}}(\mathbf{x})} dy,$$

where $f_{Y, \mathbf{X}}(y, \mathbf{x})$ is the joint density of $(Y, \mathbf{X})$, $f_{\mathbf{X}}(\mathbf{x}) = \int_{-\infty}^{\infty} f_{Y, \mathbf{X}}(y, \mathbf{x}) dy$ is the marginal density of $\mathbf{X}$, and $f_{Y|\mathbf{X}}(y; \mathbf{x}) = f_{Y, \mathbf{X}}(y, \mathbf{x})/f_{\mathbf{X}}(\mathbf{x})$ is the density of Y conditional on $\mathbf{X} = \mathbf{x}$. These definition as well as the ones that follow hold also when $\mathbf{Y}$ is a random vector.

The conditional mean has alternative interpretations. As such, $E(Y|\mathbf{X} = \mathbf{x})$ returns the mean of a conditional distribution ($f_{Y|\mathbf{X}}(y|\mathbf{x})$), a certain *non-random number*. On the other hand, $E(Y|\mathbf{X} = \mathbf{x}) = m(\mathbf{x})$ is a *non-random function* that returns the conditional mean of Y for any

given value $\mathbf{x}$ of $\mathbf{X}$. The conditional mean can also be regarded as a random variable when the value of the conditioning variable $\mathbf{X}$ is not thought as fixed. In this case, the conditional mean can be seen as a transformation $m(\mathbf{X})$ of the random variable $\mathbf{X}$, where the transformation $m(\mathbf{x}) = E(Y|\mathbf{X} = \mathbf{x})$. We may write $E(Y|\mathbf{X})$ to indicate that the conditional mean is regarded as a random variable.

In advanced probability theory conditional expectation is defined directly as a random variable and the conditioning random vector variable $\mathbf{X}$ can be of infinite dimension. This type of situation comes across when we seek to predict a random process assuming that the past values (prior to the date of forecast) of the process $\{y_t, t = 0, \pm 1, \dots\}$ are available. For us it suffices to know that the following basic properties of the conditional mean hold also in such a general situation.

- CE1 $E(aY_1 + bY_2|\mathbf{X}) = aE(Y_1|\mathbf{X}) + bE(Y_2|\mathbf{X})$, where a and b are nonrandom constants (linearity)
- CE2 $E(Y|\mathbf{X}) = E(Y)$, when Y and $\mathbf{X}$ are independent
- CE3 $E(Y) = E[E(Y|\mathbf{X})]$, (law of iterated expectation)
- CE4 $E(g(\mathbf{X})Y|\mathbf{X}) = g(\mathbf{X})E(Y|\mathbf{X})$ for any function g (such that $g(\mathbf{X})Y$ is a well defined random variable with finite expectation) \end{itemize}

Differentiation Rules

When $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable, we denote

$$\frac{\partial}{\partial \mathbf{x}} f(\mathbf{x}) = \left[\frac{\partial}{\partial x_1} f(\mathbf{x}) \cdots \frac{\partial}{\partial x_n} f(\mathbf{x}) \right]' \quad (n \times 1)$$

and

$$\frac{\partial}{\partial \mathbf{x}'} f(\mathbf{x}) = \left[\frac{\partial}{\partial x_1} f(\mathbf{x}) \cdots \frac{\partial}{\partial x_n} f(\mathbf{x}) \right]' \quad (1 \times n)$$

When $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable, we denote

$$\frac{\partial}{\partial \mathbf{x}} \mathbf{f}(\mathbf{x}) = \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_1} f_m(\mathbf{x}) \\ \vdots & & \vdots \\ \frac{\partial}{\partial x_n} f_1(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_m(\mathbf{x}) \end{bmatrix}$$

It is easy to check that $\partial \mathbf{a}' \mathbf{x} / \partial \mathbf{x} = \mathbf{a}$ ($n \times 1$) and $\partial \mathbf{A} \mathbf{x} / \partial \mathbf{x} = \mathbf{A}'$, when $\mathbf{A}$ is $m \times n$. Furthermore, when $\mathbf{A}$ is $n \times n$ we have

$$\begin{aligned} \frac{\partial}{\partial \mathbf{x}} \mathbf{x}' \mathbf{A} \mathbf{x} &= (\mathbf{A} + \mathbf{A}') \mathbf{x} \\ &= 2\mathbf{A} \mathbf{x}, \text{ if } \mathbf{A} = \mathbf{A}' \end{aligned}$$

When $\mathbf{B} = \mathbf{B}'$ is $m \times m$ we obtain using the above rules that

$$\frac{\partial}{\partial \mathbf{x}} (\mathbf{c} - \mathbf{A}\mathbf{x})' \mathbf{B} (\mathbf{c} - \mathbf{A}\mathbf{x}) = -2\mathbf{A}'\mathbf{B}\mathbf{c} + 2\mathbf{A}'\mathbf{B}\mathbf{A}\mathbf{x} \quad (\mathbf{c} \ m \times 1 \text{ and } \mathbf{A} \ m \times n)$$

Suppose now that $\mathbf{A}$ ($n \times n$) is nonsymmetric and unrestricted (that is, the value a_{ij} is unrelated to the value a_{kl} , when either $i \neq k$ or $j \neq l$). Then

$$\frac{\partial}{\partial \mathbf{A}} \mathbf{x}' \mathbf{A} \mathbf{x} \equiv \left[\frac{\partial}{\partial a_{ij}} \mathbf{x}' \mathbf{A} \mathbf{x} \right] = [x_i x_j] = \mathbf{x} \mathbf{x}' \quad (\mathbf{x} \ n \times 1 \text{ and } \mathbf{A} \ n \times n)$$

and

$$\frac{\partial}{\partial \mathbf{A}} \log \det(\mathbf{A}) = (\mathbf{A}^{-1})' = (\mathbf{A}')^{-1}$$

The latter result uses the fact that $\det(\mathbf{A}) = \sum_k a_{ik} a_{ik}^*$, where a_{ik}^* , the element of the adjoint matrix $\mathbf{A}^*$ of $\mathbf{A}$ (see Appendix ??Matrixpoly), does not depend on a_{ij} . By the chain rule we get

$$\frac{\partial}{\partial a_{ij}} \log \det(\mathbf{A}) = \frac{1}{\det(\mathbf{A})} \frac{\partial \det(\mathbf{A})}{\partial a_{ij}} = \frac{1}{\det(\mathbf{A})} a_{ij}^* = [\mathbf{A}^{-1}]_{ji} = [(\mathbf{A}^{-1})']_{ij}$$

where the third equation uses the formula of the inverse matrix $\mathbf{A}^{-1} = (\det(\mathbf{A}))^{-1} \mathbf{A}^*$ and the definition of the adjoint matrix (see Appendix 13.2).

The last two results above do not hold when $\mathbf{A}$ is symmetric. If $\mathbf{A}$ ($n \times n$) is symmetric $\mathbf{A} = \mathbf{A}'$, then $a_{ij} = a_{ji}$ and we obtain

$$\frac{\partial}{\partial a_{ij}} \mathbf{x}' \mathbf{A} \mathbf{x} = \begin{cases} x_i^2, & \text{when } i = j \\ 2x_i x_j, & \text{when } i \neq j \end{cases}$$

and

$$\frac{\partial}{\partial a_{ij}} \log(\det(\mathbf{A})) = \begin{cases} [\mathbf{A}^{-1}]_{ii}, & \text{when } i = j \\ 2[\mathbf{A}^{-1}]_{ji}, & \text{when } i \neq j \end{cases}$$

Using matrix notation these formulae can be expressed as

$$\frac{\partial}{\partial \mathbf{A}} \mathbf{x}' \mathbf{A} \mathbf{x} = 2\mathbf{x}\mathbf{x}' - \text{diag}[\mathbf{x}\mathbf{x}']$$

and

$$\frac{\partial}{\partial \mathbf{A}} \log(\det(\mathbf{A})) = 2\mathbf{A}^{-1} - \text{diag}[\mathbf{A}^{-1}]$$

where $\text{diag}[\mathbf{A}]$ indicates a diagonal matrix whose diagonal elements are those of $\mathbf{A}$ (i.e., $\text{diag}[\mathbf{A}] = [a_{ij}I(i=j)]_{ij}$, where $I(i=j) = 1$, if $i = j$ and $I(i=j) = 0$, $i \neq j$).

Apply the above results on the log likelihood function (of Section ??Linear constraints)

$$l(\boldsymbol{\pi}, \boldsymbol{\Sigma}) = \frac{T}{2} \log(\det(\boldsymbol{\Phi})) - \frac{1}{2} \sum_{t=1}^T \mathbf{u}_t(\boldsymbol{\pi})' \boldsymbol{\Phi} \mathbf{u}_t(\boldsymbol{\pi})$$

We get

$$\frac{\partial}{\partial \Phi} \mathbf{u}_t(\boldsymbol{\pi})' \Phi \mathbf{u}_t(\boldsymbol{\pi}) = 2\mathbf{u}_t(\boldsymbol{\pi})\mathbf{u}_t(\boldsymbol{\pi})' - \text{diag}[\mathbf{u}_t(\boldsymbol{\pi})\mathbf{u}_t(\boldsymbol{\pi})']$$

and (using the notation $\Phi^{-1} = \Sigma$)

$$\frac{\partial}{\partial \Phi} \log(\det(\Phi)) = 2\Phi^{-1} - \text{diag}[\Phi^{-1}] = 2\Sigma - \text{diag}[\Sigma]$$

Now, the first order condition with respect to Φ of the likelihood function is seen to be

$$\frac{\partial}{\partial \Phi} l(\boldsymbol{\pi}, \Sigma) = \frac{T}{2} \{2\Sigma - \text{diag}[\Sigma]\} - \frac{1}{2} \sum_{t=1}^T \{2\mathbf{u}_t(\boldsymbol{\pi})\mathbf{u}_t(\boldsymbol{\pi})' - \text{diag}[\mathbf{u}_t(\boldsymbol{\pi})\mathbf{u}_t(\boldsymbol{\pi})']\} = \mathbf{0}$$

By looking at this matrix equation element by element, one finds that it can be re-expressed as

$$\frac{\partial}{\partial \Phi} l(\boldsymbol{\pi}, \Sigma) = \frac{T}{2} \Sigma - \frac{1}{2} \sum_{t=1}^T \mathbf{u}_t(\boldsymbol{\pi})\mathbf{u}_t(\boldsymbol{\pi})' = \mathbf{0}$$

We obtain the same result whether or not the differentiation operation takes the symmetry restriction into account.

Appendix B: Asymptotic results

We briefly review selected results on stochastic convergence and convergence in distribution (the underlying concepts of convergence are assumed known). The first results concerns the behavior of stochastic convergence when a continuous transformation is applied. Here and later a constant means a nonrandom finite quantity.

Theorem B.1. If the sequence of random variables $\mathbf{X}_1, \mathbf{X}_2, \dots$ ($k \times 1$) converges in stochastics to a constant vector $\mathbf{c}$, i.e., $\mathbf{X}_n \xrightarrow{p} \mathbf{c}$, and if $\mathbf{g} : \mathbb{R}^k \rightarrow \mathbb{R}^l$ is continuous at $\mathbf{c}$, then $\mathbf{g}(\mathbf{X}_n) \xrightarrow{p} \mathbf{g}(\mathbf{c})$.

This result holds also for matrices. The matrix case can be handled by using vectorization (see Appendix ??). The same applies to the following variants of the continuous mapping theorem.

Theorem B.2. If the sequence of random variables $\mathbf{X}_1, \mathbf{X}_2, \dots$ ($k \times 1$) converges in distribution to a random vector $\mathbf{Z}$, i.e., $\mathbf{X}_n \xrightarrow{d} \mathbf{Z}$, and if $\mathbf{g} : \mathbb{R}^k \rightarrow \mathbb{R}^l$ is continuous, then $\mathbf{g}(\mathbf{X}_n) \xrightarrow{d} \mathbf{g}(\mathbf{Z})$.

Here the continuity condition can be weakened. It suffices that $\mathbf{g}$ is continuous within the set C_g such that $\Pr\{\mathbf{Z} \in C_g\} = 1$. Given that convergence in distribution to a constant implies stochastic convergence, Theorem B.2 implies Theorem B.1 when $\mathbf{Z} = \mathbf{c}$

- Convergence to distribution follows always from stochastic convergence but the reverse is true only when there is a constant limit for the convergence to distribution.

The following theorem contains the above two theorems as special cases.

Theorem B.3. Assume the sequences $\mathbf{X}_1, \mathbf{X}_2, \dots$ ($k \times 1$) and $\mathbf{Y}_1, \mathbf{Y}_2, \dots$ ($l \times 1$) are such that $\mathbf{X}_n \xrightarrow{d} \mathbf{Z}$ and $\mathbf{Y}_n \xrightarrow{p} \mathbf{c}$, where $\mathbf{c}$ is a constant vector. Then, $(\mathbf{X}_n, \mathbf{Y}_n) \xrightarrow{d} (\mathbf{Z}, \mathbf{c})$.

The above theorems imply the following results.

Corollary B.1. Let X_n and Y_n be as in Theorem B.3 with $k = l = 1$. Then

- (i) $X_n + Y_n \xrightarrow{d} Z + c$
- (ii) $X_n Y_n \xrightarrow{d} Zc$
- (iii) $X_n / Y_n \xrightarrow{d} Z/c$, when $c \neq 0$.

The first result holds also for vectors.

Corollary B.2. Let $\mathbf{X}_1, \mathbf{X}_2, \dots$ and $\mathbf{Z}(k \times 1)$ be as in Theorem B.3 and let $\mathbf{Y}_1, \mathbf{Y}_2, \dots (l \times 1)$ be a sequence of $l \times k$ matrices such that $\mathbf{Y}_n \xrightarrow{p} \mathbf{A}$, where $\mathbf{A}(l \times k)$ is a constant matrix. Then

(i) $\mathbf{Y}_n \mathbf{X}_n \xrightarrow{d} \mathbf{A} \mathbf{Z}$

If $k = l$, it also holds that

(ii) $\mathbf{X}_n' \mathbf{Y}_n \mathbf{X}_n \xrightarrow{d} \mathbf{Z}' \mathbf{A} \mathbf{Z}$

(iii) $\mathbf{X}_n' \mathbf{Y}_n^{-1} \mathbf{X}_n \xrightarrow{d} \mathbf{Z}' \mathbf{A}^{-1} \mathbf{Z}$, when $\mathbf{A}$ is nonsingular (i.e., invertible).

These results follow from the continuity of the functions $(\mathbf{A}, \mathbf{x}) \mapsto \mathbf{A}\mathbf{x}$, $(\mathbf{A}, \mathbf{x}) \mapsto \mathbf{x}'\mathbf{A}\mathbf{x}$ and $\mathbf{A} \mapsto \mathbf{A}^{-1}$ ($\mathbf{A}$ nonsingular) and Theorems B.2 and B.3. When $\mathbf{Z}$ is a constant, we can replace convergence in distribution by (the stronger condition of) stochastic convergence. The continuity of an inverse matrix can be justified as follows. The determinant function $\mathbf{A} \mapsto \det(\mathbf{A})$ is continuous, because the determinant can be formed by summing products of matrix elements. As the elements of $\mathbf{A}^{-1}$ can be expressed in terms of the ratios of $\mathbf{A}$ and the determinants of its submatrices (see Appendices A.4 and A.7), the function $\mathbf{A} \mapsto \mathbf{A}^{-1}$ is a composition function of continuous functions and hence continuous.

Corollary 1 (iii) assumes implicitly that $\mathbf{Y}_n \neq \mathbf{0}$ for all n (similarly Corollary 1 (iii) assumes implicitly that $\mathbf{Y}_n$ is nonsingular for all n). Although the assumption $\mathbf{Y}_n \xrightarrow{p} \mathbf{c} \neq \mathbf{0}$ implies that $\Pr\{\mathbf{Y}_n \neq \mathbf{0}\}$ tends to 1 (as $n \rightarrow \infty$), the condition $\mathbf{Y}_n \neq \mathbf{0}$ is not guaranteed to hold for finite values of n without additional assumptions (such as the continuity of $\mathbf{Y}_n$). This theoretical difficulty could be taken into account by defining $1/\mathbf{Y}_n$ by some arbitrary way without any effect on the presented result on convergence in distribution. As is usual in the literary, this issue is not taken into account explicitly. A similar interpretation can be made in the case of Corollary 2 (iii).

The above results together with the mean value theorem can be used to prove the *delta method*, which is in the real valued case as follows. Let $\theta \in \Theta$ be the parameter of interest and θ_0 its “true” value. Assume for simplicity that the parameter space Θ is an open interval. Consider an asymptotically normal estimator $\hat{\theta}_n$ for θ_0 , i.e.,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathbf{N}(0, \sigma^2(\theta_0)), \quad (\sigma^2(\theta_0) > 0)$$

If $h : \Theta \rightarrow \mathbb{R}$ is continuously differentiable and $h'(\theta_0) \neq 0$, then for the estimator $h(\hat{\theta}_n)$ of the transformation $h(\theta_0)$ it holds that

$$\sqrt{n}(h(\hat{\theta}_n) - h(\theta_0)) \xrightarrow{d} \mathbf{N}(0, [h'(\theta_0)]^2 \sigma^2(\theta_0))$$

The results can be generalized to the vector case as follows. Let $\Theta \subset \mathbb{R}^d$ be an open and convex set and $h : \Theta \rightarrow \mathbb{R}$ a continuously differentiable function. Assume that the gradient vector

$\partial h(\boldsymbol{\theta})/\partial \boldsymbol{\theta}' = [\partial h(\boldsymbol{\theta})/\partial \theta_1 \ \cdots \ \partial h(\boldsymbol{\theta})/\partial \theta_d]$ satisfies $\partial h(\boldsymbol{\theta}_0)/\partial \boldsymbol{\theta} \neq \mathbf{0}$ and that for the estimator $\hat{\boldsymbol{\theta}}_n$ ($d \times 1$) it holds that

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} \mathbf{N}(\mathbf{0}, \boldsymbol{\Sigma}(\boldsymbol{\theta}_0)), \quad \boldsymbol{\Sigma}(\boldsymbol{\theta}_0) \text{ is positive definite}$$

Then,

$$\sqrt{n}(h(\hat{\boldsymbol{\theta}}_n) - h(\boldsymbol{\theta}_0)) \xrightarrow{d} \mathbf{N}(0, V(\boldsymbol{\theta}_0)), \quad V(\boldsymbol{\theta}_0) = [\partial h(\boldsymbol{\theta})/\partial \boldsymbol{\theta}'] \boldsymbol{\Sigma}(\boldsymbol{\theta}_0) [\partial h(\boldsymbol{\theta})/\partial \boldsymbol{\theta}]$$

The delta method can be generalized further to a situation where $\mathbf{h}$ is vector valued. In this case, the mean value theorem must be applied separately to each component of $\mathbf{h}$.